Looped_Transformer

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1 Looped Transformer

1.1 理论

Looped Transformer 所有层共享相同的权重 θ，每层的输入是前一层输出与初始输入的和:

$$
\begin{aligned}
h_0 &= \text{input} \\
h_l &= \text{TransformerBlock}(h_{l-1} + h_0 | \theta) \quad \text{for } l = 1, 2, ..., L
\end{aligned}
$$

1.1.1 截断损失函数

$$
\text{Loss}(\theta) = \mathbb{E}_P \left[\frac{1}{b - b_0} \sum_{t=b_0}^{b} \frac{1}{k+1} \sum_{i=0}^{k} (Y_t(P^i|\theta) - f(x_{i+1}))^2 \right]
$$

其中 - $\min_\theta$：这是我们在寻找一组最优的物理定律（模型参数 θ）。- $\mathbb{E}_P$：对所有生成的 Prompt 分布求期望。在统计物理中，这相当于求“系综平均”，即我们在无数种不同的初始条件（数据集）下测试这个定律是否普适。 - $\frac{1}{b-b_0} \sum_{t=b_0}^{b}$：这是时间平均。$b$ 是总迭代次数，T 是窗口大小，$b_0 = \max(b - T, 0)$。我们只关心系统演化到“稳态”附近的这段时间内 (也就是 b_0 到 b 的表现)，前面瞬态的误差我们不计较。 - $\frac{1}{k+1} \sum_{i=0}^{k}$：这是空间/序列平均。也就是在给定不同长度的上下文 P^i 时，模型预测下一个点的值与真实函数 $f(x_{i+1})$ 之间的均方误差。 - $Y_t(P^i|\theta)$：在时间 t、给定历史信息 P^i 和系统参数 θ 时，系统当前的状态输出。

1.1.2 下游任务

- 数据生成方式：这篇论文解决的是纯粹的数学数据拟合问题。数据集是在训练循环中即时生成的浮点数向量。
- 以最基础的 **线性回归（Linear Regression）** 为例：
  - 维度设定：问题维度 $d = 20$，上下文样本数 $k = 40$。
  - 生成逻辑：每次迭代前，随机采样一个真实的参数 $w \sim \mathcal{N}(0, I_d)$。
  - 采样输入：生成 $k + 1$ 个输入样本（包含测试样本）$x_i \sim \mathcal{N}(0, I_d)$。

- 计算标签：$y_i = w^T x_i$。
- 拼接成 Prompt：最终喂给模型的序列 P 是一个交替的序列 $(x_1, y_1, x_2, y_2, \ldots, x_k, y_k, x_{test}, y_{dummy})$。

• 线性回归可以进一步改成非线性回归（比如 ReLU 与 Linear 叠加），或者更复杂的物理系统。

把 Prompt 映射到高维 Embedding 空间　双通道映射与“拉链式”交织

由于 x 和 y 代表不同的物理量（一个是空间坐标/特征，一个是目标观测值），我们需要赋予它们独立的投影空间：

1. 定义两个独立的线性层：

```
read_in_x = nn.Linear(20, 256)
read_in_y = nn.Linear(1, 256)
```

2. 把前 `k` 个 `x_i` 丢给 `read_in_x`，变成张量 `X_emb`，形状为 `[batch, k, 256]`。
3. 把前 `k` 个 `y_i` 丢给 `read_in_y`，变成张量 `Y_emb`，形状为 `[batch, k, 256]`。
4. 像拉拉链一样，在 `seq_len` 维度上把 `X_emb`、`Y_emb` 交织（Interleave）起来，拼接成最终长度为 `2k` 的提示向量 `P`。

在工程实现中，我们在 x_{test} 后填充了一个 y_{dummy}（如 0），使实际拼接长度变为 `2k+2`（偶数）。因此，真正对 x_{test} 的预测结果位于输出序列的倒数第二个位置，即索引 `[-2]` 处。

1.2 架构各模块代码

1.2.1 import

[1]:
```
import torch
import torch.nn as nn
import torch.nn.functional as F
import matplotlib.pyplot as plt
import time
from operator import sub
import copy
import numpy as np
import math
import concurrent.futures
import threading
import os
```

1.2.2 位置编码 (PE, Position Encoding)

绝对位置编码 (APE, Absolute Position Encoding)

[2]:
```python
class APE(nn.Module):
    def __init__(self, d_model, max_seq_len=4096, b=10000):
        super().__init__()
        theta_i = 1 / (b ** (torch.arange(0, d_model, 2).float() / d_model)) # [d_model/2]
        m = torch.arange(max_seq_len).float() # [max_seq_len]
        m_theta_i = torch.outer(m, theta_i) # [max_seq_len, d_model/2]
        # APE 核心: 直接生成一个完整的 pe 矩阵, 偶数维度放 sin, 奇数维度放 cos
        stacked=torch.stack((torch.sin(m_theta_i), torch.cos(m_theta_i)), dim=-1) # [max_seq_len, d_model/2, 2]
        pe=stacked.flatten(1,2) # [max_seq_len, d_model]
        self.register_buffer('pe', pe.unsqueeze(0)) # shape: [1, max_seq_len, d_model]

    def forward(self, x):
        # x shape: [batch_size, seq_len, d_model]
        return x + self.pe[:, :x.shape[1], :]
```

Learned APE

[3]:
```python
class LearnedAPE(nn.Module):
    def __init__(self, d_model, max_seq_len=4096):
        super().__init__()
        self.pe = nn.Embedding(max_seq_len, d_model)

    def forward(self, x):
        # x shape: [batch_size, seq_len, d_model]
        seq_len = x.shape[1]
        positions = torch.arange(seq_len, device=x.device)
        return x + self.pe(positions).unsqueeze(0)
```

ALiBi (Attention with Linear Biases) ALiBi 通过更改内积的方式引入线性位置偏置，从而实现了位置编码的功能。

$$\text{Score}(q_i, k_j) = q_i \cdot k_j - m \cdot (i - j) \quad (i \geq j)$$

其中 m 是一个与头数 (H) 相关的斜率参数，硬编码为

$$m_h = 2^{-\frac{8h}{H}} \quad h = 0, 1, ..., H - 1$$

[4]:
```python
class ALiBi(nn.Module):
    def __init__(self, num_heads, max_seq_len=4096):
        super().__init__()
        dist = torch.arange(max_seq_len) # [max_seq_len]
        dist_matrix = torch.abs(dist.view(-1, 1) - dist.view(1, -1))  #␣
↪[max_seq_len, max_seq_len]
        bias = -(2**-(8*torch.arange(1,num_heads+1) / num_heads)).view(-1, 1,␣
↪1) * dist_matrix.unsqueeze(0)  # [num_heads, max_seq_len, max_seq_len]
        # 因果掩码
        tril = torch.tril(torch.ones(max_seq_len, max_seq_len)).bool() # torch.
↪tril (triangular lower) 提取下三角元素，并把上三角元素置 0
        mask = torch.zeros(max_seq_len, max_seq_len).masked_fill(~tril,␣
↪float('-inf')) # 下三角元素置 0，上三角元素置-inf
        self.register_buffer('fused_mask', bias + mask) # [num_heads,␣
↪max_seq_len, max_seq_len]

    def forward(self, seq_len):
        # score shape: [batch_size, num_heads, seq_len, seq_len]
        return self.fused_mask[:, :seq_len, :seq_len]
```

原来的 RoPE (Rotary Position Encoding)

[5]:
```python
class RoPE(nn.Module):
    def __init__(self, d_k, max_seq_len = 4096, b=10000):
        super().__init__()
        theta_i = 1/(b**(torch.arange(0,d_k,2).float()/d_k))  # \theta_i =␣
↪b^{-\frac{2i}{d}}, \quad i \in \{0,1,\dots,\frac{d}{2}-1\}
        m = torch.arange(max_seq_len).float()  # m = [0,1,...,seq_len-1]
        m_theta_i = torch.outer(m, theta_i)  # [seq_len, d_k/2]
        cos = torch.cos(torch.cat((m_theta_i, m_theta_i), dim=-1))  # [seq_len,␣
↪d_k]
        sin = torch.sin(torch.cat((m_theta_i, m_theta_i), dim=-1))  # [seq_len,␣
↪d_k]
```

```
        self.register_buffer('cos', cos[None, None, :, :])  # 好处: cos 和 sin
不需要更新参数，注册为 buffer 后会自动放到正确的设备上
        self.register_buffer('sin', sin[None, None, :, :])  # 扩充维度以适应后续
计算

    def forward(self, x): # 适用于 Q、K (V 不需要位置编码!)
        # x: [batch_size, num_heads, seq_len, d_k]
        seq_len = x.shape[2]
        d_2 = x.shape[-1] // 2
        cos = self.cos[:, :, :seq_len, :]  # [1, 1, seq_len, d_k]
        sin = self.sin[:, :, :seq_len, :]
        x_first_half = x[..., :d_2]
        x_second_half = x[..., d_2:]
        x_flip = torch.cat((-x_second_half, x_first_half), dim=-1)
        x_out = x * cos + x_flip * sin  # 旋转位置编码
        return x_out
```

自创：**Multi-Scale Untied Positional Encoding (MS-UPE)**

[6]:
```
class MS_UPE(nn.Module):
    def __init__(self,num_heads, d_k, max_seq_len = 4096, b_0=10000,␣
↪head_ratio=2):
        super().__init__()
        b=(b_0*(head_ratio**(torch.arange(0, num_heads)))).float() # [num_heads]
        # \theta_i = b^{-\frac{2i}{d}}, \quad i \in \{0,1,\dots,\frac{d}{2}-1\}
        theta_i = 1/(b.unsqueeze(1)**(torch.arange(0,d_k,2).float()/d_k).
↪unsqueeze(0))  # [num_heads, d_k/2]
        m = torch.arange(max_seq_len).float()  # m = [0,1,...,seq_len-1]
        m_theta_i = torch.einsum('m,hk->hmk', m, theta_i)  # [num_heads,␣
↪seq_len, d_k/2]
        pe = torch.cat((torch.cos(m_theta_i), torch.sin(m_theta_i)), dim=-1)  #␣
↪[num_heads, seq_len, d_k]
        self.register_buffer('pe', pe[None, :, :, :])  # 好处: pe 不需要更新参数,
注册为 buffer 后会自动放到正确的设备上

    def forward(self, x): # 适用于 Q、K (V 不需要位置编码!)
        # x: [batch_size, num_heads, seq_len, d_k]
```

```
        seq_len = x.shape[2]
        pe = self.pe[:, :, :seq_len, :]  # [1, num_heads, seq_len, d_k]
        x_out = x + pe
        return x_out
```

1.2.3 Multi-Head Attention

注：APE 应该加在 ToyModel 上而不是 TransformerBlock 上，因为我们只需要最开头输入处编码位置，而不是叠加 num_blocks 次。

[7]:
```
class MultiHeadAttention(nn.Module):
    def __init__(self, num_heads, d_model, max_seq_len=4096,␣
↪pe_type='learned_ape', b_rope_or_upe=10000, head_ratio_upe=2):
        super().__init__()
        assert d_model % num_heads == 0, "d_model must be divisible by␣
↪num_heads"
        # 维度属性
        self.num_heads = num_heads
        self.d_model = d_model
        self.d_k = d_model // num_heads
        # 权重矩阵
        self.W_Q = nn.Linear(d_model, d_model, bias=False)
        self.W_K = nn.Linear(d_model, d_model, bias=False)
        self.W_V = nn.Linear(d_model, d_model, bias=False)
        self.W_O = nn.Linear(d_model, d_model, bias=False)
        # 位置编码模块
        self.pe_type = [pe_type] if isinstance(pe_type, str) else pe_type
        self.qk_pe_modules = nn.ModuleList()
        self.sc_pe_modules = nn.ModuleList()
        for pe in self.pe_type:
            if pe == 'rope':
                pe_module = RoPE(d_k=self.d_k, max_seq_len=max_seq_len,␣
↪b=b_rope_or_upe)
                self.qk_pe_modules.append(pe_module)  # RoPE 应用于 Q 和 K
            elif pe == 'ms_upe':
                pe_module = MS_UPE(num_heads=num_heads, d_k=self.d_k,␣
↪max_seq_len=max_seq_len, b_0=b_rope_or_upe, head_ratio=head_ratio_upe)
```

```python
                self.qk_pe_modules.append(pe_module)  # MS-UPE 应用于 Q 和 K
            elif pe == 'alibi':
                pe_module = ALiBi(num_heads=num_heads, max_seq_len=max_seq_len)
                self.sc_pe_modules.append(pe_module)  # ALiBi 应用于 score 矩阵
            elif pe in ['ape', 'learned_ape']:
                pass  # APE 和 Learned APE 在输入时直接加到 x 上，不需要单独模块
        # 因果掩码
        tril = torch.tril(torch.ones(max_seq_len, max_seq_len)).bool() # torch.
↪tril (triangular lower) 提取下三角元素，并把上三角元素置 0
        mask = torch.zeros(max_seq_len, max_seq_len).masked_fill(~tril,␣
↪float('-inf')) # 下三角元素置 0，上三角元素置-inf
        self.register_buffer('mask', mask[None, None, :, :])  # [1, 1, seq_len,␣
↪seq_len]

    def forward(self, x): # x: [batch_size, seq_len, d_model]
        # 线性投影与分头
        batch_size, seq_len, _ = x.shape
        Q = self.W_Q(x).view(batch_size, seq_len, self.num_heads, self.d_k).
↪transpose(1, 2)  # [batch_size, num_heads, seq_len, d_k]
        K = self.W_K(x).view(batch_size, seq_len, self.num_heads, self.d_k).
↪transpose(1, 2)  # 交换 (1, 2) 是因为之后 softmax 时要进行矩阵乘法 (只乘后两维)
        V = self.W_V(x).view(batch_size, seq_len, self.num_heads, self.d_k).
↪transpose(1, 2)  # 在 seq_len 维度上分配注意力
        # 根据选择的 PE 类型对 Q 和 K 进行位置编码
        for pe_module in self.qk_pe_modules:
            Q = pe_module(Q)
            K = pe_module(K)
        if not self.training:
            # 缩放点积注意力与因果掩码 Causal Masking
            ## Query-Key 点积计算注意力得分
            scores = torch.matmul(Q, K.transpose(-2, -1)) / (self.d_k ** 0.5) ␣
↪# [batch_size, num_heads, seq_len, seq_len]
            if self.sc_pe_modules:
                fused_mask = self.sc_pe_modules[0](seq_len)  # [num_heads,␣
↪seq_len, seq_len]
                scores = scores + fused_mask  # 将 ALiBi 的偏置加到得分上
```

```
                attention = F.softmax(scores, dim=-1)  # [batch_size,␣
↪num_heads, seq_len, seq_len]
            else:
                ## 因果掩码和 softmax 计算注意力权重
                attention = F.softmax(scores + self.mask[..., :seq_len, :
↪seq_len], dim=-1)  # [batch_size, num_heads, seq_len, seq_len]
            self.captured_attention = attention  # 捕获注意力矩阵, 用 hook 记录下
来, 方便后续分析
            ## 乘以 Value 完成加权求和
            out = torch.matmul(attention, V)  # [batch_size, num_heads,␣
↪seq_len, d_k]
        else:
            self.captured_attention = None  # 训练模式下不捕获注意力矩阵, 节省内存
            if self.sc_pe_modules:
                fused_mask = self.sc_pe_modules[0](seq_len)  # [num_heads,␣
↪seq_len, seq_len]
                out = F.scaled_dot_product_attention(Q, K, V,␣
↪attn_mask=fused_mask, is_causal=False)  # [batch_size, num_heads, seq_len,␣
↪d_k]
            else:
                out = F.scaled_dot_product_attention(Q, K, V, is_causal=True) ␣
↪# [batch_size, num_heads, seq_len, d_k]
        # 多头拼接并乘以输出矩阵 (必须先内存连续化 (contiguous) 再 view, 否则会报
错)
        out = out.transpose(1, 2).contiguous().view(batch_size, seq_len, -1)  #␣
↪[batch_size, seq_len, d_model]
        H = self.W_O(out)  # [batch_size, seq_len, d_model]
        return H
```

1.2.4 SwiGLU 和单个 Transformer Block 架构不变

[8]:
```
class SwiGLU(nn.Module):
    def __init__(self,d_model,d_hidden):
        super().__init__()
        self.W = nn.Linear(d_model,d_hidden,bias=False)
        self.V = nn.Linear(d_model,d_hidden,bias=False)
```

```python
        self.W2 = nn.Linear(d_hidden,d_model,bias=False)
    def forward(self,x):
        x1 = F.silu(self.W(x))
        x2 = self.V(x)
        x_out = self.W2(x1 * x2)
        return x_out
```

[9]:
```python
class TransformerBlock(nn.Module):
    def __init__(self, num_heads, d_model, max_seq_len=4096, multiplier=4,␣
↪norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape',␣
↪b_rope_or_upe=10000, head_ratio_upe=2):
        super().__init__()
        if norm_type == 'rmsnorm':
            self.norm1 = nn.RMSNorm(d_model)  # 第一次归一化
            self.norm2 = nn.RMSNorm(d_model)  # 第二次归一化
        elif norm_type == 'layernorm':
            self.norm1 = nn.LayerNorm(d_model)  # 第一次归一化
            self.norm2 = nn.LayerNorm(d_model)  # 第二次归一化
        self.attention = MultiHeadAttention(num_heads=num_heads,␣
↪d_model=d_model, max_seq_len=max_seq_len, pe_type=pe_type,␣
↪b_rope_or_upe=b_rope_or_upe, head_ratio_upe=head_ratio_upe)
        if ffn_type == 'swiglu':
            self.ffn = SwiGLU(d_model=d_model, d_hidden=d_model * multiplier)
        elif ffn_type == 'gelu':
            self.ffn = nn.Sequential(
                nn.Linear(d_model, d_model * multiplier),
                nn.GELU(),
                nn.Linear(d_model * multiplier, d_model)
            )

    def forward(self, x):
        # 第一次归一化 Pre-Norm
        x_norm1 = self.norm1(x)
        # 全局信息交互 Multi-Head Attention (MHA)
        h = self.attention(x_norm1)
        # 第一次残差连接
```

```python
        x1 = x + h
        # 第二次归一化
        x_norm2 = self.norm2(x1)
        # 前馈网络 Feed-Forward Network (FFN)
        x_out = self.ffn(x_norm2)
        # 第二次残差连接
        x2 = x1 + x_out
        return x2
```

1.2.5 Attention Probe

返回全量 attention 的 probe

[10]:
```python
class AttentionProbe:
    def __init__(self):
        self.captured_data = []
    def __call__(self, module, input, output):
        if getattr(module, 'captured_attention', None) is not None:
            attention = module.captured_attention.detach().cpu().numpy()
            self.captured_data.append(attention)
        else:
            self.captured_data.append(None)  # 如果没有捕获到注意力矩阵，记录一个 None 占位符
    def reset(self):
        self.captured_data = []
```

计算 sink score 和 sink rate 的 probe 针对层 l 的第 k 个 token 位置，sink rate 的计算公式定义如下： - Sink Score (汇聚得分)：

$$\text{sink score}_k^{(l,h)} = \frac{1}{T}\sum_{t=0}^{T-1}\alpha_{tk}^{(l,h)}$$

- Sink Rate (汇聚率)：

$$\text{sink rate}_k^{(l)} = \frac{1}{H}\sum_{h=1}^{H}\mathbb{I}(\text{sink score}_k^{(l,h)} \geq \tau)$$

公式参数说明：- $\alpha_{tk}^{(l,h)}$ 表示在层 l 的第 h 个注意力头中，从 token t 到 token k 的注意力权重。 - T 是序列的总长度。 - H 是每层注意力头的总数量。 - $\mathbb{I}$ 是指示函数（当条件 sink score$_k^{(l,h)} \geq \tau$ 成立时取值为 1，否则为 0）。 - 阈值 τ 通常被默认设置为 0.3。

[11]:
```python
class SinkMetricsProbe:
    def __init__(self, threshold=0.3):
        self.threshold = threshold
        self.captured_scores = []
        self.captured_rates = []

    def __call__(self, module, input, output):
        if getattr(module, 'captured_attention', None) is not None:
            with torch.no_grad(): # attention 矩阵仍在 GPU 上
                attention = module.captured_attention # [batch_size, num_heads,␣
↪seq_len, seq_len]
                sink_attention = attention[:, :, :, 0]  # [batch_size,␣
↪num_heads, seq_len]
                score_head = sink_attention.mean(dim=2).mean(dim=0)  #␣
↪[num_heads]
                score = score_head.mean().item()  # 平均所有头的 sink score
                rate = (score_head >= self.threshold).float().mean().item()  #␣
↪大于等于 threshold 的头占总头数的比例
                self.captured_scores.append(score)
                self.captured_rates.append(rate)
        else:
            self.captured_scores.append(None)  # 如果没有捕获到注意力矩阵，记录一
个 None 占位符
            self.captured_rates.append(None)

    def reset(self):
        self.captured_scores = []
        self.captured_rates = []
```

1.2.6 Toy Model: Looped Transformer

[12]:
```python
class ToyModel(nn.Module):
    def __init__(self, num_blocks, num_heads, d_model, max_seq_len=4096,␣
↪x_init='prompt',
                 norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape',␣
↪b_rope_or_upe=10000, head_ratio_upe=2, sink_threshold=0.3,
```

```python
                 loop=True, residual_gate=(1,1), residual_gate_type='fixed', residual_random=(1,0.1)):
        super().__init__()
        self.loop = loop
        self.x_init = x_init
        if residual_gate==(0,0):
            raise ValueError("residual_gate cannot be (0,0) because then the model would be unable to learn anything (the input would be completely blocked)")
        if residual_gate[1] ==0 and x_init=='zero':
            self.x_init = 'prompt'
            print("Warning: x_init is set to 'prompt' because residual_gate[1] is 0, which would make the model completely blind.")

        if loop:
            self.transformer_block = TransformerBlock(num_heads=num_heads, d_model=d_model, max_seq_len=max_seq_len, norm_type=norm_type, ffn_type=ffn_type, pe_type=pe_type, b_rope_or_upe=b_rope_or_upe, head_ratio_upe=head_ratio_upe) # 这里保证了各层权重始终相同
            self.probe = AttentionProbe()
            self.sink_metrics_probe = SinkMetricsProbe(threshold=sink_threshold)
            self.transformer_block.attention.register_forward_hook(self.probe)
            self.transformer_block.attention.register_forward_hook(self.sink_metrics_probe)
            self.captured_attention = None
            self.captured_sink_scores = None
            self.captured_sink_rates = None
        else:
            self.transformer_block = nn.ModuleList([
                TransformerBlock(num_heads=num_heads, d_model=d_model, max_seq_len=max_seq_len, norm_type=norm_type, ffn_type=ffn_type, pe_type=pe_type, b_rope_or_upe=b_rope_or_upe, head_ratio_upe=head_ratio_upe)
                for _ in range(num_blocks)
            ])
            self.probe = [AttentionProbe() for _ in range(num_blocks)]
```

```
            self.sink_metrics_probe = [SinkMetricsProbe(threshold=sink_threshold) for _ in range(num_blocks)]
            for block, probe, sink_probe in zip(self.transformer_block, self.probe, self.sink_metrics_probe):
                block.attention.register_forward_hook(probe)
                block.attention.register_forward_hook(sink_probe)

        residual_random = torch.tensor(residual_random, dtype=torch.float32)
        if residual_gate == 'random':
            residual_gate_tensor = torch.randn(2)*residual_random[1]+residual_random[0] # 随机初始化门控参数，初始正态分布
        else:
            residual_gate_tensor = torch.tensor(residual_gate, dtype=torch.float32)  # 将输入的元组转换为张量，方便后续计算
        if residual_gate_type == 'fixed':
            self.register_buffer('residual_gate', residual_gate_tensor)  # 固定的门控参数，注册为 buffer 使其成为模型状态的一部分，但不参与梯度更新
        elif residual_gate_type == 'learnable_scalar':
            self.residual_gate = nn.Parameter(torch.ones(2)*residual_gate_tensor)  # 可学习的标量门控参数，初始值为用户指定的值
        elif residual_gate_type == 'learnable_vector':
            if residual_gate == 'random':
                self.residual_gate = nn.Parameter(torch.randn(d_model, 2)*residual_random[1]+residual_random[0]) # 可学习的向量门控参数，初始值为随机向量
            else:
                self.residual_gate = nn.Parameter(torch.ones(d_model, 2)*residual_gate_tensor)  # 可学习的向量门控参数，初始值为全 1 向量

        pe_type = self.transformer_block.attention.pe_type
        self.ape = nn.ModuleList()
        for pe in pe_type:
            if pe == 'ape':
                self.ape.append(APE(d_model, max_seq_len=max_seq_len))
            elif pe == 'learned_ape':
```

```python
                self.ape.append(LearnedAPE(d_model, max_seq_len=max_seq_len))
            elif pe in ['rope', 'ms_upe']:
                pass # RoPE 和 MS_UPE 在 MultiHeadAttention 内部处理位置编码，不需要单独的模块在这里处理

        self.num_blocks = num_blocks
        self.num_heads = num_heads
        self.d_model = d_model
        self.max_seq_len = max_seq_len
        self.pe_type = pe_type

    def forward(self, x_0, num_eff=15, current_blocks=None):
        # num_eff 是需要加入误差计算的有效层数，也即理论公式中的 T
        # x 的形状为 [batch_size, seq_len, d_model]
        a=self.residual_gate[...,0]
        b=self.residual_gate[...,1]
        for ape_module in self.ape:
            x_0 = ape_module(x_0) # 绝对位置编码只加在最开始输入处
        if self.x_init == 'prompt':
            x = x_0
        elif self.x_init == 'zero':
            x = torch.zeros_like(x_0) # 直接用全零向量作为初始输入，测试模型是否能从零开始学习
        else:
            raise ValueError(f"Invalid x_init value: {self.x_init}. Expected 'prompt' or 'zero'.")
        if current_blocks is None:
            current_blocks = self.num_blocks
        b_0 = max(0, current_blocks - num_eff) # 计算需要加入误差计算的有效层数对应的起始层索引
        outputs = []
        if self.loop:
            self.probe.reset() # 每次前向传播前重置捕获的数据，避免混淆不同次前向传播的注意力矩阵
            self.sink_metrics_probe.reset() # 同样重置 sink metrics 的捕获数据
            for i in range(current_blocks):
```

```python
                if i == b_0:
                    x= x.detach() # 释放前面的计算图
                    x.requires_grad_(True)
                x = self.transformer_block(a*x+b*x_0)
                if i >= b_0:
                    outputs.append(x)
            self.captured_attention = list(self.probe.captured_data) # 显式 copy一份数据，避免后续被修改
            self.captured_sink_scores = list(self.sink_metrics_probe.captured_scores)
            self.captured_sink_rates = list(self.sink_metrics_probe.captured_rates)
        else:
            for probe in self.probe:
                probe.reset()
            for sink_probe in self.sink_metrics_probe:
                sink_probe.reset()
            for i, block in enumerate(self.transformer_block[:current_blocks]): # 只迭代当前有效层数，避免不必要的计算
                if i == b_0:
                    x= x.detach() # 释放前面的计算图
                    x.requires_grad_(True)
                x = block(a*x+b*x_0) # x: [batch_size, seq_len, d_model]
                if i >= b_0:
                    outputs.append(x)
            self.captured_attention = [probe.captured_data[-1] if probe.captured_data else None for probe in self.probe]
            self.captured_sink_scores = [sink_probe.captured_scores[-1] if sink_probe.captured_scores else None for sink_probe in self.sink_metrics_probe]
            self.captured_sink_rates = [sink_probe.captured_rates[-1] if sink_probe.captured_rates else None for sink_probe in self.sink_metrics_probe]
        if self.loop:
            if hasattr(self.transformer_block.attention, 'captured_attention'):
                self.transformer_block.attention.captured_attention = None
```

```python
        else:
            for block in self.transformer_block:
                if hasattr(block.attention, 'captured_attention'):
                    block.attention.captured_attention = None
        outputs = torch.stack(outputs, dim=1) # [batch_size, num_eff, seq_len,␣
↪d_model]
        return outputs
```

1.2.7 Task: Regression

转换头

[13]:
```python
class RegressionHead(nn.Module):
    def __init__(self, d_model, d_x, d_y=1, bias=False, init_scale=None):
        super().__init__()
        self.read_in_x = nn.Linear(d_x, d_model, bias=bias)
        self.read_in_y = nn.Linear(d_y, d_model, bias=bias)
        if init_scale is not None:
            nn.init.normal_(self.read_in_x.weight, mean=0.0, std=init_scale)
            nn.init.normal_(self.read_in_y.weight, mean=0.0, std=init_scale)
    def forward(self, x_data, y_data):
        # k 相当于 seq_len 的一半，因为 x 和 y 交织在一起，所以总的 seq_len 是 2k
        # x_data: [batch_size, k, d_x]
        # y_data: [batch_size, k, d_y]
        x_emb = self.read_in_x(x_data)  # [batch_size, k, d_model]
        y_emb = self.read_in_y(y_data)  # [batch_size, k, d_model]
        # 交织成 (x_1, y_1, x_2, y_2, ..., x_k, y_k)
        stacked = torch.stack((x_emb, y_emb), dim=2)  # [batch_size, k, 2,␣
↪d_model]
        Prompt = stacked.flatten(1,2)  # [batch_size, 2k, d_model]
        return Prompt
```

损失函数

[14]:
```python
class PredictionLoss(nn.Module):
    def __init__(self, d_model, d_y=1, loss_type='mse', layer_weight_decay=1.0,␣
↪seq_weight_decay=1.0, norm_type='rmsnorm'):
        super().__init__()
        if norm_type == 'rmsnorm':
```

```
            self.ln_f = nn.RMSNorm(d_model)
        elif norm_type == 'layernorm':
            self.ln_f = nn.LayerNorm(d_model)
        self.read_out = nn.Linear(d_model, d_y)
        self.layer_weight_decay = layer_weight_decay
        self.seq_weight_decay = seq_weight_decay
        if loss_type == 'mse':
            self.loss_fn = nn.MSELoss(reduction='none')
        elif loss_type == 'l1':
            self.loss_fn = nn.L1Loss(reduction='none')

    def forward(self, outputs, y_true, is_eval=False, sink_padding=None):
        # outputs:[batch_size, num_eff, seq_len, d_model]
        # y_true: [batch_size, seq_len/2, 1]
        if is_eval:
            y_outputs = outputs[:, -1, -2, :] # 只取最后一层的最后一个位置的输出作为预测值
            y_outputs = self.ln_f(y_outputs)
            y_pred_final = self.read_out(y_outputs) # [batch_size, d_y]
            return y_pred_final
        y_outputs = outputs[:, :, 0::2, :] # [batch_size, num_eff, seq_len/2, d_model]
        y_outputs = self.ln_f(y_outputs)
        y_preds = self.read_out(y_outputs) # [batch_size, num_eff, seq_len/2, d_y]
        if sink_padding is not None:
            y_preds = y_preds[:, :, sink_padding:, :]
            y_true = y_true[:, sink_padding:, :]
        y_pred_norm = torch.sqrt(torch.mean(y_preds.detach() ** 2)).item()
        y_true_norm = torch.sqrt(torch.mean(y_true.detach() ** 2)).item()
        loss_unreduced = self.loss_fn(y_preds, y_true.unsqueeze(1).expand_as(y_preds)) # [batch_size, num_eff, seq_len/2, d_y]
        if self.layer_weight_decay != 1.0:
            num_eff = outputs.shape[1]
            layer_weights = self.layer_weight_decay ** torch.arange(num_eff-1, -1, -1, device=outputs.device)  # 从最后一层到第一层递减的权重
```

```
            layer_weights = layer_weights / layer_weights.mean()
            loss_unreduced = loss_unreduced * layer_weights.view(1, num_eff, 1,␣
↪1)  # 按层加权
        if self.seq_weight_decay != 1.0:
            k=loss_unreduced.shape[2]
            seq_weights = self.seq_weight_decay ** torch.arange(k-1, -1, -1,␣
↪device=outputs.device)  # 从最后一次预测到第一次预测递减的权重
            seq_weights = seq_weights / seq_weights.mean()
            loss_unreduced = loss_unreduced * seq_weights.view(1, 1, k, 1)  # 按
位置加权
        return loss_unreduced.mean(), y_pred_norm, y_true_norm
```

拼装

[15]:
```
class RegressionSolver(nn.Module):
    def __init__(self, num_blocks, num_heads, d_model, d_x, d_y, max_seq_len,
                norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape',␣
↪b_rope_or_upe=10000, head_ratio_upe=2,
                 loop=True, loss_type='mse', sink_threshold=0.3,
                 residual_gate=(1,1), residual_gate_type='fixed',␣
↪residual_random=(1,0.1),
                 bias=False, init_scale=None, init_std=0.02, x_init='prompt',
                 layer_weight_decay=1.0, seq_weight_decay=1.0):
        super().__init__()
        if init_std == 'auto':
            init_std = d_model ** -0.5
        if init_std is not None:
            init_scale = None  # 如果用户指定了 init_std，则忽略 init_scale，使用
init_std 进行权重初始化
        self.toy_model = ToyModel(num_blocks=num_blocks, num_heads=num_heads,␣
↪d_model=d_model, max_seq_len=max_seq_len,
                                  norm_type=norm_type, ffn_type=ffn_type,␣
↪pe_type=pe_type, sink_threshold=sink_threshold,
                                  loop=loop, b_rope_or_upe=b_rope_or_upe,␣
↪head_ratio_upe=head_ratio_upe,
```

```
                                residual_gate=residual_gate,␣
↪residual_gate_type=residual_gate_type, residual_random=residual_random,␣
↪x_init=x_init)
        self.head = RegressionHead(d_model=d_model, d_x=d_x, d_y=d_y,␣
↪bias=bias, init_scale=init_scale)
        self.loss_fn = PredictionLoss(d_model=d_model, d_y=d_y,␣
↪loss_type=loss_type, layer_weight_decay=layer_weight_decay,␣
↪seq_weight_decay=seq_weight_decay, norm_type=norm_type)
        if init_std is not None:
            self.apply(lambda module: self._init_weights(module, init_std))
            std_res = init_std / math.sqrt(2 * num_blocks)
            # 遍历所有参数，通过名称匹配出残差输出层
            for name, param in self.named_parameters():
                # W_O 是 MultiHeadAttention 的输出
                # W2 是 SwiGLU 的输出
                # ffn.2 是普通 GELU 序列中的最后一个 Linear
                if name.endswith('W_O.weight') or name.endswith('W2.weight') or␣
↪name.endswith('ffn.2.weight'):
                    torch.nn.init.normal_(param, mean=0.0, std=std_res)

    def _init_weights(self, module, init_std):
        if isinstance(module, (nn.Linear, nn.Embedding)):
            nn.init.normal_(module.weight, mean=0.0, std=init_std)
            if isinstance(module, nn.Linear) and module.bias is not None:
                nn.init.zeros_(module.bias)

    def forward(self, x_data, y_data, num_eff, current_blocks=None,␣
↪is_eval=False, sink_padding=None):
        Prompt = self.head(x_data, y_data) # [batch_size, seq_len, d_model]
        outputs = self.toy_model(Prompt, num_eff=num_eff,␣
↪current_blocks=current_blocks) # [batch_size, num_eff, seq_len, d_model]
        return self.loss_fn(outputs, y_data, is_eval=is_eval,␣
↪sink_padding=sink_padding)
```

1.2.8 数据生成与加载

数据生成

线性回归数据

[16]:
```
def linear_data_generator(batch_size, seq_len, valid_d_x=None, d_x=20, d_y=1,␣
 ↪device='cpu', generator=None, ood_kwargs=None):
    with torch.no_grad():
        if ood_kwargs is None:
            ood_kwargs = {}
        if valid_d_x is None:
            valid_d_x = d_x
        w = torch.randn(batch_size, d_x, d_y, device=device,␣
 ↪generator=generator)/ (valid_d_x ** 0.5)  # 真实线性函数 w: [batch_size, d_x,␣
 ↪d_y]
        if valid_d_x < d_x:
            w[:, valid_d_x:, :] = 0.0  # 将多余的输入维度对应的权重置零，保证它们
对输出没有影响
        x_distribution = ood_kwargs.get('x_distribution', 'gaussian')
        if x_distribution == 'gaussian':
            x_data = torch.randn(batch_size, seq_len//2, d_x, device=device,␣
 ↪generator=generator)  # 输入特征 x: [batch_size, seq_len//2, d_x]
        elif x_distribution == 'uniform':
            x_data = (torch.rand(batch_size, seq_len//2, d_x, device=device,␣
 ↪generator=generator) * 2 - 1)/(3**0.5)  # 均匀分布，方差归一化为 1
        elif x_distribution == 'laplace':
            x_data = torch.distributions.Laplace(0, 1/(2**0.5)).
 ↪sample((batch_size, seq_len//2, d_x)).to(device)  # 拉普拉斯分布，方差归一化为
1

        if 'x_scale' in ood_kwargs:
            x_data = x_data * ood_kwargs['x_scale']  # 调整输入特征的尺度，制造
OOD 数据
        if 'x_mean_shift' in ood_kwargs:
            x_data = x_data + ood_kwargs['x_mean_shift']  # 平移输入特征的均值，
制造 OOD 数据
        if 'x_noise_std' in ood_kwargs:
            x_data = x_data + torch.randn_like(x_data) *␣
 ↪ood_kwargs['x_noise_std']  # 添加输入特征的噪声，制造 OOD 数据
```

```python
        if valid_d_x < d_x:
            x_data[..., valid_d_x:] = 0.0  # 将多余的输入维度对应的数据置零，保证
它们对输出没有影响

        y_data = x_data @ w  # 计算标签 y = x @ w: [batch_size, seq_len//2, d_y]

        if 'y_scale' in ood_kwargs:
            y_data = y_data * ood_kwargs['y_scale']  # 调整标签的尺度，制造 OOD
数据
        if 'y_mean_shift' in ood_kwargs:
            y_data = y_data + ood_kwargs['y_mean_shift']  # 平移标签的均值，制造
OOD 数据
        if 'y_noise_std' in ood_kwargs:
            y_data = y_data + torch.randn_like(y_data) *␣
 ↪ood_kwargs['y_noise_std']  # 添加标签的噪声，制造 OOD 数据

        return x_data, y_data
```

非线性回归数据

[17]:
```python
def nonlinear_data_generator(batch_size, seq_len, valid_d_x=None, d_x=20,␣
 ↪d_y=1, d_hidden=None, function_callable=None, device='cpu', generator=None,␣
 ↪ood_kwargs=None):
    '''
    linear(d_x->d_hidden) + nonlinear_func + linear(d_hidden->d_y) 的形式生成数
据
    function_callable: 非线性函数，如 lambda x: F.relu(2*torch.sin(x))+torch.
 ↪cos(x)
    d_hidden: 隐藏层维度，控制非线性函数的输入输出维度。
              若 d_hidden=d_x，则第一层线性变换默认设定为恒等映射，数据生成过程变为
nonlinear_func + linear 的形式；
              若 d_hidden=d_y，则第二层线性变换默认设定为恒等映射，数据生成过程变为
linear + nonlinear_func 的形式。
    '''
    with torch.no_grad():
        if ood_kwargs is None:
            ood_kwargs = {}
```

```
        if valid_d_x is None:
            valid_d_x = d_x

        w1 = torch.randn(batch_size, d_x, d_hidden, device=device,␣
↪generator=generator)/ (valid_d_x ** 0.5)  # 第一层权重 w1: [batch_size, d_x,␣
↪d_hidden]
        w2 = torch.randn(batch_size, d_hidden, d_y, device=device,␣
↪generator=generator)/ (d_hidden ** 0.5)  # 第二层权重 w2: [batch_size,␣
↪d_hidden, d_y]
        if valid_d_x < d_x:
            w1[:, valid_d_x:, :] = 0.0  # 将多余的输入维度对应的权重置零，保证它们
对输出没有影响
        x_distribution = ood_kwargs.get('x_distribution', 'gaussian')
        if x_distribution == 'gaussian':
            x_data = torch.randn(batch_size, seq_len//2, d_x, device=device,␣
↪generator=generator)  # 输入特征 x: [batch_size, seq_len//2, d_x]
        elif x_distribution == 'uniform':
            x_data = (torch.rand(batch_size, seq_len//2, d_x, device=device,␣
↪generator=generator) * 2 - 1)/(3**0.5)  # 均匀分布，方差归一化为 1
        elif x_distribution == 'laplace':
            x_data = torch.distributions.Laplace(0, 1/(2**0.5)).
↪sample((batch_size, seq_len//2, d_x)).to(device)  # 拉普拉斯分布，方差归一化为
1

        if 'x_scale' in ood_kwargs:
            x_data = x_data * ood_kwargs['x_scale']  # 调整输入特征的尺度，制造
OOD 数据
        if 'x_mean_shift' in ood_kwargs:
            x_data = x_data + ood_kwargs['x_mean_shift']  # 平移输入特征的均值，
制造 OOD 数据
        if 'x_noise_std' in ood_kwargs:
            x_data = x_data + torch.randn_like(x_data) *␣
↪ood_kwargs['x_noise_std']  # 添加输入特征的噪声，制造 OOD 数据

        if valid_d_x < d_x:
```

```
            x_data[..., valid_d_x:] = 0.0  # 将多余的输入维度对应的数据置零，保证它们对输出没有影响

        if d_x==d_hidden:
            w1=torch.eye(d_x, device=device).unsqueeze(0).expand(batch_size, -1, -1) # 如果输入维度和隐藏层维度相同，变为 nonlinear_func + linear 的形式
            if not getattr(nonlinear_data_generator, '_has_warned_w1_identity', False):
                print("Warning: w1 is set to identity matrix because d_x equals d_hidden.")
                nonlinear_data_generator._has_warned_w1_identity = True
        if d_hidden==d_y:
            w2=torch.eye(d_hidden, device=device).unsqueeze(0).expand(batch_size, -1, -1) # 如果隐藏层维度和输出维度相同，变为 linear + nonlinear_func 的形式
            if not getattr(nonlinear_data_generator, '_has_warned_w2_identity', False):
                print("Warning: w2 is set to identity matrix because d_hidden equals d_y.")
                nonlinear_data_generator._has_warned_w2_identity = True

        hidden = x_data @ w1  # [batch_size, seq_len//2, d_hidden]

        func=ood_kwargs.get('function_callable', function_callable) # Concept Shift
        # 应用非线性函数
        hidden = func(hidden)  # [batch_size, seq_len//2, d_hidden]
        # 第二层线性变换
        y_data = hidden @ w2  # [batch_size, seq_len//2, d_y]

        if 'y_scale' in ood_kwargs:
            y_data = y_data * ood_kwargs['y_scale']  # 调整标签的尺度，制造 OOD 数据
        if 'y_mean_shift' in ood_kwargs:
```

```
            y_data = y_data + ood_kwargs['y_mean_shift']  # 平移标签的均值，制造
OOD 数据
        if 'y_noise_std' in ood_kwargs:
            y_data = y_data + torch.randn_like(y_data) *␣
↪ood_kwargs['y_noise_std']  # 添加标签的噪声，制造 OOD 数据

        return x_data, y_data
```

lorenz attractor 数据　理论公式

系统的演化由以下三个一阶非线性微分方程决定：

$$\frac{dx}{dt} = \sigma(y - x)$$
$$\frac{dy}{dt} = x(\rho - z) - y$$
$$\frac{dz}{dt} = xy - \beta z$$

参数物理意义 * $x(t)$：表示流体运动的强度（对流速度）。* $y(t)$：表示上升和下降流体的温度差。* $z(t)$：表示垂直温度剖面的非线性畸变程度。* σ：Prandtl 数，影响流动特性 * ρ：Rayleigh 数，决定对流强度 * β：几何物理尺寸相关参数

经典的混沌参数 $\sigma = 10, \rho = 28, \beta = \frac{8}{3}$

此时，系统的轨迹会收敛到一个分形结构上，表现出极度敏感的混沌特性。任何微小的初始条件差异，都会导致后续轨迹的巨大分歧。

[18]:
```
def lorenz_derivative(state, sigma=10.0, rho=28.0, beta=8.0/3.0):
    # state: [batch_size, 3]
    x, y, z = state[:, 0], state[:, 1], state[:, 2]
    dx = sigma * (y - x)
    dy = x * (rho - z) - y
    dz = x * y - beta * z
    return torch.stack((dx, dy, dz), dim=1)  # [batch_size, 3]
```

四阶龙格-库塔法（简称 RK4）

- 核心思想：**多看几步，求个平均**
  - k1：先看一眼当前位置的速度。
  - k2：试探性地往前走半步，看看那里的速度。
  - k3：根据上一步的结果，再在半步的位置看一眼速度。

    – k4：试探性地往前走一步，看看终点附近的速度。
- 最后，把这四个速度按照不同的权重（通常是两头轻、中间重）加权平均，作为最终的 “靠谱速度” 来更新位置。

[19]:
```python
def rk4_step(state, dt=0.01, sigma=10.0, rho=28.0, beta=8.0/3.0):
    k1 = lorenz_derivative(state, sigma, rho, beta)
    k2 = lorenz_derivative(state + 0.5 * dt * k1, sigma, rho, beta)
    k3 = lorenz_derivative(state + 0.5 * dt * k2, sigma, rho, beta)
    k4 = lorenz_derivative(state + dt * k3, sigma, rho, beta)
    return state + (dt / 6.0) * (k1 + 2*k2 + 2*k3 + k4)
'''try:
    rk4_step = torch.compile(rk4_step)
except Exception:
    pass'''
```

[19]:
```
'try:\n    rk4_step = torch.compile(rk4_step)\nexcept Exception:\n    pass'
```

lorenz 数据计算

[20]:
```python
def lorenz_kernel(init_state, total_steps:int, dt:float, sigma:float, rho:
 ↪float, beta:float):
    # init_state: [batch_size, 3]
    batch_size = init_state.shape[0]
    device = init_state.device
    dtype = init_state.dtype
    trajectory = torch.empty((total_steps, batch_size, 3), device=device,␣
 ↪dtype=dtype)
    current_state = init_state
    for t in range(total_steps):
        current_state = rk4_step(state=current_state, dt=dt, sigma=sigma,␣
 ↪rho=rho, beta=beta)
        trajectory[t] = current_state
    trajectory = trajectory.transpose(0, 1)  # 转置为 [batch_size, total_steps,␣
 ↪3]
    return trajectory  # [batch_size, total_steps, 3]
'''
try:
    lorenz_kernel = torch.compile(lorenz_kernel)
```

```
except Exception:
    pass
'''
```

[20]: '\ntry:\n   lorenz_kernel = torch.compile(lorenz_kernel)\nexcept Exception:\n   pass\n'

把数据变成 data_generator

[21]:
```
_lorenz_cache_lock = threading.Lock()  # 定义一个锁对象，确保线程安全
def lorenz_data_generator(batch_size, seq_len, valid_d_x=None, d_x=3, d_y=3,␣
↪lorenz_kwargs=None, device='cpu', generator=None, ood_kwargs=None,␣
↪load_path=None):
    with torch.no_grad():  # 数据生成过程中不需要计算梯度，使用 no_grad 加速
        if ood_kwargs is None:
            ood_kwargs = {}
        if valid_d_x is None:
            valid_d_x = d_x
        if lorenz_kwargs is None:
            lorenz_kwargs = {}
        burn_in = lorenz_kwargs.get('burn_in', 500) # 预热步数，默认为 500 步，确
保系统状态进入混沌吸引子
        if load_path is not None and not os.path.exists(load_path):
            print(f"Warning: load_path {load_path} does not exist. Generating␣
↪new Lorenz data.")
            load_path = None
        if load_path is not None:
            with _lorenz_cache_lock:
                if not hasattr(lorenz_data_generator, "_cache") or␣
↪lorenz_data_generator._cache_path != load_path:
                    print(f"Loading Lorenz data from {load_path}...")
                    checkpoint = torch.load(load_path, map_location='cpu')
                    lorenz_data_generator._cache = checkpoint['trajectory']
                    lorenz_data_generator._cache_path = load_path
```

```python
                    if ood_kwargs.get('init_distribution', 'gaussian') != 'gaussian' or ood_kwargs.get('lorenz_noise_std', 0.0) > 0.0 or ood_kwargs.get('sigma_shift', 0.0) != 0.0 or ood_kwargs.get('rho_shift', 0.0) != 0.0 or ood_kwargs.get('beta_shift', 0.0) != 0.0:
                        print("Warning: The loaded Lorenz data was generated with a Gaussian distribution , no noise and default parameters. If you want to generate OOD data with different distribution or noise, please set load_path to None.")
            pool = lorenz_data_generator._cache # [pool_size, traj_len, 3]
            pool_size = pool.shape[0]
            indices = torch.randint(0, pool_size, (batch_size,), device='cpu')
            batch_traj = pool[indices]  # [batch_size, traj_len, 3]
            if pool.shape[1] < burn_in + seq_len//2 + 1:
                old_burn_in = burn_in
                burn_in = max(0, pool.shape[1] - seq_len//2 - 1)
                print(f"Warning: The trajectories in the pool have length {pool.shape[1]}, which is less than burn_in + seq_len//2 + 1 = {old_burn_in + seq_len//2 + 1}. Burn-in will be automatically reduced to {burn_in}.")
            batch_traj = batch_traj[:, burn_in:burn_in+seq_len//2+1, :].to(device)  # [batch_size, seq_len//2+1, 3]
            x_data = batch_traj[:, :seq_len//2, :]  # [batch_size, seq_len//2, 3]
            y_data = batch_traj[:, 1:seq_len//2+1, :]  # [batch_size, seq_len//2, 3]


        else:
            dt = lorenz_kwargs.get('dt', 0.01) # 时间步长默认为 0.01
            # 初始化
            total_steps = burn_in + seq_len//2 + 1
            init_distribution = ood_kwargs.get('init_distribution', 'gaussian')
            if init_distribution == 'gaussian':
                init_state = torch.randn(batch_size, 3, device=device, generator=generator) * 15 + torch.tensor([0, 0, 25.0], device=device)  # 初始状态（放大 15 倍并平移 (0, 0, 25) 以确保在洛伦兹吸引子中心附近)
            elif init_distribution == 'uniform':
```

```
                init_state = (torch.rand(batch_size, 3, device=device, generator=generator) * 30 - 15) + torch.tensor([0, 0, 25.0], device=device) # 均匀分布，范围 [-15, 15]，并平移 (0, 0, 25)
            elif init_distribution == 'laplace':
                u = torch.rand(batch_size, 3, device=device, generator=generator) - 0.5
                init_state = torch.tensor([0, 0, 25.0], device=device) - (15 / (2**0.5)) * torch.sign(u) * torch.log(1 - 2 * torch.abs(u))
            noise_std = ood_kwargs.get('lorenz_noise_std', 0.0)
            sigma = ood_kwargs.get('sigma_shift', 0.0)+10.0
            rho = ood_kwargs.get('rho_shift', 0.0)+28.0
            beta = ood_kwargs.get('beta_shift', 0.0)+8.0/3.0
            # 生成轨迹
            trajectory = lorenz_kernel(init_state=init_state, total_steps=total_steps, dt=dt, sigma=sigma, rho=rho, beta=beta)[:, burn_in:, :]  # [batch_size, seq_len//2+1, 3]
            if noise_std > 0.0:
                trajectory = trajectory + torch.randn_like(trajectory) * noise_std  # 在观测轨迹上添加噪声，制造 OOD 数据
            x_data = trajectory[:, :-1, :]  # x_data: [batch_size, seq_len//2, 3]
            y_data = trajectory[:, 1:, :]   # y_data: [batch_size, seq_len//2, 3]

        # 裁剪或填充至给定维度
        if valid_d_x > 3:
            x_data = F.pad(x_data, (0, valid_d_x - 3), value=0.0)  # 将输入特征维度扩展到 valid_d_x，新增的维度填充为 0
        elif valid_d_x < 3:
            x_data = x_data[:, :, :valid_d_x]  # 将输入特征维度裁剪到 valid_d_x
        if d_y > 3:
            y_data = F.pad(y_data, (0, d_y - 3), value=0.0)  # 将标签维度扩展到 d_y，新增的维度填充为 0
        elif d_y < 3:
            y_data = y_data[:, :, :d_y]  # 将标签维度裁剪到 d_y
```

```
        return x_data, y_data
```

[22]:
```
def clear_lorenz_cache():
    if hasattr(lorenz_data_generator, "_cache"):
        del lorenz_data_generator._cache
        del lorenz_data_generator._cache_path
        print("Lorenz data cache cleared.")
    else:
        print("No Lorenz data cache found.")
```

由于 lorenz 数据生成较慢，故预先生成离线数据池，训练时从中随机采样，评估时再在线生成。

[23]:
```
def create_lorenz_pool(pool_size=10000, traj_len=500, dt=0.01, sigma=10.0,␣
↪beta=8/3, rho=28.0, save_path='auto'):
    if save_path == 'auto':
        save_path = f'data/lorenz/length_{traj_len}_dt{dt}_sigma{sigma:.
↪1f}_beta{beta:.1f}_rho{rho:.1f}.pth'
    with torch.no_grad():
        # 初始化
        if torch.cuda.is_available():
            device = 'cuda'
        elif torch.backends.mps.is_available():
            device = 'mps'
        else:
            device = 'cpu'
        generator = torch.Generator(device=device)
        init_state = torch.randn(pool_size, 3, device=device,␣
↪generator=generator) * 15 + torch.tensor([0, 0, 25], device=device)  # 初始状
态（放大 15 倍并平移 (0, 0, 25) 以确保在洛伦兹吸引子中心附近）
        # 生成轨迹
        trajectory = lorenz_kernel(init_state=init_state, total_steps=traj_len,␣
↪dt=dt, sigma=sigma, rho=rho, beta=beta)  # [pool_size, traj_len, 3]
        trajectory = trajectory.cpu()  # 将数据移回 CPU，准备保存

        if save_path and os.path.exists(save_path):
            print(f"检测到已有数据文件，正在读取并进行增量拼接...")
            old_data = torch.load(save_path, map_location='cpu')
```

```
            old_trajectory = old_data['trajectory']
            if old_trajectory.shape[1] != trajectory.shape[1]:
                raise ValueError(f"拼接失败！本地老数据的时序长度为␣
↪{old_trajectory.shape[1]}，而你当前设定的时序长度要求为 {trajectory.shape[1]}。
请保持参数一致！")
            # 在 Batch 维度 (dim=0) 进行无限追加拼接
            final_trajectory = torch.cat([old_trajectory, trajectory], dim=0)
            print(f"增量拼接成功！数据池规模由 {old_trajectory.shape[0]} 条扩大至␣
↪{final_trajectory.shape[0]} 条。")
        else:
            print(f"未检测到历史文件，正在创建全新的离线数据池...")
            final_trajectory = trajectory

        if save_path:
            os.makedirs(os.path.dirname(save_path), exist_ok=True)
            torch.save({'trajectory': final_trajectory}, save_path)
            print(f"数据池已安全写入本地：{save_path}")
```

数据加载器

[24]:
```
def dataloader(batch_size=64, seq_len=80, valid_d_x=None, d_x=20, d_y=1,␣
↪device='cpu', data_type='linear', d_hidden=None, function_callable=None,␣
↪sink_padding=1, generator=None, ood_kwargs=None, lorenz_kwargs=None,␣
↪load_lorenz_from=None):
    if valid_d_x is None:
        valid_d_x = d_x
    while True:
        if data_type == 'linear':
            x_data, y_data = linear_data_generator(batch_size=batch_size,␣
↪seq_len=seq_len, valid_d_x=valid_d_x, d_x=d_x, d_y=d_y, device=device,␣
↪generator=generator, ood_kwargs=ood_kwargs)
        elif data_type == 'nonlinear':
            x_data, y_data = nonlinear_data_generator(batch_size=batch_size,␣
↪seq_len=seq_len, valid_d_x=valid_d_x, d_x=d_x, d_y=d_y, d_hidden=d_hidden,␣
↪function_callable=function_callable, device=device, generator=generator,␣
↪ood_kwargs=ood_kwargs)
        elif data_type == 'lorenz':
```

```
            x_data, y_data = lorenz_data_generator(batch_size=batch_size,␣
↪seq_len=seq_len, valid_d_x=valid_d_x, d_x=d_x, d_y=d_y, device=device,␣
↪generator=generator, ood_kwargs=ood_kwargs, lorenz_kwargs=lorenz_kwargs,␣
↪load_path=load_lorenz_from)
        if sink_padding is not None:
            x_data = F.pad(x_data, (0, 0, sink_padding, 0), value=0)  # 在
seq_len 维度最前面添加 sink_padding 个全 0 向量
            y_data = F.pad(y_data, (0, 0, sink_padding, 0), value=0)
        yield x_data, y_data
```

1.2.9 显存检测

[25]:
```
def print_vram_usage(tag="", peak=None):
    if torch.backends.mps.is_available():
        # MPS 专用 API (Apple Silicon)
        allocated = torch.mps.current_allocated_memory() / (1024 ** 3)
        driver_alloc = torch.mps.driver_allocated_memory() / (1024 ** 3)
        if peak is not None:
            print(f"[{tag}] 显存占用 -> PyTorch 分配: {peak[0]:.2f} GB | Metal
驱动分配: {peak[1]:.2f} GB")
        else:
            print(f"[{tag}] 显存占用 -> PyTorch 分配: {allocated:.2f} GB | Metal
驱动分配: {driver_alloc:.2f} GB")

    elif torch.cuda.is_available():
        # CUDA 专用 API
        allocated = torch.cuda.memory_allocated() / (1024 ** 3)
        peak = torch.cuda.max_memory_allocated() / (1024 ** 3)
        reserved = torch.cuda.memory_reserved() / (1024 ** 3)
        total = torch.cuda.get_device_properties(0).total_memory / (1024 ** 3)
        free = total - reserved
        print(f"[{tag}] 显存占用 -> 已分配: {allocated:.2f} GB | 峰值: {peak:.
↪2f} GB | 缓存池: {reserved:.2f} GB | 剩余可用: {free:.2f} GB")

    else:
        print(f"[{tag}] 当前运行在 CPU, 占用主板内存。")
```

1.3 实验模块拼装

[26]:
```
class LoopedTransformerExperiment:
    def __init__(self, num_blocks, num_heads=8, d_model=256, lr=1e-4,␣
↪gate_lr_ratio=100, max_seq_len=4096, d_x=20, d_y=1,
                 seed=None, experiment_name='Experiment',
                 norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape',␣
↪b_rope_or_upe=10000, head_ratio_upe=2,
                 optimizer_type='adam',wd_adamw=0.01,
                 loop=True, loss_type='mse', bias=False, init_scale=None,␣
↪init_std=0.02, x_init='prompt', sink_threshold=0.3,
                 residual_gate=(1,1), residual_gate_type='fixed',␣
↪residual_random=(1,0.1),
                 task='regression', timing=True, load_path=None,
                 layer_weight_decay=1.0, seq_weight_decay=1.0):
        if timing:
            start_time = time.time()
        # 设备选择：优先使用 MPS（适用于 Apple Silicon），其次是 CUDA（适用于␣
↪NVIDIA GPU），最后退回 CPU
        if torch.backends.mps.is_available():
            self.device = torch.device("mps")
        elif torch.cuda.is_available():
            self.device = torch.device("cuda")
        else:
            self.device = torch.device("cpu")
        self.is_offloaded = False
        self.generator = torch.Generator(device=self.device)
        if seed is not None:
            torch.manual_seed(seed)
            self.generator.manual_seed(seed)
        # 模型和优化器
        self.task = task
        if task == 'regression':
            self.model = RegressionSolver(num_blocks=num_blocks,␣
↪num_heads=num_heads, d_model=d_model, d_x=d_x, d_y=d_y,␣
↪max_seq_len=max_seq_len,
```

```
                                          norm_type=norm_type,␣
↪ffn_type=ffn_type, pe_type=pe_type, b_rope_or_upe=b_rope_or_upe,
                                          head_ratio_upe=head_ratio_upe,␣
↪loop=loop, loss_type=loss_type,
                                          bias=bias, init_scale=init_scale,␣
↪init_std=init_std, x_init=x_init, sink_threshold=sink_threshold,
                                          residual_gate=residual_gate,␣
↪residual_gate_type=residual_gate_type, residual_random=residual_random,
                                         ␣
↪layer_weight_decay=layer_weight_decay, seq_weight_decay=seq_weight_decay
                                          ).to(self.device)
        gate_params=[]
        base_params=[]
        self.eval_results={}
        for name, param in self.model.named_parameters():
            if 'residual_gate' in name:
                gate_params.append(param)
            else:
                base_params.append(param)
        param_groups = [{'params': base_params},
                        {'params': gate_params, 'lr': lr * gate_lr_ratio}]
        if optimizer_type == 'adam':
            self.optimizer = torch.optim.Adam(param_groups, lr=lr)
        elif optimizer_type == 'sgd':
            self.optimizer = torch.optim.SGD(param_groups, lr=lr)
        elif optimizer_type == 'adamw':
            self.optimizer = torch.optim.AdamW(param_groups, lr=lr,␣
↪weight_decay=wd_adamw)
        self.d_x = d_x
        self.d_y = d_y
        self.num_blocks = num_blocks
        self.init_time = None
        self.loss_history = []
        self.y_pred_norm_history = []
        self.y_true_norm_history = []
        self.residual_gate_history = []
```

```
        self.sink_score_history = []
        self.sink_rate_history = []
        self.experiment_name = experiment_name
        if load_path is not None:
            self.load_checkpoint(load_path)
        if experiment_name is not None:
            if timing:
                self.init_time = time.time() - start_time
                print(f"{experiment_name} initialized in {self.init_time:.2f}␣
↪seconds. Using device: {self.device}")
            else:
                print(f"{experiment_name} initialized. Using device: {self.
↪device}")

    def train(self, batch_size=64, seq_len=80, epochs=20, steps_per_epoch=1,
              data_type='linear', d_hidden=None, function_callable=None,␣
↪lorenz_kwargs=None, load_lorenz_from=None,
              num_eff=15, sink_padding=None, scheduled_training=False,
              curriculum=None, scheduler_type=None, lr_scale=0.1,␣
↪step_size_scheduler=10,
              print_every=None, timing=False, save_path=None):
        # x_data: [batch_size, k, d_x]
        # y_data: [batch_size, k, d_y]
        self.model.train()
        self.max_torch_vram = 0
        self.max_metal_vram = 0
        self.train_time = None
        dataloader_static_kwargs = dict(batch_size=batch_size, d_x=self.d_x,␣
↪d_y=self.d_y, device=self.device, data_type=data_type,␣
↪sink_padding=sink_padding, generator=self.generator, d_hidden=d_hidden,␣
↪function_callable=function_callable, ood_kwargs=None,␣
↪lorenz_kwargs=lorenz_kwargs, load_lorenz_from=load_lorenz_from)
        active_seq_len = seq_len
        active_d_x = self.d_x
        active_data_loader = dataloader(seq_len=active_seq_len,␣
↪valid_d_x=active_d_x, **dataloader_static_kwargs)
```

```
        if scheduler_type == 'cosine':
            scheduler = torch.optim.lr_scheduler.CosineAnnealingLR(self.
↪optimizer, T_max=epochs, eta_min=lr_scale*self.optimizer.
↪param_groups[0]['lr'])
        elif scheduler_type == 'step':
            scheduler = torch.optim.lr_scheduler.StepLR(self.optimizer,␣
↪step_size=step_size_scheduler, gamma=lr_scale)
        num_eff = min(num_eff, self.num_blocks) # 确保 num_eff 不超过总层数
        if curriculum is None:
            curriculum={}
        total_steps = epochs * steps_per_epoch
        if timing:
            start_time = time.time()
        for epoch in range(epochs):
            epoch_loss = 0.0
            epoch_y_pred_norm = 0.0
            epoch_y_true_norm = 0.0
            if scheduled_training:
                current_blocks = min(num_eff + epoch, self.num_blocks) # 随着训
练的进行，逐渐增加参与误差计算的有效层数
            else:
                current_blocks = self.num_blocks
            for step in range(steps_per_epoch):
                self.optimizer.zero_grad()
                global_step = epoch * steps_per_epoch + step
                if curriculum:
                    progress = min(1.0, global_step / (total_steps*curriculum.
↪get('duration_ratio',0.5)))
                    if progress==1.0:
                        curriculum={} # 课程结束，重置为普通训练
                        print(f"Curriculum ended at epoch {epoch+1}, step␣
↪{step+1}.")
                    else:
                        d_x_start = curriculum.get('d_x', self.d_x)
                        seq_len_start = curriculum.get('seq_len', seq_len)
```

```
                        current_d_x = int(d_x_start + progress * (self.d_x - d_x_start))
                        current_seq_len = int(seq_len_start + progress * (seq_len - seq_len_start))
                else:
                    current_d_x = self.d_x
                    current_seq_len = seq_len
                if current_seq_len != active_seq_len or current_d_x != active_d_x:
                    active_seq_len = current_seq_len
                    active_d_x = current_d_x
                    active_data_loader = dataloader(seq_len=active_seq_len, valid_d_x=active_d_x, **dataloader_static_kwargs)
                x_data, y_data = next(active_data_loader) # 从数据加载器中获取一个批次的数据
                loss, y_pred_norm, y_true_norm = self.model(x_data, y_data, num_eff=num_eff, current_blocks=current_blocks, is_eval=False, sink_padding=sink_padding) # 前向传播计算损失 [batch_size, num_eff, seq_len/2, d_y]
                loss = loss.mean()  # 对所有维度求平均得到一个标量损失值
                loss.backward()  # 反向传播计算梯度
                if torch.backends.mps.is_available():
                    self.max_metal_vram = max(self.max_metal_vram, torch.mps.driver_allocated_memory() / (1024 ** 3))
                    self.max_torch_vram = max(self.max_torch_vram, torch.mps.current_allocated_memory() / (1024 ** 3))
                self.optimizer.step()  # 更新模型参数
                epoch_loss += loss.item()
                epoch_y_pred_norm += y_pred_norm
                epoch_y_true_norm += y_true_norm

            self.model.eval()  # 切换到评估模式，记录注意力分数 (这样不会进行 FlashAttention 跳过 attention 矩阵生成)
            with torch.no_grad():
```

```
            _ = self.model(x_data, y_data, num_eff=num_eff,␣
↪current_blocks=current_blocks, is_eval=False, sink_padding=sink_padding)  #␣
↪再次前向传播以获取当前的残差门控参数和注意力分数
            scores = self.model.toy_model.captured_sink_scores
            rates = self.model.toy_model.captured_sink_rates
            if scores and rates:
                valid_scores = [score for score in scores if score is not␣
↪None]
                valid_rates = [rate for rate in rates if rate is not None]
                avg_sink_score = np.mean(valid_scores) if valid_scores else␣
↪0.0
                avg_sink_rate = np.mean(valid_rates) if valid_rates else 0.0
            else:
                avg_sink_score, avg_sink_rate = 0.0, 0.0
        self.model.train()  # 切换回训练模式

        if scheduler_type is not None:
            scheduler.step()  # 按 epoch 级别更新学习率
        avg_loss = epoch_loss / steps_per_epoch
        avg_y_pred_norm = epoch_y_pred_norm / steps_per_epoch
        avg_y_true_norm = epoch_y_true_norm / steps_per_epoch
        avg_y_cos = (1 + (avg_y_pred_norm / avg_y_true_norm) ** 2 -␣
↪(avg_loss / (avg_y_true_norm ** 2))) / (2 * (avg_y_pred_norm /␣
↪avg_y_true_norm)) if avg_y_true_norm != 0 and avg_y_pred_norm != 0 else 0.0
        self.y_pred_norm_history.append(avg_y_pred_norm)
        self.y_true_norm_history.append(avg_y_true_norm)
        self.loss_history.append(avg_loss)  # 记录损失值
        self.sink_score_history.append(avg_sink_score)
        self.sink_rate_history.append(avg_sink_rate)
        self.residual_gate_history.append(self.model.toy_model.
↪residual_gate.detach().cpu().numpy() if hasattr(self.model.toy_model,␣
↪'residual_gate') else None)  # 记录门控参数值
        if print_every is not None and (epoch + 1) % print_every == 0:
            avg_y_cos = (1 + (avg_y_pred_norm / avg_y_true_norm) ** 2 -␣
↪(avg_loss / (avg_y_true_norm ** 2))) / (2 * (avg_y_pred_norm /␣
↪avg_y_true_norm)) if avg_y_true_norm != 0 and avg_y_pred_norm != 0 else 0.0
```

```python
                print(f"Epoch {epoch+1}/{epochs}, Loss: {avg_loss:.4f}, Norm Ratio: {avg_y_pred_norm/avg_y_true_norm if avg_y_true_norm != 0 else 0.0:.4f}, Norm Cosine Similarity: {avg_y_cos:.4f}, Avg Sink Score: {avg_sink_score:.4f}, Avg Sink Rate: {avg_sink_rate:.4f}")
        if timing:
            self.train_time = time.time()-start_time
            print(f"Training completed in {self.train_time:.2f} seconds.")
        self.residual_gate_values = self.residual_gate_history[-1]
        self.final_loss = self.loss_history[-1]
        if save_path is not None:
            self.save_checkpoint(save_path)

    def evaluate(self, eval_name='eval', batch_size=64, seq_len=80, data_type='linear', loss_type='mse', sink_padding=1, d_hidden=None, function_callable=None, ood_kwargs=None, lorenz_kwargs=None):
        self.model.eval()
        if ood_kwargs is None:
            ood_kwargs = {}
        if 'seq_len_scale' in ood_kwargs:
            seq_len = int(seq_len * ood_kwargs['seq_len_scale'])  # 调整序列长度，制造 OOD 数据
        with torch.no_grad():
            # 在评估模式下，直接使用所有层的输出进行预测（因为已经 no_grad 了，所以不需要防止梯度爆炸了）
            # 生成一个评估用的数据批次，注意这里的 seq_len 要比训练时大 2，因为我们需要预测最后一个位置的 y 值
            x_eval, y_eval = next(dataloader(batch_size=batch_size, seq_len=seq_len+2, d_x=self.d_x, d_y=self.d_y, device=self.device, data_type=data_type, sink_padding=sink_padding,
                                             generator=self.generator, d_hidden=d_hidden, function_callable=function_callable, ood_kwargs=ood_kwargs, lorenz_kwargs=lorenz_kwargs, load_lorenz_from=None)) # Lorenz 在评估时不使用离线数据池，而是直接生成新的数据
            y_pred = self.model(x_eval, y_eval, num_eff=self.num_blocks, current_blocks=self.num_blocks, is_eval=True, sink_padding=sink_padding) # [batch_size, d_y]
```

```
            # 计算 loss
            if loss_type == 'mse':
                loss=F.mse_loss(y_pred, y_eval[:, -1, :]).item()
            elif loss_type == 'l1':
                loss=F.l1_loss(y_pred, y_eval[:, -1, :]).item()
            # 计算预测值和真实值的范数以及它们的比值
            y_pred_norm = torch.sqrt(torch.mean(y_pred ** 2)).item()
            y_true_norm = torch.sqrt(torch.mean(y_eval[:, -1, :] ** 2)).item()
            y_norm_ratio = y_pred_norm / y_true_norm if y_true_norm != 0 else 0.
↪0

            y_cos = (1 + y_norm_ratio**2 - loss/(y_true_norm**2))/ (2 *␣
↪y_norm_ratio) if y_true_norm != 0 and y_norm_ratio !=0 else 0.0
            print(f"[{eval_name}] [{self.experiment_name}]Loss: {loss:.4f},␣
↪Norm Ratio: {y_norm_ratio:.4f}, Norm Cosine Similarity: {y_cos:.4f}")
            self.eval_results.update({
                f'{eval_name}_loss': loss,
                f'{eval_name}_y_pred_norm': y_pred_norm,
                f'{eval_name}_y_true_norm': y_true_norm,
                f'{eval_name}_y_norm_ratio': y_norm_ratio,
                f'{eval_name}_y_cos': y_cos,
                f'{eval_name}_sink_scores': self.model.toy_model.
↪captured_sink_scores, # list[float]: 长度为 num_blocks, 每个元素是对应的一层中,
所有注意力头分配给第 0 个位置的平均注意力权重
                f'{eval_name}_sink_rates': self.model.toy_model.
↪captured_sink_rates,   # list[float]: 长度为 num_blocks, 每个元素是对应的一层
中, 分配给第 0 个位置的注意力权重比例 >sink_threshold 的注意力头的比例
                f'{eval_name}_captured_attention': self.model.toy_model.
↪captured_attention
            })


    def get_results(self, result_list:list[str]):
        all_results = {
            'init_time': getattr(self, 'init_time', None),                    ␣
↪               # float: 实验初始化时间 (秒)
```

```python
            'train_time': getattr(self, 'train_time', None),                    # float: 训练时间（秒）
            'loss_history': getattr(self, 'loss_history', None),                # list[float]: 训练过程中每个 epoch 的损失值
            'y_pred_norm_history': getattr(self, 'y_pred_norm_history', None),  # list[float]: 训练过程中每个 epoch 的预测值范数
            'y_true_norm_history': getattr(self, 'y_true_norm_history', None),  # list[float]: 训练过程中每个 epoch 的真实值范数
            'y_norm_error_history': [pred - true for pred, true in zip(getattr(self, 'y_pred_norm_history', []), getattr(self, 'y_true_norm_history', []))] if getattr(self, 'y_pred_norm_history', None) is not None and getattr(self, 'y_true_norm_history', None) is not None else None,                          # list[float]: 训练过程中每个 epoch 的预测值范数与真实值范数之差的绝对值
            'y_norm_ratio_history': [pred / true if true != 0 else None for pred, true in zip(getattr(self, 'y_pred_norm_history', []), getattr(self, 'y_true_norm_history', []))] if getattr(self, 'y_pred_norm_history', None) is not None and getattr(self, 'y_true_norm_history', None) is not None else None, # list[float]: 训练过程中每个 epoch 的预测值范数与真实值范数之比
            'y_cos_history': [(1 + (pred / true)**2 - (loss / (true**2))) / (2 * (pred / true)) if true != 0 and pred != 0 else None for pred, true, loss in zip(getattr(self, 'y_pred_norm_history', []), getattr(self, 'y_true_norm_history', []), getattr(self, 'loss_history', []))] if getattr(self, 'y_pred_norm_history', None) is not None and getattr(self, 'y_true_norm_history', None) is not None and getattr(self, 'loss_history', None) is not None else None, # list[float]: 训练过程中每个 epoch 的预测值与真实值的余弦相似度
            'sink_score_history': getattr(self, 'sink_score_history', None),    # list[float]: 长度为 epochs，每个元素为每个 epoch 的平均（对 num_blocks,steps_per_epoch 取平均）sink score
            'sink_rate_history': getattr(self, 'sink_rate_history', None),      # list[float]: 长度为 epochs，每个元素为每个 epoch 的平均（对 num_blocks,steps_per_epoch 取平均）sink rate
            'final_loss': self.loss_history[-1] if getattr(self, 'loss_history', None) else None,# float: 最终的损失值
```

```
            'final_y_pred_norm': self.y_pred_norm_history[-1] if getattr(self,␣
↪'y_pred_norm_history', None) else None,               # float: 最终的预测值范
数
            'final_y_true_norm': self.y_true_norm_history[-1] if getattr(self,␣
↪'y_true_norm_history', None) else None,               # float: 最终的真实值范
数
            'residual_gate_history': getattr(self, 'residual_gate_history',␣
↪None),                # list[np.array[2] 或 np.array[d_model, 2]]: 训练过程中每
个 epoch 的残差门控参数值
            'final_residual_gate': getattr(self, 'residual_gate_values', None),␣
↪                # np.array[2] 或 np.array[d_model, 2]: 最终的残差门控参数值
            'final_sink_score': self.sink_score_history[-1] if getattr(self,␣
↪'sink_score_history', None) else None,                # float: 最后一个 epoch
的 sink score
            'final_sink_rate': self.sink_rate_history[-1] if getattr(self,␣
↪'sink_rate_history', None) else None,                 # float: 最后一个
epoch 的 sink rate
            'captured_sink_scores': getattr(self.model.toy_model,␣
↪'captured_sink_scores', None), # list[float]: 同 eval_results 中的
sink_scores, 只不过取自训练过程中最后一次前向传播的捕获值
            'captured_sink_rates': getattr(self.model.toy_model,␣
↪'captured_sink_rates', None),   # list[float]: 同 eval_results 中的
sink_rates, 只不过取自训练过程中最后一次前向传播的捕获值
            'captured_attention': getattr(self.model.toy_model,␣
↪'captured_attention', None)      # list[np.array[batch_size, num_heads,␣
↪seq_len, seq_len]]: 训练时最后一次前向传播捕获的注意力权重
            }
        if hasattr(self, 'eval_results'):
            all_results.update(self.eval_results)  # 将评估结果添加到␣
↪all_results 中
        if any("residual_gate" in key for key in result_list) and self.
↪residual_gate_values is not None:
            if len(self.residual_gate_values.shape) == 1:
                all_results['residual_gate_history_a'] = [gate[0] for gate in␣
↪self.residual_gate_history]
```

```
                all_results['residual_gate_history_b'] = [gate[1] for gate in self.residual_gate_history]
                all_results['residual_gate_history_a_relative'] = [gate[0]-self.residual_gate_history[0][0] for gate in self.residual_gate_history]
                all_results['residual_gate_history_b_relative'] = [gate[1]-self.residual_gate_history[0][1] for gate in self.residual_gate_history]
                # 标量门控的 _mean 就是自身，便于与向量门控混合绘图
                all_results['residual_gate_history_a_mean'] = all_results['residual_gate_history_a']
                all_results['residual_gate_history_b_mean'] = all_results['residual_gate_history_b']
                all_results['residual_gate_history_a_mean_relative'] = all_results['residual_gate_history_a_relative']
                all_results['residual_gate_history_b_mean_relative'] = all_results['residual_gate_history_b_relative']
                all_results['final_residual_gate_a'] = self.residual_gate_values[0]
                all_results['final_residual_gate_b'] = self.residual_gate_values[1]
                all_results['final_residual_gate_a_mean'] = all_results['final_residual_gate_a']
                all_results['final_residual_gate_b_mean'] = all_results['final_residual_gate_b']
            elif len(self.residual_gate_values.shape) == 2:
                all_results['residual_gate_history_a_mean'] = [gate[:,0].mean() for gate in self.residual_gate_history]
                all_results['residual_gate_history_b_mean'] = [gate[:,1].mean() for gate in self.residual_gate_history]
                all_results['residual_gate_history_a_mean_relative'] = [gate[:,0].mean()-self.residual_gate_history[0][:,0].mean() for gate in self.residual_gate_history]
                all_results['residual_gate_history_b_mean_relative'] = [gate[:,1].mean()-self.residual_gate_history[0][:,1].mean() for gate in self.residual_gate_history]
```

```
                all_results['residual_gate_history_a_std'] = [gate[:,0].std()␣
↪for gate in self.residual_gate_history]
                all_results['residual_gate_history_b_std'] = [gate[:,1].std()␣
↪for gate in self.residual_gate_history]
                all_results['final_residual_gate_a_mean'] = self.
↪residual_gate_values[:,0].mean()
                all_results['final_residual_gate_b_mean'] = self.
↪residual_gate_values[:,1].mean()
                all_results['final_residual_gate_a_std'] = self.
↪residual_gate_values[:,0].std()
                all_results['final_residual_gate_b_std'] = self.
↪residual_gate_values[:,1].std()
        if result_list is not None:
            return {key: all_results.get(key) for key in result_list}

    def offload_to_cpu(self):
        self.model.to('cpu')
        for state in self.optimizer.state.values():
            for k, v in state.items():
                if isinstance(v, torch.Tensor):
                    state[k] = v.cpu()
        self.is_offloaded = True

    def load_to_device(self):
        self.model.to(self.device)
        for state in self.optimizer.state.values():
            for k, v in state.items():
                if isinstance(v, torch.Tensor):
                    state[k] = v.to(self.device)
        self.is_offloaded = False

    def save_checkpoint(self, path='auto'):
        if path == 'auto':
            exp_name = self.experiment_name if self.experiment_name is not None␣
↪else "Experiment"
```

```
            safe_name = exp_name.replace(' ', '_').replace('(', '').replace(')', '')
            path = f'saved_checkpoints/{safe_name}.pth'
        """ 保存实验的完整状态"""
        os.makedirs(os.path.dirname(path) if os.path.dirname(path) else '.', exist_ok=True)
        checkpoint = {
            'model_state_dict': self.model.state_dict(),
            'loss_history': getattr(self, 'loss_history', []),
            'y_pred_norm_history': getattr(self, 'y_pred_norm_history', []),
            'y_true_norm_history': getattr(self, 'y_true_norm_history', []),
            'sink_score_history': getattr(self, 'sink_score_history', []),
            'sink_rate_history': getattr(self, 'sink_rate_history', []),
            'residual_gate_history': getattr(self, 'residual_gate_history', []),
        }
        if hasattr(self, 'optimizer'):
            checkpoint['optimizer_state_dict'] = self.optimizer.state_dict()
        torch.save(checkpoint, path)
        print(f" Checkpoint 已安全保存至: {path}")

    def load_checkpoint(self, path='auto'):
        """ 加载实验的完整状态"""
        if path == 'auto':
            exp_name = self.experiment_name if self.experiment_name is not None else "Experiment"
            safe_name = exp_name.replace(' ', '_').replace('(', '').replace(')', '')
            path = f'saved_checkpoints/{safe_name}.pth'
        if not os.path.exists(path):
            print(f"  找不到文件 {path}，跳过加载。")
            return
        # 统一先加载到 CPU，后续再通过 load_to_device() 转移
        checkpoint = torch.load(path, map_location='cpu', weights_only=False)
        if isinstance(checkpoint, dict) and 'model_state_dict' in checkpoint:
            self.model.load_state_dict(checkpoint['model_state_dict'])
            # 恢复历史记录（方便画图断点续联）
```

```
            self.loss_history = checkpoint.get('loss_history', [])
            self.y_pred_norm_history = checkpoint.get('y_pred_norm_history', [])
            self.y_true_norm_history = checkpoint.get('y_true_norm_history', [])
            self.sink_score_history = checkpoint.get('sink_score_history', [])
            self.sink_rate_history = checkpoint.get('sink_rate_history', [])
            self.residual_gate_history = checkpoint.
↪get('residual_gate_history', [])
            # 恢复优化器状态
            if 'optimizer_state_dict' in checkpoint and hasattr(self,␣
↪'optimizer'):
                self.optimizer.
↪load_state_dict(checkpoint['optimizer_state_dict'])
        else:
            print("检测到纯权重 state_dict 文件（或历史版本），直接加载␣
↪model_state_dict...")
            self.model.load_state_dict(checkpoint)
            self.loss_history = []
            self.y_pred_norm_history = []
            self.y_true_norm_history = []
            self.sink_score_history = []
            self.sink_rate_history = []
            self.residual_gate_history = []
        print(f" 成功从 {path} 恢复实验状态。")
```

1.4 默认配置参数

[27]:
```
def default_setup(manual=None):
    init_parameters = dict(
    # RoPE/MS-UPE
        b_rope_or_upe=10000,            # float: RoPE 或 UPE 的基数
        head_ratio_upe=2,               # float: UPE 头比例

    # MultiHeadAttention
        num_heads=8,                    # int: 注意力头数 H
        d_model=256,                    # int: Transformer 的维度 D
```

```
        max_seq_len=100,                # int: 模型支持的最大序列长度（位置编码相关）
        pe_type=['learned_ape'],        # list: 位置编码类型 ('ape', 'learned_ape', 'rope', 'ms_upe', 'alibi')

    # SinkMetricsProbe
        sink_threshold=0.3,             # float: 判断阈值，sink rate 表示所有 token 在第 0 个位置的注意力的平均值超过这个阈值的头数比例

    # TransformerBlock
        norm_type='layernorm',          # str: transformer_block 中的归一化类型 ('layernorm' 或 'rmsnorm')
        ffn_type='gelu',                # str: transformer_block 中前馈网络激活函数类型 ('gelu' 或 'swiglu')

    # ToyModel
        num_blocks=20,                  # int: Transformer 的层数 b
        loop=True,                      # bool: 是否权重共享
        residual_gate=(1,1),            # tuple or str: transformer_block 之间传递残差的门控参数初始值，使 `x=a*x+b*x_0` ((a,b) 或'random')
        residual_gate_type='fixed',     # str: 残差门控类型 ('fixed', 'learnable_scalar', 'learnable_vector')
        residual_random=(1,0.1),        # tuple (mean, std): 当 residual_gate='random' 时满足高斯分布 N(mean, std)
        x_init='zero',                  # str: x 在所有 block 之前的初始化方式 ('prompt'(x=x_0) 或 'zero'(x=0))

    # RegressionHead
        d_x=20,                         # int: 数据输入的特征维度
        d_y=1,                          # int: 数据输出的特征维度
        bias=False,                     # bool: Regression Head 是否使用偏置
        init_scale=None,                # float or None: Regression Head 的初始化缩放因子 (None 表示使用默认初始化)。仅当 init_std 为 None 时有效，如果 init_std 不为 None，则使用 init_std 进行权重初始化，并忽略 init_scale。
```

```
    # RegressionSolver
        init_std=0.02,                  # float or 'auto' or None: 所有 nn.
    ↪Linear 和 nn.Embedding 权重初始化的标准差 ('auto'表示直接取 1/sqrt(d_model),
None 表示使用默认初始化)

    # PredictionLoss
        loss_type='mse',                # str: 损失函数类型 ('mse' 或 'l1')
        layer_weight_decay=0.8,         # float: 层权重衰减因子, 控制不同层的损失
权重 (默认 0.8 表示从最后一层到第一层递减的权重, 1.0 表示不加权)
        seq_weight_decay=0.8,           # float: 序列权重衰减因子, 控制不同时间步
的损失权重 (默认 0.8 表示从最后一次预测到第一次预测递减的权重, 1.0 表示不加权)

    # LoopedTransformerExperiment
        lr=1e-4,                        # float: 学习率
        gate_lr_ratio=100,              # float: 当使用残差门控时, 门控参数的学习
率相对于模型其他参数的倍数
        seed=42,                        # int or None: 手动随机种子, 确保实验可复
现
        experiment_name='Experiment',   # str or None: 实验名称 (None 表示不打印
设备和实验初始化信息)
        timing=True,                    # bool: 是否打印初始化时间
        optimizer_type='adamw',         # str: 优化器类型 ('adam' 或 'sgd'或␣
    ↪'adamw')
        wd_adamw=0.01,                  # float: AdamW 优化器的权重衰减系数, 仅当
optimizer_type='adamw'时有效
        task='regression',              # str: 任务类型 ('regression')
        load_path=None,                 # str or 'auto'(指'saved_checkpoints/
    ↪<safe_experiment_name>.pth') or None: 预训练模型路径 (None 表示不加载预训练模
型)
    )

    train_parameters = dict(
    # MultiHeadAttention
        batch_size=64,                  # int: 数据批次大小
        seq_len=80,                     # int: 数据序列长度
```

```
    # ToyModel
        num_eff=15,                      # int: 有效层数，即参与误差计算、参与梯度
反向传播的层数 T=b-b_0

    # dataloader
        sink_padding=None,               # int or None: seq 最前面填充的 sink␣
↪token 组数 (None 表示不使用 sink padding)
        d_hidden=64,                     # int: 非线性数据生成器中隐藏层的维度，仅
当 data_type='nonlinear'时有效
        function_callable=lambda x: 2**0.5 * F.relu(x), # function: 非线性数据生
成器中使用的非线性函数，仅当 data_type='nonlinear'时有效 (系数 2**0.5 是为了使方差
期望变为 1)
        lorenz_kwargs=dict(dt=0.01,burn_in=500),         # dict: 洛伦兹系统数据生
成的额外参数，仅当 data_type='lorenz'时有效
                                         # lorenz_kwargs 的可选键包括:
                                         # 'dt': float, 洛伦兹系统的时间步长
                                         # 'burn_in': int, 洛伦兹系统的预热步数，
即在正式生成训练数据之前，先运行系统 burn_in 步以达到吸引子区域
        load_lorenz_from=None,          # str or None: 洛伦兹系统数据生成的初始状
态加载路径 (None 表示随机初始化)

    # LoopedTransformerExperiment
        epochs=100,                      # int: 训练轮数
        steps_per_epoch=1,               # int: 每轮训练的步数
        data_type='linear',              # str: 回归数据类型␣
↪('linear'或'nonlinear'或'lorenz')
                                         # 注意: 当 data_type='lorenz'时，务必令
d_x=3,d_y=3 以匹配洛伦兹系统的状态空间维度
        scheduler_type=None,             # str or None: 学习率调节器类型␣
↪('cosine' 或 'step' 或 None)
        lr_scale=0.1,                    # float: 学习率调节器的缩放因子，仅当
scheduler_type 不为 None 时有效
        step_size_scheduler=10,          # int: StepLR 调度器的步长，仅当
scheduler_type='step'时有效
```

```
        scheduled_training=False,       # bool: 是否使用 kick-start，即随着训练的进行逐渐增加参与的层数 b=epoch+num_eff ，直到达到最大层数 num_blocks（用处并不大）
        curriculum=None,                # dict or None: 课程学习配置 (None 表示不使用课程学习)
                                        # curriculum 的可选键包括:
                                        # 'd_x': int, curriculum learning 的初始有效 d_x
                                        # 'seq_len': int, curriculum learning 的初始有效 seq_len
                                        # 'duration_ratio': float, curriculum learning 的持续时间占比，取值范围为 0 到 1，表示在训练的前 duration_ratio 比例的时间内逐渐增加有效 d_x 或 seq_len，之后保持不变。
        print_every=20,                 # int or None: 打印频率/epoch (None 表示不打印训练过程中的损失信息)
        timing=True,                    # bool: 是否打印训练时间
        save_path=None                  # str or 'auto'(指'saved_checkpoints/<safe_experiment_name>.pth') or None: 训练完成后模型的保存路径 (None 表示不保存模型)
    )

    eval_config=dict(
        eval_name='eval',               # str: 评估名称
        ood_kwargs=None,                # dict or None: 用于 OOD 数据生成的额外参数，仅 evaluate 时有效。
                                        # ood_kwargs 的可选键包括:
                                        # 'x_distribution': str, 输入特征的分布类型 ('gaussian', 'uniform', 'laplace') # x 分布偏移
                                        # 'x_scale': float, 输入特征的尺度调整因子                # x 尺度偏移
                                        # 'x_mean_shift': float, 输入特征的均值平移量            # x 均值偏移
                                        # 'x_noise_std': float, 输入特征的噪声标准差              # x 噪声鲁棒性
                                        # 'y_scale': float, 标签的尺度调整因子                   # y 尺度偏移
```

```
                                          # 'y_mean_shift': float, 标签的均值平移
量                                         # y 均值偏移
                                          # 'y_noise_std': float, 标签的噪声标准差␣
↪                                          # y 噪声鲁棒性
                                          # 'function_callable': function, 数据生
成函数，仅当 data_type='nonlinear'时有效   # 函数偏移
                                          # 'seq_len_scale': float, evaluate 时序
列长度缩放因子                              # 序列长度外推
                                          # 'init_distribution': str,          ␣
↪                      # (for lorenz attractor) 初始状态分布类型
                                          # 'lorenz_noise_std': float, 洛伦兹系统
噪声标准差              # (for lorenz attractor) 噪声鲁棒性
                                          # 'sigma_shift': float, 洛伦兹系统的
sigma 参数偏移量            # (for lorenz attractor) 参数偏移
                                          # 'rho_shift': float, 洛伦兹系统的 rho
参数偏移量                # (for lorenz attractor) 参数偏移
                                          # 'beta_shift': float, 洛伦兹系统的 beta
参数偏移量              # (for lorenz attractor) 参数偏移
    )

    if manual is not None:
        for key,value in manual.items():
            if key in init_parameters:
                init_parameters[key]=value
            elif key in train_parameters:
                train_parameters[key]=value
            else:
                print(f"Warning:key {key} do not exist in init_parameters and␣
↪train_parameters.")
    return init_parameters, train_parameters
```

1.5 实验台搭建

[28]:
```
class ExperimentTable:
    '''
    Attributes:
        num_experiments: int, 实验数量
        init_parameters: list of dict, 在 __init__() 方法中传入，用于指定每个实验
的初始化参数，可以覆盖默认参数
        train_parameters: list of dict, 在 __init__() 方法中传入，用于指定每个实
验的训练参数，可以覆盖默认参数
        result_lists: list of tuple of list, 在 run() 方法中传入，用于指定需要展
示的结果名称和对应的横坐标类型。

        results: list of dict，所有实验的结果，每个 dict 是一个实验的结果，为
metric 和 value 的键值对。
        如 [{'loss_history': [...], 'final_loss': 0.123,␣
↪'residual_gate_history_a_mean': [...], 'final_residual_gate_a_mean': 0.45, ..
↪.}, {...}, ...]
    '''
    def __init__(self, params_groups:list[dict], manual=None):
        '''
        Args:
            params_groups: list of dict, 每个 dict 包含一个实验的参数设置，可以覆
盖默认参数，
            如 [{residual_gate: (0.5, 0.5), experiment_name: 'Experiment 1'},␣
↪{residual_gate: (0.8, 0.2), experiment_name: 'Experiment 2'}]

            manual: dict, 全局默认参数修改

        Attrbutes:
            num_experiments: int, 实验数量
            init_parameters: list of dict, 每个 dict 包含一个实验的初始化参数设置
            train_parameters: list of dict, 每个 dict 包含一个实验的训练参数设置
        '''
        init, train = default_setup(manual=manual)
        self.num_experiments = len(params_groups)
```

```
        self.init_parameters = [copy.deepcopy(init) for _ in range(self.num_experiments)]
        self.train_parameters = [copy.deepcopy(train) for _ in range(self.num_experiments)]
        # 自动分类传入的参数以更新默认参数
        for i, params in enumerate(params_groups):
            for key, value in params.items():
                if key in self.init_parameters[i]:
                    self.init_parameters[i][key] = value
                elif key in self.train_parameters[i]:
                    self.train_parameters[i][key] = value
                else:
                    raise ValueError(f"Unknown parameter: {key}")
        self.experiments=[None]*self.num_experiments
        for i in range(self.num_experiments):
            experiment = LoopedTransformerExperiment(**self.init_parameters[i])
            experiment.offload_to_cpu() # 初始化后将模型移回 CPU，节省 GPU 显存
            self.experiments[i] = experiment

    def run(self, result_lists:list[tuple[list[str], str, int]], modes=['train'], parallel_workers=1, eval_configs=None):
        '''
        Args:
            result_lists: list of tuple of list，每个 tuple 包含一组实验测量的数据、对应的横坐标类型和 baseline 索引（不写表示不使用 baseline），
            每个 list 中的字符串表示需要展示的结果名称，必须是 LoopedTransformerExperiment.get_results() 中返回的结果名称，用'|'作为双 y 轴分割符
            横坐标类型可以从 'epoch'（表示以训练轮数为横坐标的折线图，metric 必须是列表）、'block'（表示以模型深度为横坐标的折线图，metric 必须是列表，仅限 eval 和 captured 的 sink 指标）、'experiment'（表示以实验编号为横坐标的柱状图，metric 是单个值）中选择。
            如 [(['loss_history', '|', 'residual_gate_history_a_mean', 'residual_gate_history_b_mean'], 'epoch', 0), (['final_residual_gate_a_mean', 'final_residual_gate_b_mean','|'], 'experiment')]
```

```
            modes: list of str, 表示需要运行的模式，可以是 'train' 和 'evaluate'␣
↪的任意组合

            parallel_workers: int, 表示并行运行实验的工作线程数。

            eval_configs: list of dict (长度最好与 modes 中 'evaluate' 的数量相
同)，每个 dict 的形式为 {'eval_name': str, 'ood_kwargs': dict} (详见␣
↪default_setup() 中 的 eval_config 注释)
            如 [{'eval_name': 'id'}, {'eval_name': 'ood_x_scale', 'ood_kwargs':␣
↪{'x_scale': 2.0}}]

            需要注意的是，eval_configs 中的 eval_name 将直接作为评估结果中损失值、
范数等指标的前缀，如 id_loss、ood_x_scale_y_pred_norm、ood_x_scale_y_norm_ratio

        Attributes:
            result_lists: 同 Args
            results: list of dict，所有实验的结果，每个 dict 是一个实验的结果，为
metric 和 value 的键值对。
            experiments: list of LoopedTransformerExperiment 对象，已转移到 CPU
以节省 GPU 显存
        '''
        # 创建实验对象并运行
        result_set = set(key for item in result_lists for key in item[0] if key␣
↪!= '|') # 兼容长度为 2 或 3 的 tuple，获取所有需要展示的结果名称的集合
        self.result_lists = result_lists
        self.results = [None]*self.num_experiments
        if eval_configs is None:
            eval_configs = []
        print_lock = threading.Lock()  # 用于在线程中安全地打印日志
        vram_printed = False
        def single_train(i, parallel=False):
            nonlocal vram_printed
            eval_count=0
            experiment = self.experiments[i]
            experiment.load_to_device() # 训练前将模型加载到 GPU
```

```python
            if parallel:
                self.train_parameters[i]['timing']=False
                self.train_parameters[i]['print_every']=None
            if parallel:
                with print_lock:
                    if not vram_printed:
                        print_vram_usage(tag=f"Parallel 峰值显存检测")
                        vram_printed = True
            else:
                print_vram_usage(tag=f"Experiment {i+1} 训练前")
            for mode in modes:
                if mode == 'train':
                    experiment.train(**self.train_parameters[i])
                    self.results[i] = experiment.get_results(result_set)
                    if parallel:
                        with print_lock:
                            print_vram_usage(tag=f"Experiment {i+1} 训练峰值", peak=(experiment.max_torch_vram, experiment.max_metal_vram))
                    else:
                        print_vram_usage(tag=f"Experiment {i+1} 训练峰值", peak=(experiment.max_torch_vram, experiment.max_metal_vram))
                elif mode == 'evaluate':
                    eval_parameters = {k:v for k,v in self.train_parameters[i].items() if k in ['batch_size', 'seq_len', 'data_type', 'sink_padding', 'd_hidden', 'function_callable', 'lorenz_kwargs']}
                    if 'loss_type' in self.init_parameters[i]:
                        eval_parameters['loss_type'] = self.init_parameters[i]['loss_type'] # 确保评估使用与训练相同类型的损失函数
                    if eval_configs and len(eval_configs) > 0:
                        for k,current_config in enumerate(eval_configs):
                            current_eval_params = eval_parameters.copy()
                            current_eval_params.update(current_config)
                            experiment.evaluate(**current_eval_params)
                    else: # 万一没有传入 eval_configs 的兜底
                        print(f"  No eval_configs provided for evaluation, using default eval parameters for Experiment {i+1}.")
```

```python
                        default_config = {'eval_name': f'eval_{eval_count}'}
                        current_eval_params = eval_parameters.copy()
                        current_eval_params.update(default_config)
                        experiment.evaluate(**current_eval_params)
                        eval_count+=1
                    self.results[i] = experiment.get_results(result_set)
            experiment.offload_to_cpu() # 训练完成后将模型和结果移回 CPU，节省 GPU 显存
            self.experiments[i] = experiment
            if torch.cuda.is_available():
                torch.cuda.empty_cache()
            if torch.backends.mps.is_available():
                torch.mps.empty_cache()
            if not parallel:
                print_vram_usage(tag=f"Experiment {i+1} 清理后")
                print(f"Experiment {i+1}/{self.num_experiments} completed.\n")

        print_vram_usage(tag=f"本底显存检测")
        if parallel_workers == 1:
            for i in range(self.num_experiments):
                single_train(i, parallel=False)
        elif isinstance(parallel_workers, int) and parallel_workers > 1:
            print(f"Running {self.num_experiments} experiments in parallel with {parallel_workers} workers...")
            with concurrent.futures.ThreadPoolExecutor(max_workers=parallel_workers) as executor:
                executor.map(lambda i: single_train(i, parallel=True), range(self.num_experiments))
        else:
            raise ValueError("parallel_workers must be a positive integer.")

        if self.results[0] is not None:
            invalid_keys = [key for key in result_set if key not in self.results[0]]
            if invalid_keys:
```

```
            print(f"\n[Final Check Warning] 以下请求的指标在所有流程结束后仍
未找到，请检查拼写是否正确或数据是否生成失败：\n{invalid_keys}\n")

    def plot(self, compare_experiments=True,
             subplot_shape=(1,-1),figure_size=None,suptitle='Looped Transformer␣
↪Experiment Results',
             colors=['blue', 'red', 'green', 'yellow', 'cyan', 'orange',␣
↪'purple', 'brown']):
        '''
        Args:
            compare_experiments: bool, True 表示在一张图中比较不同实验的结果，每张
图仅比较一个指标；False 表示每个实验单独成图，每张图仅比较一个实验的多个指标（第一个
指标使用左轴，第二个及以后的指标使用共享 x 轴的右轴）

            subplot_shape: tuple of int, 表示子图的 (行数, 列数), (-1 表示自动计
算), 如 (1, -1) 表示所有图在一行, (-1, 2) 表示每行两个图, 自动计算行数

            figure_size: tuple of float, 画布大小, 单位为英寸, 如 (15, 10)。默认
值 None 表示使用 (8*subplot_shape[1], 6*subplot_shape[0]) 的大小

            suptitle: str, 整体图的标题

            colors: list of str, 画图使用的颜色列表，每张图从头开始循环使用这些颜色
        '''
        # 画图
        num_plots = 0
        for item in self.result_lists:
            x_type = item[1]
            clean_list = [m for m in item[0] if m != '|']
            if x_type in ['epoch', 'block']:
                if compare_experiments:
                    num_plots += len(clean_list)
                else:
                    num_plots += self.num_experiments
            elif x_type == 'experiment':
                num_plots += 1
```

```
        if subplot_shape == (-1, -1):
            raise ValueError("Invalid subplot_shape: both dimensions cannot be␣
↪-1. Please specify at least one dimension or provide a valid shape.")
        if subplot_shape[1] == -1:
            subplot_shape = (subplot_shape[0], math.ceil(num_plots /␣
↪subplot_shape[0]))  # 自动计算列数
        elif subplot_shape[0] == -1:
            subplot_shape = (math.ceil(num_plots / subplot_shape[1]),␣
↪subplot_shape[1])  # 自动计算行数
        if subplot_shape[0] * subplot_shape[1] < num_plots:
            raise ValueError(f"subplot_shape {subplot_shape} is too small for␣
↪the number of plots {num_plots}. Please increase the dimensions.")
        if figure_size is None:
            figure_size = (8*subplot_shape[1], 6*subplot_shape[0])

        if self.results[0] is None:
            print("No results to plot. Please run the experiments first.")
            return
        results = self.results
        fig,axs = plt.subplots(*subplot_shape, figsize=figure_size)
        axs_flat = np.atleast_1d(axs).flatten()  # 将 axs 转换为一维数组，方便索
引

        index=0
        for item in self.result_lists:
            result_list = item[0]  # 需要展示的结果名称列表
            x = item[1]            # 横坐标类型 ('epoch' 或 'experiment')
            baseline_index = item[2] if len(item) > 2 else None  # 基线索引（可
选），表示用于计算相对值的基线实验索引
            baseline_name=None
            if x in ['epoch', 'block']: # 表示画横坐标是 epoch 或 block 的曲线
                if compare_experiments:
                    clean_result_list = [m for m in result_list if m != '|']
                    for metric in clean_result_list:
                        if baseline_index is not None and 0 <= baseline_index <␣
↪self.num_experiments:
```

```
                        baseline_name = self.
↪init_parameters[baseline_index].get('experiment_name', f'Experiment␣
↪{baseline_index+1}')
                        baseline_data = np.
↪array(results[baseline_index][metric])
                    else:
                        baseline_name = None
                        baseline_data = None
                    for i in range(self.num_experiments):
                        exp_name = self.init_parameters[i].
↪get('experiment_name', f'Experiment {i+1}')
                        y_data = np.array(results[i][metric])
                        label_name = exp_name  # 专门给图例起名字，保护原变
量
                        if baseline_name is not None:
                            min_length = min(len(y_data),␣
↪len(baseline_data))
                            y_data = y_data[:min_length] - baseline_data[:
↪min_length]  # 计算相对于基线的差值
                            if i == baseline_index:
                                label_name += ' (Baseline)'
                        axs_flat[index].plot(y_data, label=label_name,␣
↪color=colors[i%len(colors)])
                    axs_flat[index].set_xlabel('Epoch' if x == 'epoch' else␣
↪'Block (Depth)')
                    if baseline_name is not None:
                        # 加上 \n 换行，防止 ylabel 太长挤出画布边界
                        axs_flat[index].set_ylabel(f"{metric}\n(relative to␣
↪{baseline_name})")
                    else:
                        axs_flat[index].set_ylabel(metric)
                    axs_flat[index].tick_params(axis='y')
                    axs_flat[index].set_title('Comparison of ' + metric)
                    axs_flat[index].legend(loc='upper right')
                    axs_flat[index].grid(True)
```

```
                        # 为顶部留出 35% 的空间防止遮挡图例
                        y_min, y_max = axs_flat[index].get_ylim()
                        axs_flat[index].set_ylim(y_min, y_max + 0.35 * (y_max -␣
↪y_min))
                        index += 1
                else:
                    if baseline_index is not None and 0 <= baseline_index <␣
↪self.num_experiments:
                        baseline_name = self.init_parameters[baseline_index].
↪get('experiment_name', f'Experiment {baseline_index+1}')
                        baseline_data_dict = {m: np.
↪array(results[baseline_index][m]) for m in result_list if m!='|'}
                    else:
                        baseline_name = None
                        baseline_data_dict = None
                    for i in range(self.num_experiments):
                        color_index = 0
                        exp_name = self.init_parameters[i].
↪get('experiment_name', f'Experiment {i+1}')
                        if '|' in result_list:
                            split_idx = result_list.index('|')
                            clean_result_list = [metric for metric in␣
↪result_list if metric != '|']
                        else:
                            split_idx = len(result_list)
                            clean_result_list = result_list
                        num_metrics = len(clean_result_list)
                        for j in range(split_idx):
                            metric_j = clean_result_list[j]
                            y_data = np.array(results[i][metric_j])
                            label_name = metric_j
                            if baseline_name is not None:
                                baseline_data = baseline_data_dict[metric_j]
                                min_length = min(len(y_data),␣
↪len(baseline_data))
```

```
                                y_data = y_data[:min_length] - baseline_data[:
↪min_length]  # 计算相对于基线的差值
                                if i == baseline_index:
                                    label_name += ' (Baseline)'
                            axs_flat[index].plot(y_data, label=label_name,␣
↪color=colors[color_index%len(colors)])
                            color_index += 1
                        axs_flat[index].set_xlabel('Epoch' if x == 'epoch' else␣
↪'Block (Depth)')
                        ylabel_left = '\n'.join(clean_result_list[:split_idx])
                        if baseline_name is not None:
                            ylabel_left += f"\n(relative to {baseline_name})"
                        axs_flat[index].set_ylabel(ylabel_left, color=colors[0])
                        axs_flat[index].tick_params(axis='y',␣
↪labelcolor=colors[0])
                        if num_metrics > split_idx:
                            ax_twin = axs_flat[index].twinx()  # 创建共享 x 轴但
独立 y 轴的第二个坐标轴
                            primary_color = colors[split_idx%len(colors)]
                            for j in range(split_idx, num_metrics):
                                metric_j = clean_result_list[j]
                                y_data = np.array(results[i][metric_j])
                                label_name = metric_j
                                if baseline_name is not None:
                                    baseline_data = baseline_data_dict[metric_j]
                                    min_length = min(len(y_data),␣
↪len(baseline_data))
                                    y_data = y_data[:min_length] -␣
↪baseline_data[:min_length]  # 计算相对于基线的差值
                                    if i == baseline_index:
                                        label_name += ' (Baseline)'
                                ax_twin.plot(y_data, label=label_name,␣
↪color=colors[color_index%len(colors)])
                                color_index += 1
                            ax_twin.set_ylabel('\n'.
↪join(clean_result_list[split_idx:]), color=primary_color)
```

```
                            ax_twin.tick_params(axis='y',␣
↪labelcolor=primary_color)
                            ax_twin.legend(loc='upper right')
                        axs_flat[index].set_title(exp_name + ' - ' + ' and\n'.
↪join(clean_result_list))
                        axs_flat[index].legend(loc='upper left')
                        axs_flat[index].grid(True)
                        # 为顶部留出 35% 的空间防止遮挡图例
                        y1_min, y1_max = axs_flat[index].get_ylim()
                        axs_flat[index].set_ylim(y1_min, y1_max + 0.35 *␣
↪(y1_max - y1_min))
                        if num_metrics > split_idx:
                            y2_min, y2_max = ax_twin.get_ylim()
                            ax_twin.set_ylim(y2_min, y2_max + 0.35 * (y2_max -␣
↪y2_min))
                        index += 1
            elif x == 'experiment': # 表示画不同实验对比的柱状图
                exp_names = [self.init_parameters[i].get('experiment_name',␣
↪f'Experiment {i+1}') for i in range(self.num_experiments)]
                if baseline_index is not None and 0 <= baseline_index < self.
↪num_experiments:
                    baseline_name = self.init_parameters[baseline_index].
↪get('experiment_name', f'Experiment {baseline_index+1}')
                    exp_names[baseline_index] += '\n(Baseline)'

                x_pos = np.arange(len(exp_names))
                if '|' in result_list:
                    split_idx = result_list.index('|')
                    clean_result_list = [metric for metric in result_list if␣
↪metric != '|']
                else:
                    split_idx = len(result_list)
                    clean_result_list = result_list
                num_metrics = len(clean_result_list)
                width = 0.8 / num_metrics # 确保所有柱子加起来不超过 0.8 的宽度,
留下 0.2 的组间空白
```

```
                color_index = 0
                # 左侧坐标轴
                for j in range(split_idx):
                    metric_j = clean_result_list[j]
                    y_data_j = [results[i][metric_j] for i in range(self.
↪num_experiments)]
                    if baseline_name is not None:
                        base_val = y_data_j[baseline_index]
                        y_data_j = [val - base_val for val in y_data_j]
                    offset_j = (j - num_metrics / 2 + 0.5) * width
                    axs_flat[index].bar(x_pos + offset_j, y_data_j, width,␣
↪label=metric_j, color=colors[color_index % len(colors)])
                    color_index += 1
                ylabel_left = '\n'.join(clean_result_list[:split_idx])
                if baseline_name is not None:
                    ylabel_left += f"\n(relative to {baseline_name})"
                axs_flat[index].set_ylabel(ylabel_left, color=colors[0])
                axs_flat[index].tick_params(axis='y', labelcolor=colors[0])

                if num_metrics > split_idx:
                    ax_twin = axs_flat[index].twinx() # 创建共享 X 轴但独立 Y 轴
的坐标系
                    primary_color = colors[split_idx % len(colors)]
                    for j in range(split_idx, num_metrics):
                        metric_j = clean_result_list[j]
                        y_data_j = [results[i][metric_j] for i in range(self.
↪num_experiments)]
                        if baseline_name is not None:
                            base_val = y_data_j[baseline_index]
                            y_data_j = [val - base_val for val in y_data_j]

                        offset_j = (j - num_metrics / 2 + 0.5) * width
                        ax_twin.bar(x_pos + offset_j, y_data_j, width,␣
↪label=metric_j, color=colors[color_index % len(colors)])
                        color_index += 1
                    ylabel_right = '\n'.join(clean_result_list[split_idx:])
```

```
                if baseline_name is not None:
                    ylabel_right += f"\n(relative to {baseline_name})"
                ax_twin.set_ylabel(ylabel_right, color=primary_color)
                ax_twin.tick_params(axis='y', labelcolor=primary_color)
                ax_twin.legend(loc='upper right')
                axs_flat[index].legend(loc='upper left')
            else:
                axs_flat[index].legend(loc='best')
            y1_min, y1_max = axs_flat[index].get_ylim()
            pos1 = max(y1_max, 0)
            neg1 = max(-y1_min, 0)
            s1 = max(pos1, neg1) if max(pos1, neg1) > 0 else 1.0
            pos_norm1 = pos1 / s1
            neg_norm1 = neg1 / s1
            if num_metrics > split_idx:
                y2_min, y2_max = ax_twin.get_ylim()
                pos2 = max(y2_max, 0)
                neg2 = max(-y2_min, 0)
                s2 = max(pos2, neg2) if max(pos2, neg2) > 0 else 1.0
                pos_norm2 = pos2 / s2
                neg_norm2 = neg2 / s2
                max_pos_norm = max(pos_norm1, pos_norm2)
                max_neg_norm = max(neg_norm1, neg_norm2)
            else:
                max_pos_norm = pos_norm1
                max_neg_norm = neg_norm1
                s2 = 1.0 # 占位
            final_pos_norm = max(max_pos_norm * 1.35, 0.35)
            needs_zero_alignment = (baseline_name is not None) or (y1_min <
 ↪0) or (num_metrics > split_idx and y2_min < 0)
            if needs_zero_alignment:
                # 按照统一计算好的归一化比例，乘以各自的真实尺度
                axs_flat[index].set_ylim(-max_neg_norm * s1, final_pos_norm
 ↪* s1)
                if num_metrics > split_idx:
```

```
                        ax_twin.set_ylim(-max_neg_norm * s2, final_pos_norm *␣
↪s2)
                else:
                    axs_flat[index].set_ylim(0, final_pos_norm * s1)
                    if num_metrics > split_idx:
                        ax_twin.set_ylim(0, final_pos_norm * s2)
                axs_flat[index].axhline(0, color='gray', linewidth=1,␣
↪linestyle='-', zorder=0) # 画一条贯穿整个图表的辅助零线，充当视觉上的“地平线”
                axs_flat[index].set_xticks(x_pos)
                axs_flat[index].set_xticklabels(exp_names, rotation=45,␣
↪ha='right')
                axs_flat[index].set_title(f"Comparison of {' and\n'.
↪join(clean_result_list)}")
                axs_flat[index].grid(axis='y', linestyle='--', alpha=0.7)
                index += 1
        clear_lorenz_cache()  # 清理洛伦兹系统数据缓存，释放内存
        # 如果绘制的图的数量少于子图的总数量，则隐藏多余的子图
        for i in range(index, len(axs_flat)):
            axs_flat[i].set_visible(False)
        if suptitle is not None:
            plt.suptitle(suptitle+"\n")
        plt.tight_layout()
        plt.show()
```

2 Linear Regression Experiments

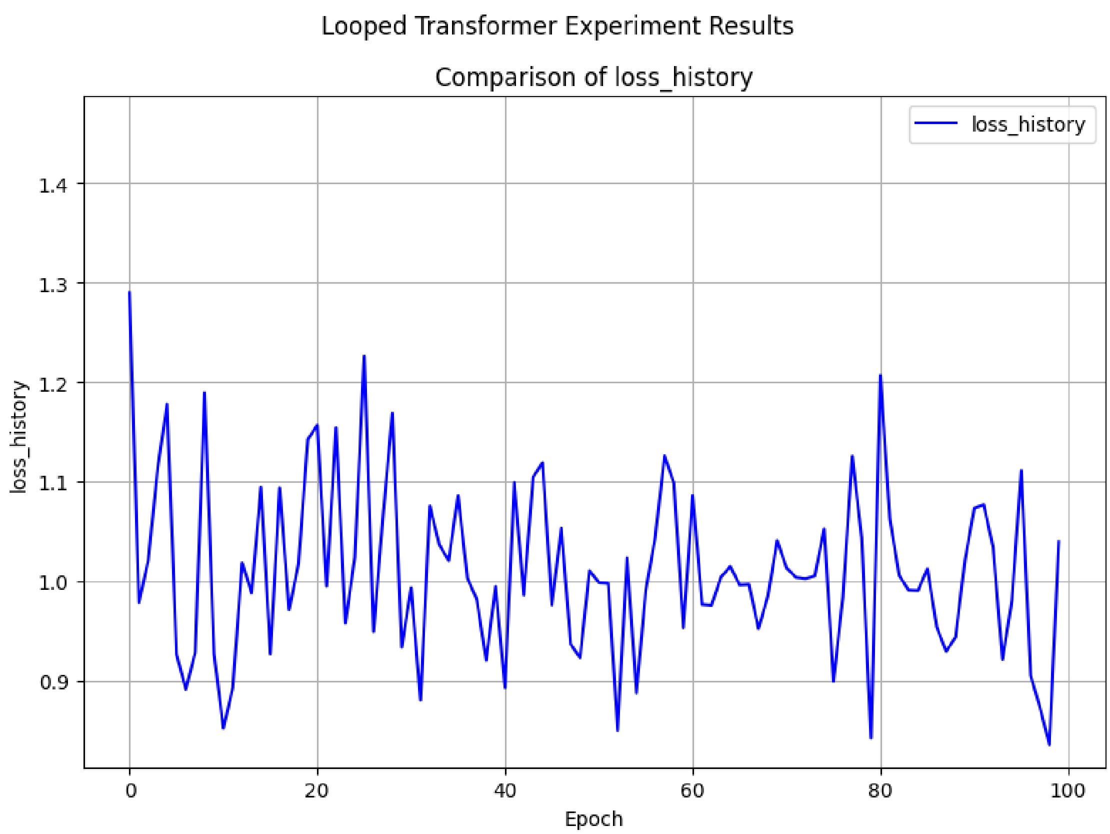
2.1 实验数据观测

2.1.1 绘制 Loss 下降曲线

[182]:
```
loss_table = ExperimentTable(params_groups=[{'experiment_name':␣
↪'loss_history','curriculum': {'d_x': 5, 'seq_len': 10, 'duration_ratio': 0.
↪5}}])
loss_table.run(result_lists=[(['loss_history'], 'epoch')])
loss_table.plot(figure_size=(8,6))
```

```
loss_history initialized in 0.03 seconds. Using device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.41 GB | Metal 驱动分配: 2.06 GB
```

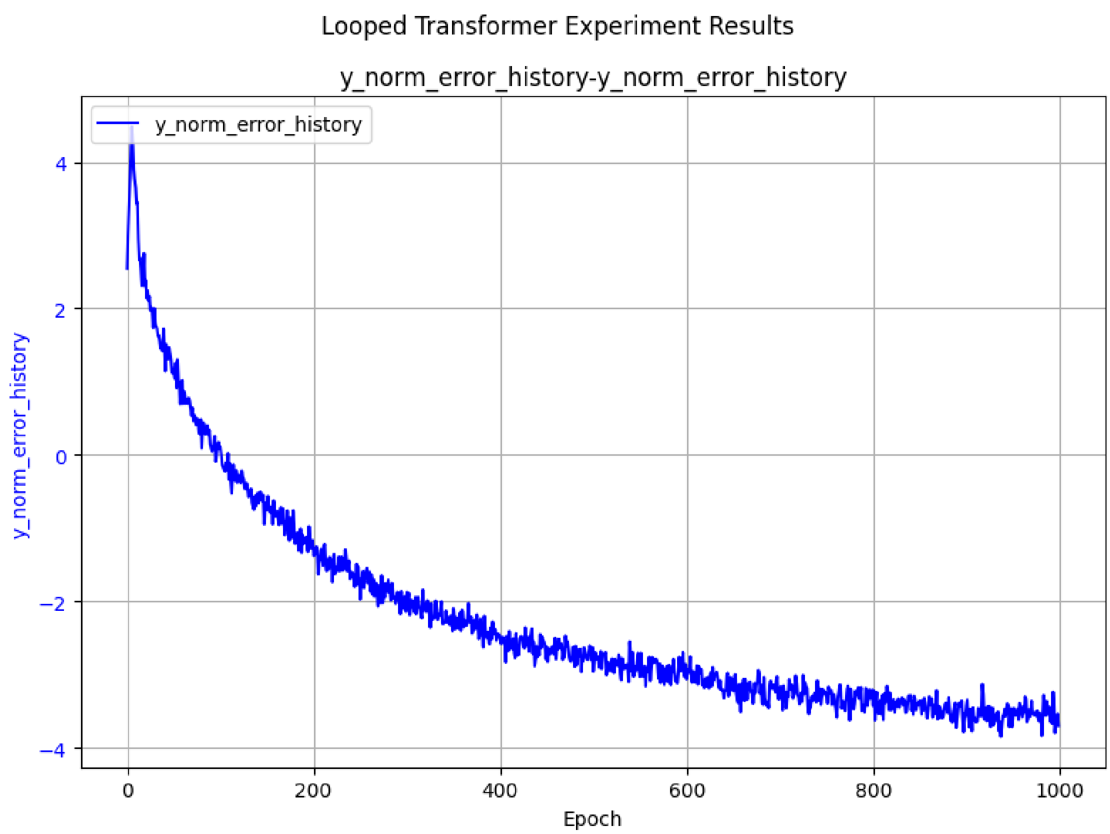
```
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.42 GB | Metal 驱动分配: 2.06 GB
Epoch 20/100, Loss: 1.1422, Norm Ratio: 0.1630, Avg Sink Score: 0.1154, Avg Sink
Rate: 0.0000
Epoch 40/100, Loss: 0.9943, Norm Ratio: 0.0928, Avg Sink Score: 0.0737, Avg Sink
Rate: 0.0000
Curriculum ended at epoch 51, step 1.
Epoch 60/100, Loss: 0.9526, Norm Ratio: 0.0989, Avg Sink Score: 0.0618, Avg Sink
Rate: 0.0000
Epoch 80/100, Loss: 0.8416, Norm Ratio: 0.1008, Avg Sink Score: 0.0618, Avg Sink
Rate: 0.0000
Epoch 100/100, Loss: 1.0393, Norm Ratio: 0.1031, Avg Sink Score: 0.0617, Avg
Sink Rate: 0.0000
Training completed in 48.43 seconds.
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.44 GB | Metal 驱动分配: 6.72 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.43 GB | Metal 驱动分配: 2.11 GB
Experiment 1/1 completed.
```



2.1.2 绘制 y_pred 和 y_true 的变化曲线

[75]:
```
norm_table = ExperimentTable(params_groups=[{'epochs': 1000, 'experiment_name':␣
↪'y_norm_error_history', 'print_every': 200}])
norm_table.run(result_lists=[(['y_norm_error_history'], 'epoch')])
norm_table.plot(figure_size=(8,6),compare_experiments=False)
```

```
y_norm_error_history initialized in 0.02 seconds. Using device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Epoch 200/1000, Loss: 18.8248, current_blocks: 20
Epoch 400/1000, Loss: 19.7988, current_blocks: 20
Epoch 600/1000, Loss: 17.7491, current_blocks: 20
Epoch 800/1000, Loss: 20.9858, current_blocks: 20
Epoch 1000/1000, Loss: 20.9834, current_blocks: 20
Training completed in 451.35 seconds.
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Experiment 1/1 completed.
```



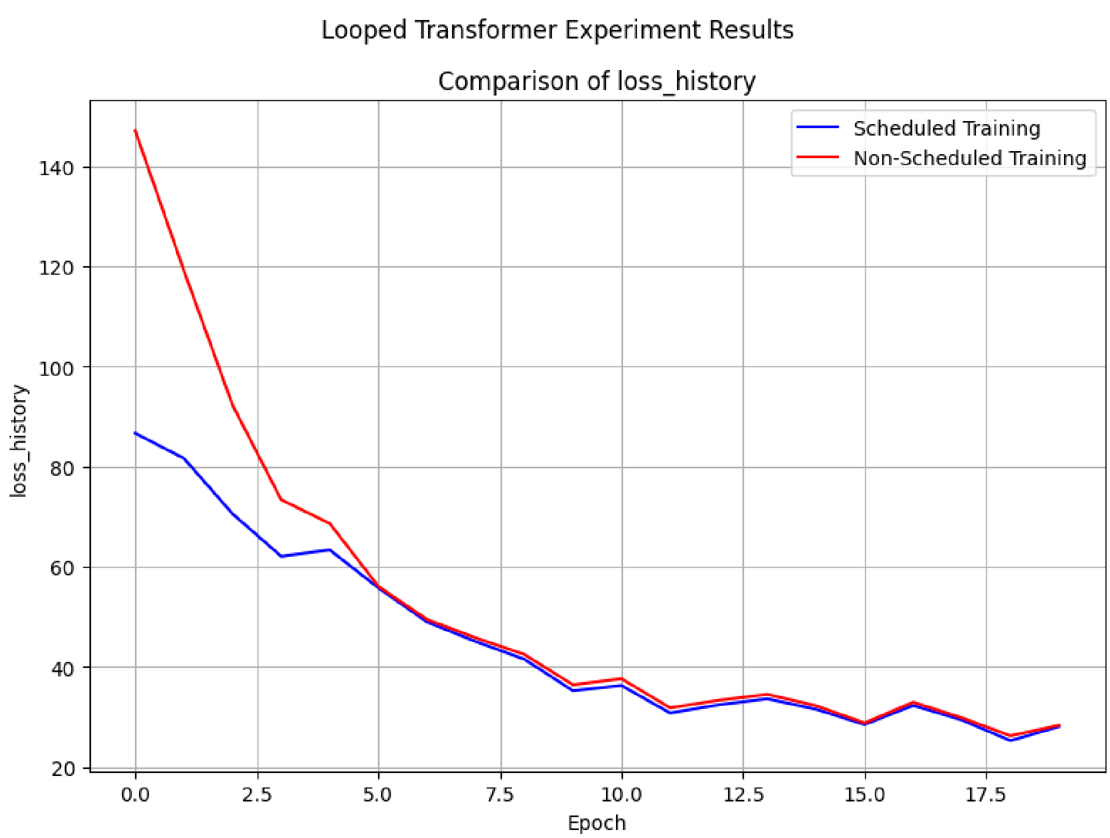
2.2 对比实验

2.2.1 Scheduled Training vs Non-Scheduled Training 对比实验

[76]:
```
table = ExperimentTable(params_groups=[{'scheduled_training': True,␣
↪'experiment_name': 'Scheduled Training'},
                                       {'scheduled_training': False,␣
↪'experiment_name': 'Non-Scheduled Training'}])
table.run(result_lists=[(['loss_history'], 'epoch')])
table.plot(figure_size=(8, 6), compare_experiments=True)
```

```
Scheduled Training initialized in 0.02 seconds. Using device: mps
Non-Scheduled Training initialized in 0.01 seconds. Using device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.05 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.05 GB
Epoch 5/20, Loss: 63.3171, current_blocks: 19
```

```
Epoch 10/20, Loss: 35.1716, current_blocks: 20
Epoch 15/20, Loss: 31.4720, current_blocks: 20
Epoch 20/20, Loss: 28.0077, current_blocks: 20
Training completed in 9.75 seconds.
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Experiment 1/2 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.05 GB
Epoch 5/20, Loss: 68.5955, current_blocks: 20
Epoch 10/20, Loss: 36.3633, current_blocks: 20
Epoch 15/20, Loss: 32.1849, current_blocks: 20
Epoch 20/20, Loss: 28.2660, current_blocks: 20
Training completed in 9.81 seconds.
[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.13 GB
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Experiment 2/2 completed.
```



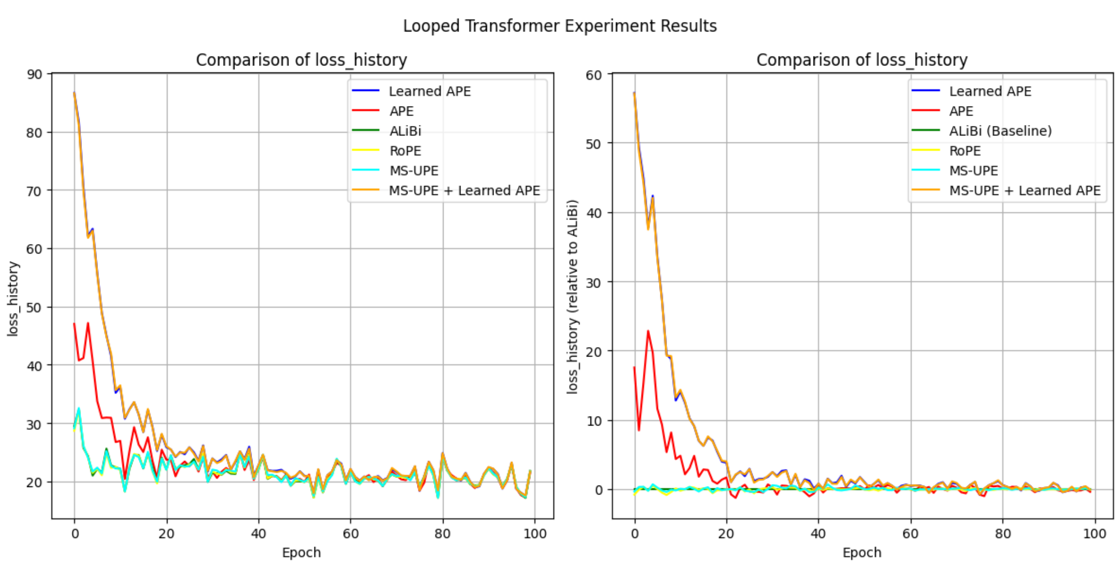
2.2.2 不同 Positional Embedding 对比实验

[79]:
```python
pe_table = ExperimentTable(params_groups=[
    {'pe_type': ['learned_ape'], 'experiment_name': 'Learned APE'},
    {'pe_type': ['ape'], 'experiment_name': 'APE'},
    {'pe_type': ['alibi'], 'experiment_name': 'ALiBi'},
    {'pe_type': ['rope'], 'experiment_name': 'RoPE'},
    {'pe_type': ['ms_upe'], 'experiment_name': 'MS-UPE'},
    {'pe_type': ['ms_upe', 'learned_ape'], 'experiment_name': 'MS-UPE + Learned␣
↪APE'},
], manual={'epochs': 100, 'print_every': 20})
pe_table.run(result_lists=[
    (['loss_history'], 'epoch'),
    (['loss_history'], 'epoch', 2)  # 以 ALiBi 为基线
])
pe_table.plot(figure_size=(12, 6))
```

```
Learned APE initialized in 0.02 seconds. Using device: mps
APE initialized in 0.01 seconds. Using device: mps
ALiBi initialized in 0.01 seconds. Using device: mps
RoPE initialized in 0.01 seconds. Using device: mps
MS-UPE initialized in 0.01 seconds. Using device: mps
MS-UPE + Learned APE initialized in 0.01 seconds. Using device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.19 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.19 GB
Epoch 20/100, Loss: 28.0077, current_blocks: 20
Epoch 40/100, Loss: 20.8999, current_blocks: 20
Epoch 60/100, Loss: 20.2006, current_blocks: 20
Epoch 80/100, Loss: 18.3025, current_blocks: 20
Epoch 100/100, Loss: 21.6682, current_blocks: 20
Training completed in 47.76 seconds.
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.22 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
Experiment 1/6 completed.
```

```
[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 25.4152, current_blocks: 20
Epoch 40/100, Loss: 20.2102, current_blocks: 20
Epoch 60/100, Loss: 19.7146, current_blocks: 20
Epoch 80/100, Loss: 17.6053, current_blocks: 20
Epoch 100/100, Loss: 21.3824, current_blocks: 20
Training completed in 48.05 seconds.
[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.21 GB
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
Experiment 2/6 completed.

[Experiment 3 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 24.0522, current_blocks: 20
Epoch 40/100, Loss: 20.8352, current_blocks: 20
Epoch 60/100, Loss: 19.5621, current_blocks: 20
Epoch 80/100, Loss: 17.1478, current_blocks: 20
Epoch 100/100, Loss: 21.7769, current_blocks: 20
Training completed in 55.24 seconds.
[Experiment 3 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.23 GB
[Experiment 3 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
Experiment 3/6 completed.

[Experiment 4 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 23.8768, current_blocks: 20
Epoch 40/100, Loss: 20.4678, current_blocks: 20
Epoch 60/100, Loss: 19.6902, current_blocks: 20
Epoch 80/100, Loss: 17.1698, current_blocks: 20
Epoch 100/100, Loss: 21.6961, current_blocks: 20
Training completed in 68.12 seconds.
[Experiment 4 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.22 GB
[Experiment 4 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
Experiment 4/6 completed.

[Experiment 5 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 23.8994, current_blocks: 20
Epoch 40/100, Loss: 20.4356, current_blocks: 20
Epoch 60/100, Loss: 19.6087, current_blocks: 20
```

```
Epoch 80/100, Loss: 17.1837, current_blocks: 20
Epoch 100/100, Loss: 21.6859, current_blocks: 20
Training completed in 52.83 seconds.
[Experiment 5 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.22 GB
[Experiment 5 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
Experiment 5/6 completed.

[Experiment 6 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 28.1398, current_blocks: 20
Epoch 40/100, Loss: 20.6823, current_blocks: 20
Epoch 60/100, Loss: 20.1465, current_blocks: 20
Epoch 80/100, Loss: 18.4156, current_blocks: 20
Epoch 100/100, Loss: 21.6613, current_blocks: 20
Training completed in 54.00 seconds.
[Experiment 6 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.23 GB
[Experiment 6 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
Experiment 6/6 completed.
```

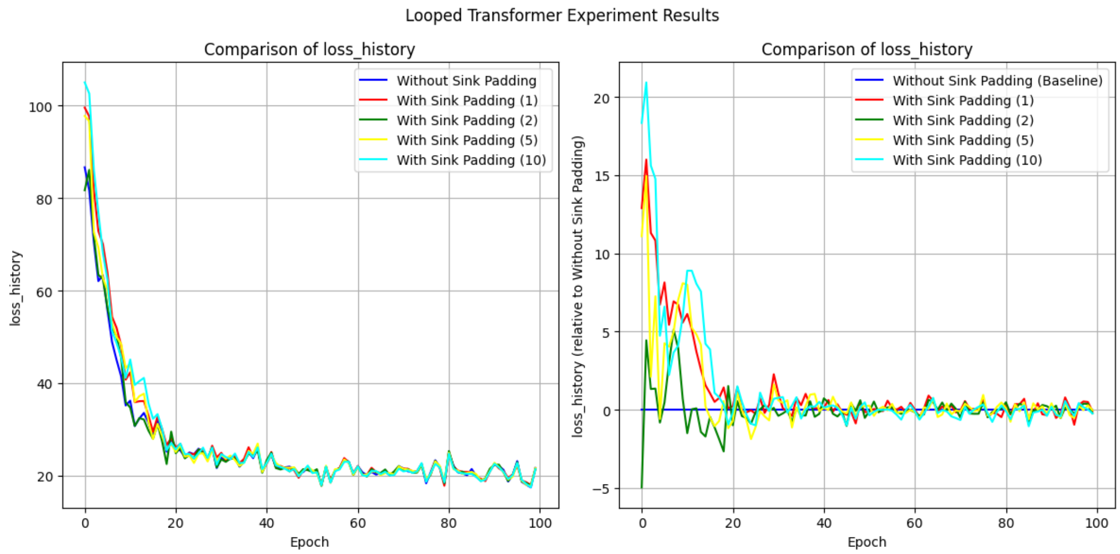


2.2.3 sink padding 的有无对比实验

[80]:
```python
sink_table = ExperimentTable(params_groups=[
    {'experiment_name': 'Without Sink Padding'},
    {'experiment_name': 'With Sink Padding (1)', 'sink_padding': 1},
    {'experiment_name': 'With Sink Padding (2)', 'sink_padding': 2},
    {'experiment_name': 'With Sink Padding (5)', 'sink_padding': 5},
    {'experiment_name': 'With Sink Padding (10)', 'sink_padding': 10},
], manual={'epochs': 100, 'print_every': 20})
sink_table.run(result_lists=[
    (['loss_history'], 'epoch'),
    (['loss_history'], 'epoch', 0),  # 以 Without Sink Padding 为基线
])
sink_table.plot(figure_size=(12, 6))
```

```
Without Sink Padding initialized in 0.02 seconds. Using device: mps
With Sink Padding (1) initialized in 0.01 seconds. Using device: mps
With Sink Padding (2) initialized in 0.01 seconds. Using device: mps
With Sink Padding (5) initialized in 0.01 seconds. Using device: mps
With Sink Padding (10) initialized in 0.01 seconds. Using device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 28.0077, current_blocks: 20
Epoch 40/100, Loss: 20.8999, current_blocks: 20
Epoch 60/100, Loss: 20.2006, current_blocks: 20
Epoch 80/100, Loss: 18.3025, current_blocks: 20
Epoch 100/100, Loss: 21.6682, current_blocks: 20
Training completed in 50.96 seconds.
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.22 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
Experiment 1/5 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 28.1106, current_blocks: 20
Epoch 40/100, Loss: 20.9680, current_blocks: 20
Epoch 60/100, Loss: 20.6434, current_blocks: 20
Epoch 80/100, Loss: 17.8341, current_blocks: 20
Epoch 100/100, Loss: 21.5738, current_blocks: 20
```

```
Training completed in 53.92 seconds.
[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.22 GB
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
Experiment 2/5 completed.

[Experiment 3 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 29.5266, current_blocks: 20
Epoch 40/100, Loss: 20.6409, current_blocks: 20
Epoch 60/100, Loss: 20.1771, current_blocks: 20
Epoch 80/100, Loss: 18.3634, current_blocks: 20
Epoch 100/100, Loss: 21.4438, current_blocks: 20
Training completed in 53.87 seconds.
[Experiment 3 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.22 GB
[Experiment 3 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
Experiment 3/5 completed.

[Experiment 4 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 26.8264, current_blocks: 20
Epoch 40/100, Loss: 20.8156, current_blocks: 20
Epoch 60/100, Loss: 20.2012, current_blocks: 20
Epoch 80/100, Loss: 18.7726, current_blocks: 20
Epoch 100/100, Loss: 21.6547, current_blocks: 20
Training completed in 57.91 seconds.
[Experiment 4 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 6.22 GB
[Experiment 4 清理后] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Experiment 4/5 completed.

[Experiment 5 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 27.0638, current_blocks: 20
Epoch 40/100, Loss: 21.4043, current_blocks: 20
Epoch 60/100, Loss: 20.5190, current_blocks: 20
Epoch 80/100, Loss: 18.5852, current_blocks: 20
Epoch 100/100, Loss: 21.4375, current_blocks: 20
Training completed in 66.44 seconds.
[Experiment 5 训练峰值] 显存占用 -> PyTorch 分配: 3.14 GB | Metal 驱动分配: 6.26 GB
[Experiment 5 清理后] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Experiment 5/5 completed.
```



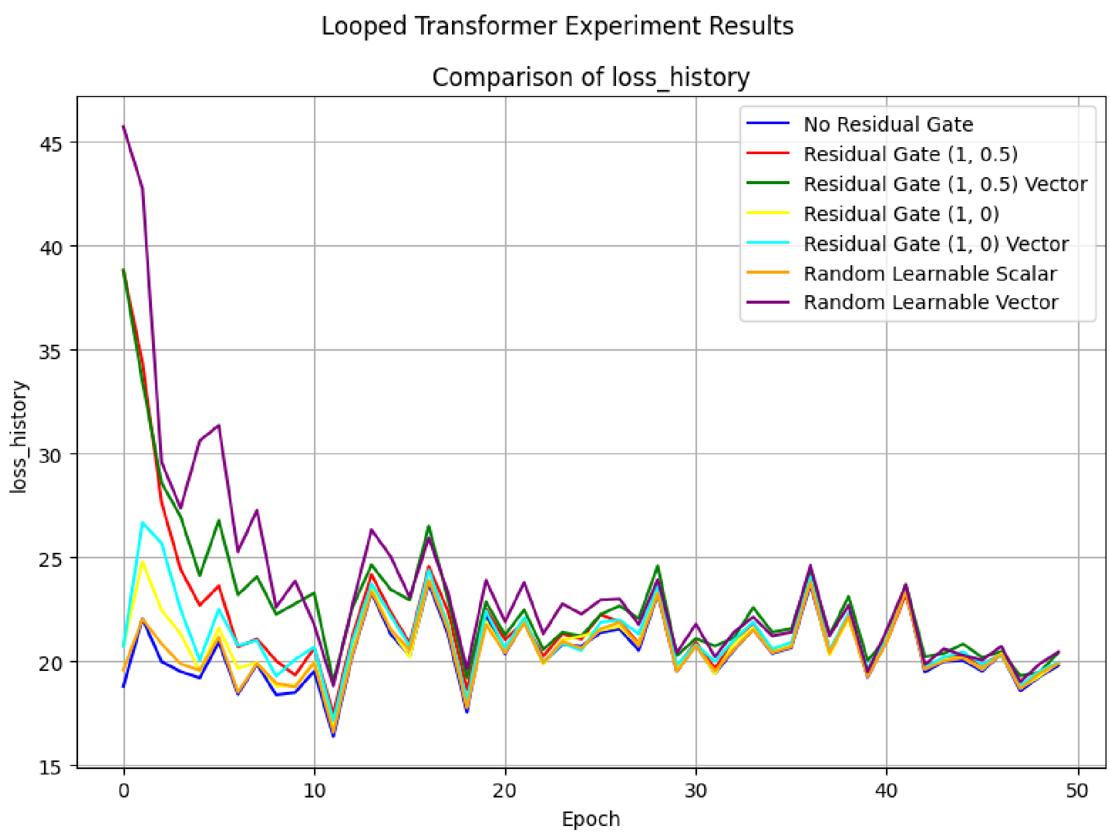
2.2.4 Residual Gate 对比实验

[27]:
```
gate_table = ExperimentTable(params_groups=[
    {'experiment_name': 'No Residual Gate', 'residual_gate': (0, 0),␣
 ↪'residual_gate_type': 'fixed'},
    {'experiment_name': 'Residual Gate (1, 0.5)', 'residual_gate': (1, 0.5),␣
 ↪'residual_gate_type': 'learnable_scalar'},
    {'experiment_name': 'Residual Gate (1, 0.5) Vector', 'residual_gate': (1, 0.
 ↪5), 'residual_gate_type': 'learnable_vector'},
    {'experiment_name': 'Residual Gate (1, 0)', 'residual_gate': (1, 0),␣
 ↪'residual_gate_type': 'learnable_scalar'},
    {'experiment_name': 'Residual Gate (1, 0) Vector', 'residual_gate': (1, 0),␣
 ↪'residual_gate_type': 'learnable_vector'},
    {'experiment_name': 'Random Learnable Scalar', 'residual_gate': 'random',␣
 ↪'residual_gate_type': 'learnable_scalar', 'residual_random': (0.9, 0.1)},
    {'experiment_name': 'Random Learnable Vector', 'residual_gate': 'random',␣
 ↪'residual_gate_type': 'learnable_vector', 'residual_random': (0.9, 0.1)},
], manual={'gate_lr_ratio': 100, 'scheduler_type': 'cosine', 'epochs': 50,␣
 ↪'print_every': 25})
```

```python
gate_table.run(result_lists=[
    (['loss_history'], 'epoch')
])
gate_table.plot(figure_size=(8, 6))
```

```
No Residual Gate initialized in 0.01 seconds. Using device: mps
Residual Gate (1, 0.5) initialized in 0.01 seconds. Using device: mps
Residual Gate (1, 0.5) Vector initialized in 0.01 seconds. Using device: mps
Residual Gate (1, 0) initialized in 0.01 seconds. Using device: mps
Residual Gate (1, 0) Vector initialized in 0.01 seconds. Using device: mps
Random Learnable Scalar initialized in 0.01 seconds. Using device: mps
Random Learnable Vector initialized in 0.01 seconds. Using device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/50, Loss: 20.7067, current_blocks: 20
Epoch 50/50, Loss: 19.7848, current_blocks: 20
Training completed in 19.79 seconds.
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 1/7 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Epoch 25/50, Loss: 21.0445, current_blocks: 20
Epoch 50/50, Loss: 19.9295, current_blocks: 20
Training completed in 20.28 seconds.
[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.15 GB
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 2/7 completed.

[Experiment 3 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Epoch 25/50, Loss: 21.1970, current_blocks: 20
Epoch 50/50, Loss: 20.4095, current_blocks: 20
Training completed in 20.98 seconds.
[Experiment 3 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.15 GB
[Experiment 3 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 3/7 completed.
```

```
[Experiment 4 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/50, Loss: 21.1870, current_blocks: 20
Epoch 50/50, Loss: 19.9217, current_blocks: 20
Training completed in 20.64 seconds.
[Experiment 4 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.15 GB
[Experiment 4 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 4/7 completed.

[Experiment 5 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/50, Loss: 20.5020, current_blocks: 20
Epoch 50/50, Loss: 19.9380, current_blocks: 20
Training completed in 21.49 seconds.
[Experiment 5 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB
[Experiment 5 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 5/7 completed.

[Experiment 6 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/50, Loss: 20.6064, current_blocks: 20
Epoch 50/50, Loss: 19.8823, current_blocks: 20
Training completed in 21.05 seconds.
[Experiment 6 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB
[Experiment 6 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 6/7 completed.

[Experiment 7 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/50, Loss: 22.2538, current_blocks: 20
Epoch 50/50, Loss: 20.4417, current_blocks: 20
Training completed in 21.80 seconds.
[Experiment 7 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB
[Experiment 7 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 7/7 completed.
```

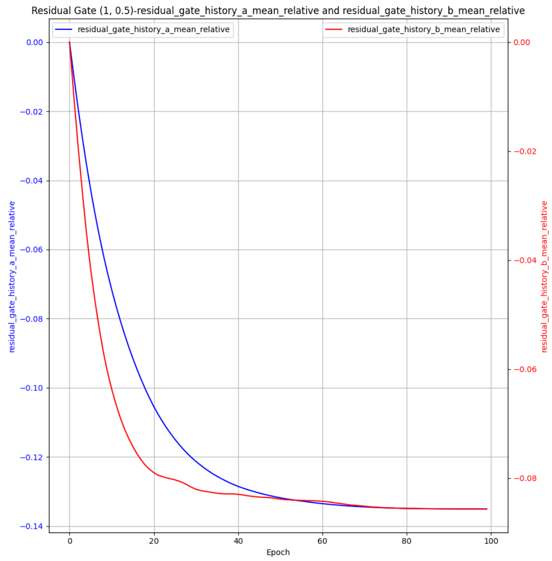


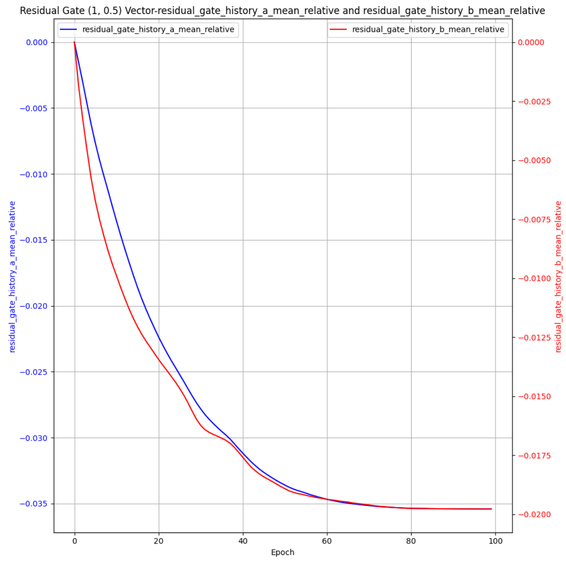
[26]:
```
gate_table = ExperimentTable(params_groups=[
    {'experiment_name': 'Residual Gate (1, 0.5)', 'residual_gate': (1, 0.5),␣
 ↪'residual_gate_type': 'learnable_scalar'},
    {'experiment_name': 'Residual Gate (1, 0.5) Vector', 'residual_gate': (1, 0.
 ↪5), 'residual_gate_type': 'learnable_vector'},
    {'experiment_name': 'Residual Gate (1, 0)', 'residual_gate': (1, 0),␣
 ↪'residual_gate_type': 'learnable_scalar'},
    {'experiment_name': 'Residual Gate (1, 0) Vector', 'residual_gate': (1, 0),␣
 ↪'residual_gate_type': 'learnable_vector'},
    {'experiment_name': 'Random Learnable Scalar', 'residual_gate': 'random',␣
 ↪'residual_gate_type': 'learnable_scalar', 'residual_random': (0.9, 0.1)},
    {'experiment_name': 'Random Learnable Vector', 'residual_gate': 'random',␣
 ↪'residual_gate_type': 'learnable_vector', 'residual_random': (0.9, 0.1)},
], manual={'gate_lr_ratio': 100, 'scheduler_type': 'cosine', 'epochs': 100,␣
 ↪'print_every': 25})
```

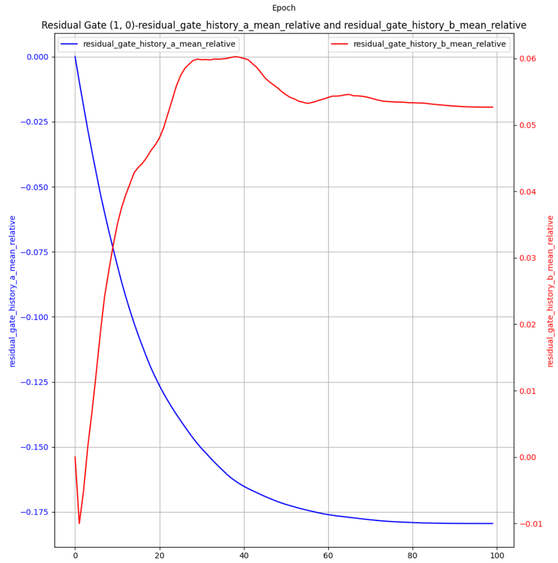
```python
gate_table.run(result_lists=[
    (['residual_gate_history_a_mean_relative',␣
 ↪'residual_gate_history_b_mean_relative'], 'epoch')
])
gate_table.plot(figure_size=(20,30),compare_experiments=False,␣
 ↪subplot_shape=(-1,2), suptitle='')
```

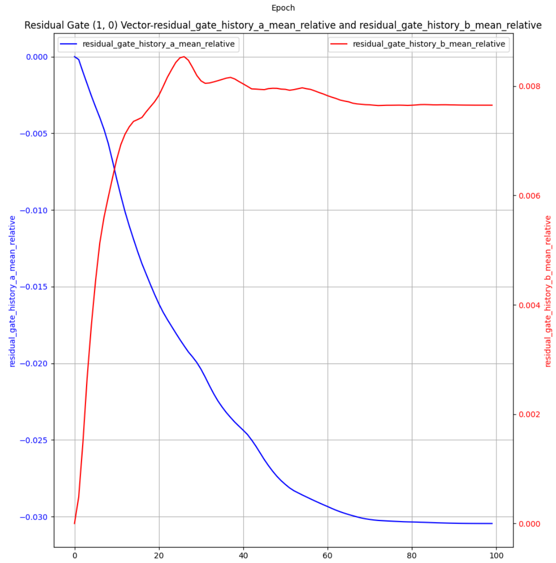
```
Residual Gate (1, 0.5) initialized in 0.02 seconds. Using device: mps
Residual Gate (1, 0.5) Vector initialized in 0.01 seconds. Using device: mps
Residual Gate (1, 0) initialized in 0.01 seconds. Using device: mps
Residual Gate (1, 0) Vector initialized in 0.01 seconds. Using device: mps
Random Learnable Scalar initialized in 0.01 seconds. Using device: mps
Random Learnable Vector initialized in 0.01 seconds. Using device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.02 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.02 GB
Epoch 25/100, Loss: 21.1192, current_blocks: 20
Epoch 50/100, Loss: 19.8862, current_blocks: 20
Epoch 75/100, Loss: 21.3765, current_blocks: 20
Epoch 100/100, Loss: 20.4947, current_blocks: 20
Training completed in 34.69 seconds.
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 1/6 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/100, Loss: 21.3562, current_blocks: 20
Epoch 50/100, Loss: 20.4467, current_blocks: 20
Epoch 75/100, Loss: 21.5648, current_blocks: 20
Epoch 100/100, Loss: 20.8265, current_blocks: 20
Training completed in 35.89 seconds.
[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.15 GB
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 2/6 completed.

[Experiment 3 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/100, Loss: 21.0529, current_blocks: 20
Epoch 50/100, Loss: 19.8676, current_blocks: 20
```

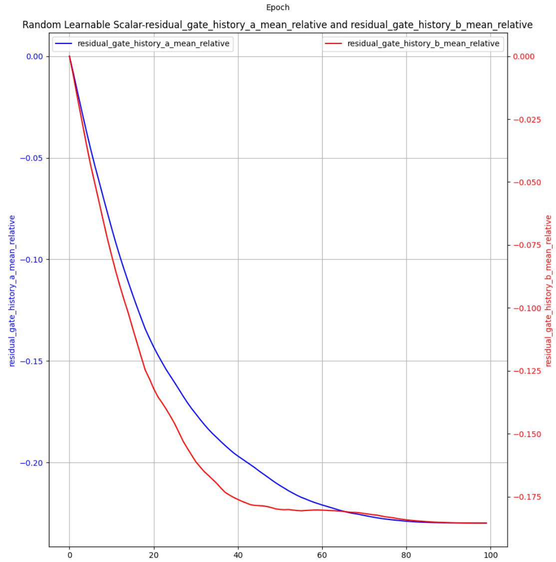
```
Epoch 75/100, Loss: 21.1713, current_blocks: 20
Epoch 100/100, Loss: 20.4512, current_blocks: 20
Training completed in 35.34 seconds.
[Experiment 3 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.15 GB
[Experiment 3 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 3/6 completed.

[Experiment 4 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/100, Loss: 20.5109, current_blocks: 20
Epoch 50/100, Loss: 20.1726, current_blocks: 20
Epoch 75/100, Loss: 21.4211, current_blocks: 20
Epoch 100/100, Loss: 20.5051, current_blocks: 20
Training completed in 36.48 seconds.
[Experiment 4 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.15 GB
[Experiment 4 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 4/6 completed.

[Experiment 5 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/100, Loss: 20.5986, current_blocks: 20
Epoch 50/100, Loss: 19.8349, current_blocks: 20
Epoch 75/100, Loss: 21.2072, current_blocks: 20
Epoch 100/100, Loss: 20.3681, current_blocks: 20
Training completed in 37.95 seconds.
[Experiment 5 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB
[Experiment 5 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 5/6 completed.

[Experiment 6 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/100, Loss: 22.4328, current_blocks: 20
Epoch 50/100, Loss: 20.2457, current_blocks: 20
Epoch 75/100, Loss: 21.3175, current_blocks: 20
Epoch 100/100, Loss: 20.7065, current_blocks: 20
Training completed in 41.02 seconds.
[Experiment 6 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB
[Experiment 6 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 6/6 completed.
```

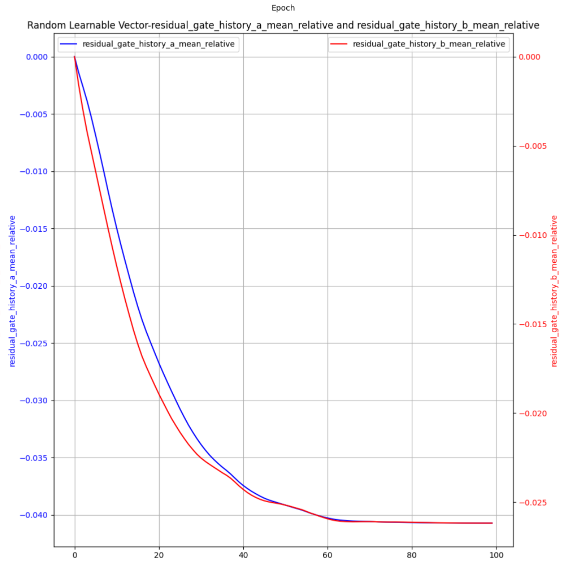

Residual Gate (1, 0.5)-residual_gate_history_a_mean_relative and residual_gate_history_b_mean_relative
residual_gate_history_a_mean_relative
residual_gate_history_b_mean_relative
Epoch


Residual Gate (1, 0.5) Vector-residual_gate_history_a_mean_relative and residual_gate_history_b_mean_relative
residual_gate_history_a_mean_relative
residual_gate_history_b_mean_relative
Epoch


Residual Gate (1, 0)-residual_gate_history_a_mean_relative and residual_gate_history_b_mean_relative
residual_gate_history_a_mean_relative
residual_gate_history_b_mean_relative
Epoch


Residual Gate (1, 0) Vector-residual_gate_history_a_mean_relative and residual_gate_history_b_mean_relative
residual_gate_history_a_mean_relative
residual_gate_history_b_mean_relative
Epoch


Random Learnable Scalar-residual_gate_history_a_mean_relative and residual_gate_history_b_mean_relative
residual_gate_history_a_mean_relative
residual_gate_history_b_mean_relative
Epoch

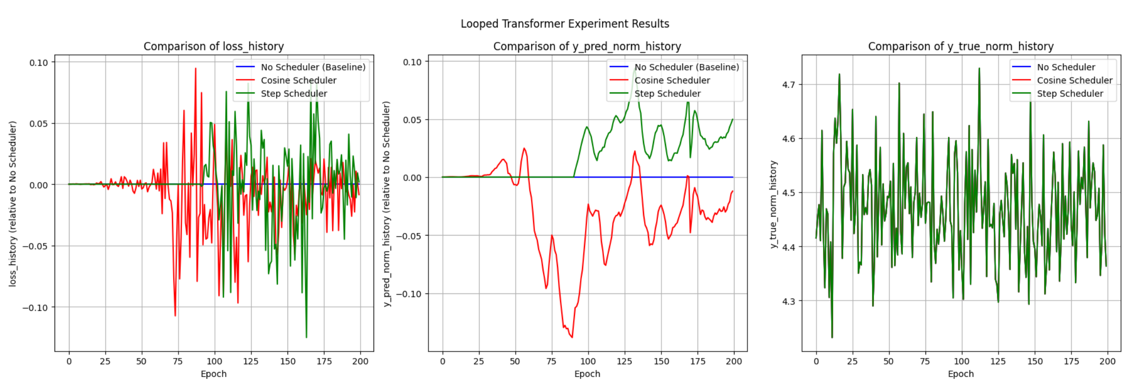

Random Learnable Vector-residual_gate_history_a_mean_relative and residual_gate_history_b_mean_relative
residual_gate_history_a_mean_relative
residual_gate_history_b_mean_relative
Epoch

2.2.5 不同 scheduler 对比实验

[28]:
```
experiment_table = ExperimentTable(params_groups=[{'experiment_name': 'No␣
↪Scheduler'},
                                                  {'scheduler_type': 'cosine',␣
↪'experiment_name': 'Cosine Scheduler', 'lr_scale': 0},
                                                  {'scheduler_type': 'step',␣
↪'experiment_name': 'Step Scheduler', 'lr_scale': 0.1, 'step_size_scheduler':␣
↪90}],
    manual={'epochs':200, 'lr': 1e-3, 'optimizer_type': 'adamw', 'pe_type':␣
↪['alibi'],
    'wd_adamw': 0.2, 'd_model': 32, 'num_heads': 1, 'max_seq_len': 500,␣
↪'seq_len': 400, 'batch_size': 64, 'd_x': 20,
    'num_blocks': 20, 'num_eff': 15,
    'residual_gate': (1,1), 'residual_gate_type': 'learnable_scalar',␣
↪'gate_lr_ratio': 100,
    'scheduler_type': None, 'lr_scale': 0.01, 'step_size_scheduler': 10,
    'print_every': 50,
    'layer_weight_decay': 0.8, 'seq_weight_decay': 0.9
    })
experiment_table.run(result_lists=[(['loss_history'], 'epoch', 0),␣
↪(['y_pred_norm_history'], 'epoch', 0), (['y_true_norm_history'], 'epoch')])
experiment_table.plot(figure_size=(18, 6))
```

```
No Scheduler initialized in 0.02 seconds. Using device: mps
Cosine Scheduler initialized in 0.01 seconds. Using device: mps
Step Scheduler initialized in 0.01 seconds. Using device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 50/200, Loss: 19.8760, current_blocks: 20
Epoch 100/200, Loss: 21.0343, current_blocks: 20
Epoch 150/200, Loss: 19.6615, current_blocks: 20
Epoch 200/200, Loss: 18.4375, current_blocks: 20
Training completed in 66.28 seconds.
```

```
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.02 GB | Metal 驱动分配: 2.13 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.07 GB
Experiment 1/3 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.08 GB
Epoch 50/200, Loss: 19.8793, current_blocks: 20
Epoch 100/200, Loss: 21.0404, current_blocks: 20
Epoch 150/200, Loss: 19.6585, current_blocks: 20
Epoch 200/200, Loss: 18.4290, current_blocks: 20
Training completed in 64.98 seconds.
[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 0.02 GB | Metal 驱动分配: 2.13 GB
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.08 GB
Experiment 2/3 completed.

[Experiment 3 训练前] 显存占用 -> PyTorch 分配: 0.01 GB | Metal 驱动分配: 0.09 GB
Epoch 50/200, Loss: 19.8760, current_blocks: 20
Epoch 100/200, Loss: 21.0657, current_blocks: 20
Epoch 150/200, Loss: 19.6840, current_blocks: 20
Epoch 200/200, Loss: 18.4393, current_blocks: 20
Training completed in 65.11 seconds.
[Experiment 3 训练峰值] 显存占用 -> PyTorch 分配: 0.02 GB | Metal 驱动分配: 2.13 GB
[Experiment 3 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.08 GB
Experiment 3/3 completed.
```



3 Non-linear Regression Experiments

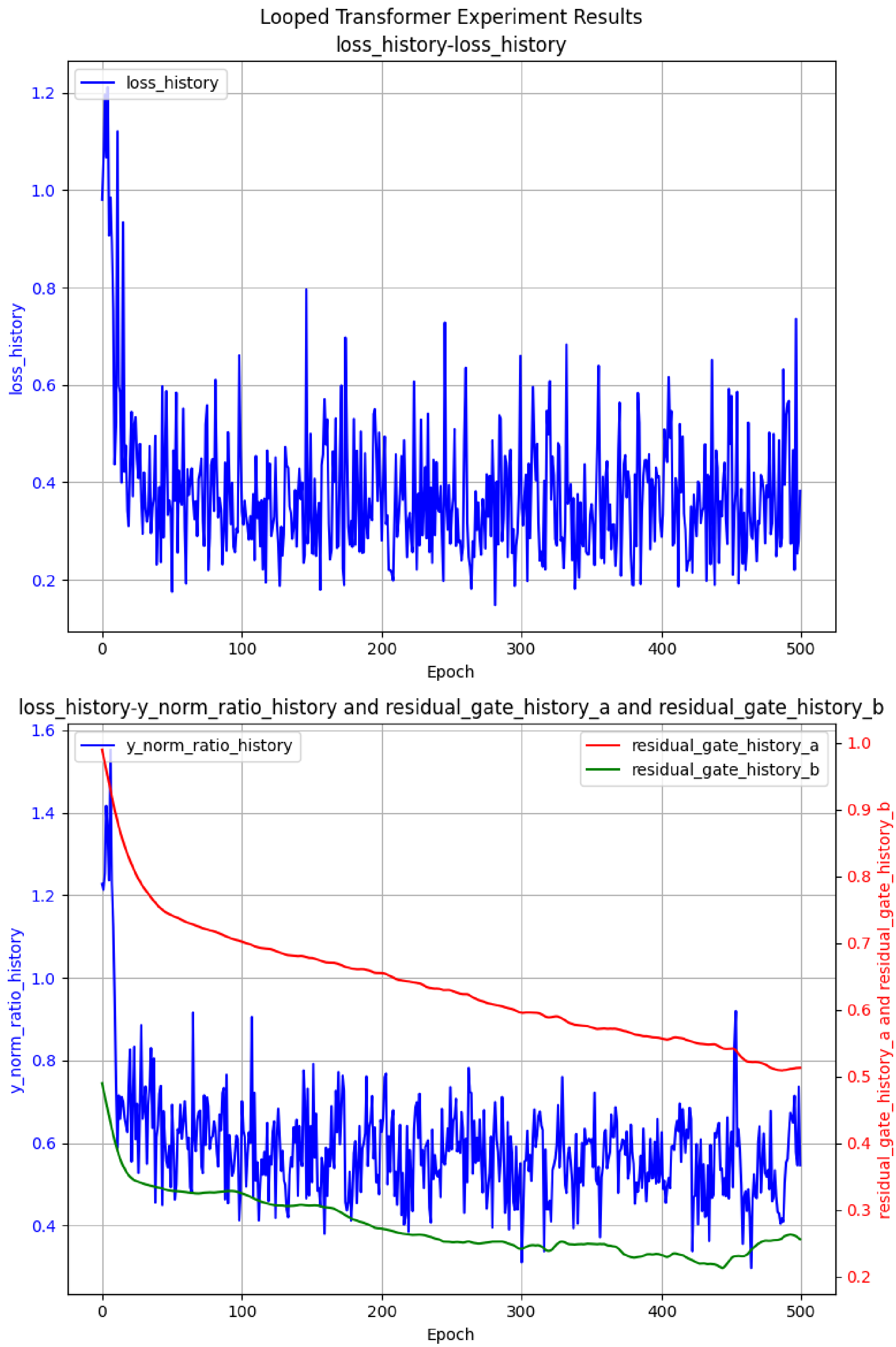
3.1 实验数据观测

[70]:
```
manual=dict(
    data_type='nonlinear',
    function_callable=lambda x: F.relu(x),
    max_seq_len=500,
    scheduled_training=True,
    pe_type=['alibi'],
    residual_gate=(1, 0.5),
    residual_gate_type='learnable_scalar',
    num_eff=5,
    num_blocks=20,
    batch_size=8,
    num_heads=2,
    d_model=128,
    seq_len=400,
    d_hidden=8,
    epochs=100)
```

[31]:
```
loss_table = ExperimentTable(params_groups=[{'experiment_name': 'loss_history',
                                             'epochs': 500,
                                             'print_every': 100}],␣
 ↪manual=manual)
loss_table.run(result_lists=[(['loss_history'], 'epoch'),␣
 ↪(['y_norm_ratio_history', 'residual_gate_history_a',␣
 ↪'residual_gate_history_b'], 'epoch')])
loss_table.plot(figure_size=(8,12), compare_experiments=False,␣
 ↪subplot_shape=(-1,1))
```

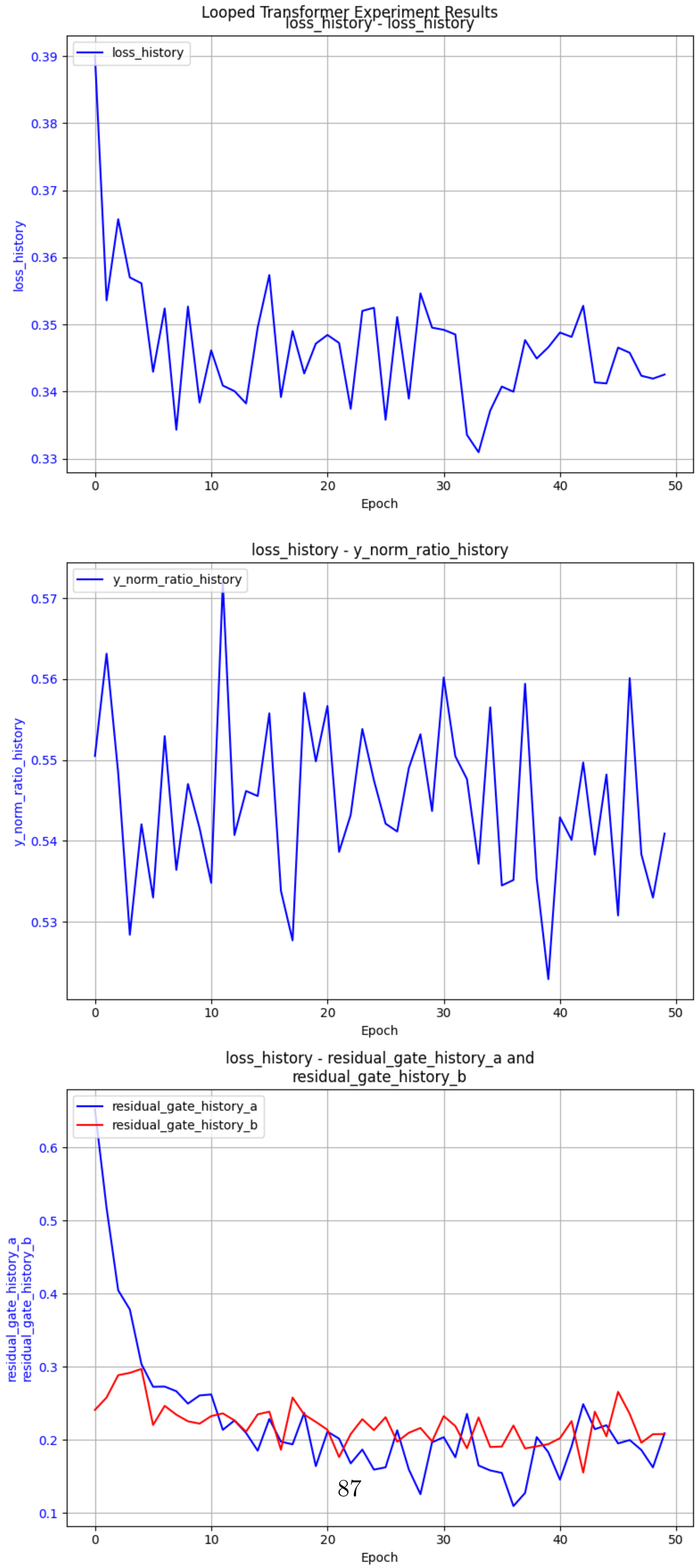
```
loss_history initialized in 0.02 seconds. Using device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.22 GB | Metal 驱动分配: 0.28 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.22 GB | Metal 驱动分配: 0.28 GB
Epoch 100/500, Loss: 0.4739, Norm Ratio: 0.4741, Current Blocks: 20
Epoch 200/500, Loss: 0.3862, Norm Ratio: 0.6557, Current Blocks: 20
Epoch 300/500, Loss: 0.6595, Norm Ratio: 0.5505, Current Blocks: 20
Epoch 400/500, Loss: 0.3205, Norm Ratio: 0.5234, Current Blocks: 20
```

```
Epoch 500/500, Loss: 0.3823, Norm Ratio: 0.5465, Current Blocks: 20
Training completed in 51.75 seconds.
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.23 GB | Metal 驱动分配: 0.93 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.22 GB | Metal 驱动分配: 0.29 GB
Experiment 1/1 completed.
```


Looped Transformer Experiment Results
loss_history-loss_history
loss_history
loss_history
Epoch
loss_history-y_norm_ratio_history and residual_gate_history_a and residual_gate_history_b
y_norm_ratio_history
residual_gate_history_a
residual_gate_history_b
y_norm_ratio_history
residual_gate_history_a and residual_gate_history_b
Epoch

[95]:
```python
loss_table = ExperimentTable(params_groups=[{'experiment_name': 'loss_history',
                                             'epochs': 50,
                                             'steps_per_epoch': 200,
                                             'init_std': 'auto',
                                             'print_every': 5}], manual=manual)
loss_table.run(result_lists=[(['loss_history'], 'epoch'),␣
 ↪(['y_norm_ratio_history'], 'epoch'), (['residual_gate_history_a',␣
 ↪'residual_gate_history_b', '|'], 'epoch')])
loss_table.plot(compare_experiments=False, subplot_shape=(-1,1))
```

```
loss_history initialized in 0.02 seconds. Using device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.28 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.28 GB
Epoch 5/50, Loss: 0.3561, Norm Ratio: 0.5421, Current Blocks: 9
Epoch 10/50, Loss: 0.3383, Norm Ratio: 0.5416, Current Blocks: 14
Epoch 15/50, Loss: 0.3495, Norm Ratio: 0.5456, Current Blocks: 19
Epoch 20/50, Loss: 0.3471, Norm Ratio: 0.5498, Current Blocks: 20
Epoch 25/50, Loss: 0.3525, Norm Ratio: 0.5475, Current Blocks: 20
Epoch 30/50, Loss: 0.3495, Norm Ratio: 0.5437, Current Blocks: 20
Epoch 35/50, Loss: 0.3371, Norm Ratio: 0.5565, Current Blocks: 20
Epoch 40/50, Loss: 0.3466, Norm Ratio: 0.5229, Current Blocks: 20
Epoch 45/50, Loss: 0.3412, Norm Ratio: 0.5482, Current Blocks: 20
Epoch 50/50, Loss: 0.3425, Norm Ratio: 0.5409, Current Blocks: 20
Training completed in 1039.72 seconds.
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.02 GB | Metal 驱动分配: 0.70 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.02 GB
Experiment 1/1 completed.
```



Looped Transformer Experiment Results
loss_history - loss_history
loss_history
Epoch
loss_history - y_norm_ratio_history
y_norm_ratio_history
Epoch
loss_history - residual_gate_history_a and residual_gate_history_b
residual_gate_history_a
residual_gate_history_b
Epoch

实验数据分析 1. 参数量分析 2. norm_ratio 与 loss 的关系分析 - 对于 Linear+ReLU+Linear，由于 Linear 矩阵归一化后方差为 1，ReLU 又会丢掉一半的激活，因此 y_true 的二阶矩（能量）期望为 $\mathbb{E}[y^2] \approx 0.5$ - 模型在进行 In-Context Learning (ICL) 时，遇到了认知瓶颈。它把复杂的真实信号 y 拆解成了两个部分：$y = y_{learnable} + y_{noise}$ - $y_{learnable}$：模型通过现有的上下文能够学明白的"低频/线性"规律。- y_{noise}：由于非线性过于复杂或上下文信息仍显不足，模型无法解开的"高频/非线性"残差。- 在 MSE 损失函数的逼迫下，模型最聪明的做法就是精准预测自己能看懂的部分，对看不懂的部分直接输出 0。因此它的预测值 $\hat{y} \approx y_{learnable}$。- 根据 Norm Ratio 倒推 loss：- 观察到 Norm Ratio 在 0.54 附近震荡，即

$$NormRatio = \sqrt{\frac{\mathbb{E}[\hat{y}^2]}{\mathbb{E}[y^2]}} \approx \sqrt{\frac{\mathbb{E}[y_{learnable}^2]}{\mathbb{E}[y_{learnable}^2] + \mathbb{E}[y_{noise}^2]}} \approx 0.54$$

- 代入 $\mathbb{E}[y^2] \approx 0.5$，可以推断出 $\mathbb{E}[y_{learnable}^2] \approx 0.14$，$\mathbb{E}[y_{noise}^2] \approx 0.36$。- 由于 $\hat{y} \to y_{learnable}$，因此最终的 loss 约为

$$\mathbb{E}[(y-\hat{y})^2] = \mathbb{E}[y_{noise}^2] + \mathbb{E}[(y_{learnable} - \hat{y})^2] \approx 0.36$$

与实际观察到的 loss 值相近。

4 Curriculum Learning 实验

4.1 linear

4.1.1 训练 10000 步

[ ]:
```
manual_config = dict(
    # 任务基础配置
    data_type='linear',
    d_x=20,
    seq_len=80,
    batch_size=64,

    # 模型架构 (完全对齐论文)
    num_blocks=20,
    num_eff=15,
    d_model=256,
    num_heads=8,
    pe_type=['learned_ape'],

```

```
    # 核心物理机制配置
    x_init='zero',              # 绝对零起点
    init_std='auto',            # 方差自适应 (1/sqrt(256))
    residual_gate=(1, 1),       # 标准注入
    residual_gate_type='fixed',

    # 优化器剥离外挂
    optimizer_type='adam',
    lr=1e-4,
    layer_weight_decay=1.0,     # 关闭层级惩罚
    seq_weight_decay=1.0,       # 关闭时间步惩罚
    scheduler_type=None,        # 恒定学习率

    # 运行控制
    epochs=50,
    steps_per_epoch=200,
    print_every=1,

    scheduled_training=True     # 开启 b 的 Kick-start
)

# === 定义并执行实验表 ===
final_table = ExperimentTable(params_groups=[
    {
        'experiment_name': 'Without Curriculum (Hard Mode)',
        'curriculum': {} # 空字典关闭课程学习
    },
    {
        'experiment_name': 'With Curriculum (Perfect Path)',
        'curriculum': {
            'd_x': 5,
            'seq_len': 10,
            'duration_ratio': 0.8 # 前 40 个 epoch 完成课程
        }
    }
], manual=manual_config)
```

```
final_table.run(result_lists=[(['loss_history'], 'epoch'),␣
 ↪(['y_norm_ratio_history'], 'epoch')])
final_table.plot(figure_size=(10, 12), compare_experiments=True,␣
 ↪subplot_shape=(-1, 1), suptitle='Curriculum Learning Comparison on Looped␣
 ↪Transformer')
```

```
Without Curriculum (Hard Mode) initialized in 0.03 seconds. Using device: mps
With Curriculum (Perfect Path) initialized in 0.02 seconds. Using device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.04 GB | Metal 驱动分配: 1.03 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.05 GB | Metal 驱动分配: 1.03 GB
Epoch 1/50, Loss: 1.0582, Norm Ratio: 0.1946, Avg Sink Score: 0.0655, Avg Sink
Rate: 0.0000
Epoch 2/50, Loss: 1.0064, Norm Ratio: 0.0913, Avg Sink Score: 0.0636, Avg Sink
Rate: 0.0000
Epoch 3/50, Loss: 1.0058, Norm Ratio: 0.0701, Avg Sink Score: 0.0629, Avg Sink
Rate: 0.0000
Epoch 4/50, Loss: 1.0001, Norm Ratio: 0.0598, Avg Sink Score: 0.0622, Avg Sink
Rate: 0.0000
Epoch 5/50, Loss: 1.0026, Norm Ratio: 0.0482, Avg Sink Score: 0.0622, Avg Sink
Rate: 0.0000
Epoch 6/50, Loss: 0.9974, Norm Ratio: 0.0422, Avg Sink Score: 0.0621, Avg Sink
Rate: 0.0000
Epoch 7/50, Loss: 1.0029, Norm Ratio: 0.0469, Avg Sink Score: 0.0620, Avg Sink
Rate: 0.0000
Epoch 8/50, Loss: 0.9991, Norm Ratio: 0.0409, Avg Sink Score: 0.0618, Avg Sink
Rate: 0.0000
Epoch 9/50, Loss: 1.0000, Norm Ratio: 0.0467, Avg Sink Score: 0.0619, Avg Sink
Rate: 0.0000
Epoch 10/50, Loss: 1.0101, Norm Ratio: 0.0459, Avg Sink Score: 0.0619, Avg Sink
Rate: 0.0000
Epoch 11/50, Loss: 1.0005, Norm Ratio: 0.0469, Avg Sink Score: 0.0618, Avg Sink
Rate: 0.0000
Epoch 12/50, Loss: 1.0032, Norm Ratio: 0.0422, Avg Sink Score: 0.0617, Avg Sink
Rate: 0.0000
Epoch 13/50, Loss: 1.0008, Norm Ratio: 0.0384, Avg Sink Score: 0.0617, Avg Sink
Rate: 0.0000
```

```
Epoch 14/50, Loss: 1.0042, Norm Ratio: 0.0402, Avg Sink Score: 0.0616, Avg Sink
Rate: 0.0000
Epoch 15/50, Loss: 1.0014, Norm Ratio: 0.0380, Avg Sink Score: 0.0616, Avg Sink
Rate: 0.0000
Epoch 16/50, Loss: 0.9992, Norm Ratio: 0.0409, Avg Sink Score: 0.0616, Avg Sink
Rate: 0.0000
Epoch 17/50, Loss: 1.0021, Norm Ratio: 0.0352, Avg Sink Score: 0.0616, Avg Sink
Rate: 0.0000
Epoch 18/50, Loss: 0.9987, Norm Ratio: 0.0407, Avg Sink Score: 0.0615, Avg Sink
Rate: 0.0000
Epoch 19/50, Loss: 1.0048, Norm Ratio: 0.0314, Avg Sink Score: 0.0616, Avg Sink
Rate: 0.0000
Epoch 20/50, Loss: 1.0034, Norm Ratio: 0.0404, Avg Sink Score: 0.0615, Avg Sink
Rate: 0.0000
Epoch 21/50, Loss: 1.0021, Norm Ratio: 0.0342, Avg Sink Score: 0.0615, Avg Sink
Rate: 0.0000
Epoch 22/50, Loss: 1.0008, Norm Ratio: 0.0289, Avg Sink Score: 0.0615, Avg Sink
Rate: 0.0000
Epoch 23/50, Loss: 1.0047, Norm Ratio: 0.0295, Avg Sink Score: 0.0614, Avg Sink
Rate: 0.0000
Epoch 24/50, Loss: 1.0013, Norm Ratio: 0.0320, Avg Sink Score: 0.0614, Avg Sink
Rate: 0.0000
Epoch 25/50, Loss: 1.0034, Norm Ratio: 0.0348, Avg Sink Score: 0.0614, Avg Sink
Rate: 0.0000
Epoch 26/50, Loss: 0.9986, Norm Ratio: 0.0308, Avg Sink Score: 0.0613, Avg Sink
Rate: 0.0000
Epoch 27/50, Loss: 1.0008, Norm Ratio: 0.0267, Avg Sink Score: 0.0613, Avg Sink
Rate: 0.0000
Epoch 28/50, Loss: 1.0033, Norm Ratio: 0.0325, Avg Sink Score: 0.0613, Avg Sink
Rate: 0.0000
Epoch 29/50, Loss: 1.0068, Norm Ratio: 0.0361, Avg Sink Score: 0.0613, Avg Sink
Rate: 0.0000
Epoch 30/50, Loss: 1.0005, Norm Ratio: 0.0325, Avg Sink Score: 0.0613, Avg Sink
Rate: 0.0000
Epoch 31/50, Loss: 1.0023, Norm Ratio: 0.0311, Avg Sink Score: 0.0612, Avg Sink
Rate: 0.0000
Epoch 32/50, Loss: 1.0016, Norm Ratio: 0.0284, Avg Sink Score: 0.0611, Avg Sink
```

```
Rate: 0.0000
Epoch 33/50, Loss: 1.0026, Norm Ratio: 0.0264, Avg Sink Score: 0.0611, Avg Sink
Rate: 0.0000
Epoch 34/50, Loss: 1.0028, Norm Ratio: 0.0295, Avg Sink Score: 0.0611, Avg Sink
Rate: 0.0000
Epoch 35/50, Loss: 1.0044, Norm Ratio: 0.0264, Avg Sink Score: 0.0610, Avg Sink
Rate: 0.0000
Epoch 36/50, Loss: 1.0022, Norm Ratio: 0.0256, Avg Sink Score: 0.0609, Avg Sink
Rate: 0.0000
Epoch 37/50, Loss: 1.0007, Norm Ratio: 0.0263, Avg Sink Score: 0.0608, Avg Sink
Rate: 0.0000
Epoch 38/50, Loss: 1.0035, Norm Ratio: 0.0268, Avg Sink Score: 0.0607, Avg Sink
Rate: 0.0000
Epoch 39/50, Loss: 1.0021, Norm Ratio: 0.0262, Avg Sink Score: 0.0605, Avg Sink
Rate: 0.0000
Epoch 40/50, Loss: 1.0019, Norm Ratio: 0.0252, Avg Sink Score: 0.0604, Avg Sink
Rate: 0.0000
Epoch 41/50, Loss: 1.0015, Norm Ratio: 0.0256, Avg Sink Score: 0.0603, Avg Sink
Rate: 0.0000
Epoch 42/50, Loss: 0.9988, Norm Ratio: 0.0248, Avg Sink Score: 0.0602, Avg Sink
Rate: 0.0000
Epoch 43/50, Loss: 1.0032, Norm Ratio: 0.0256, Avg Sink Score: 0.0601, Avg Sink
Rate: 0.0000
Epoch 44/50, Loss: 0.9974, Norm Ratio: 0.0256, Avg Sink Score: 0.0600, Avg Sink
Rate: 0.0000
Epoch 45/50, Loss: 1.0036, Norm Ratio: 0.0226, Avg Sink Score: 0.0599, Avg Sink
Rate: 0.0000
Epoch 46/50, Loss: 0.9997, Norm Ratio: 0.0225, Avg Sink Score: 0.0598, Avg Sink
Rate: 0.0000
Epoch 47/50, Loss: 1.0008, Norm Ratio: 0.0271, Avg Sink Score: 0.0597, Avg Sink
Rate: 0.0000
Epoch 48/50, Loss: 1.0028, Norm Ratio: 0.0248, Avg Sink Score: 0.0595, Avg Sink
Rate: 0.0000
Epoch 49/50, Loss: 0.9979, Norm Ratio: 0.0251, Avg Sink Score: 0.0593, Avg Sink
Rate: 0.0000
Epoch 50/50, Loss: 1.0009, Norm Ratio: 0.0250, Avg Sink Score: 0.0592, Avg Sink
Rate: 0.0000
```

```
Training completed in 5862.61 seconds.
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.07 GB | Metal 驱动分配: 2.16 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.06 GB | Metal 驱动分配: 1.08 GB
Experiment 1/2 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 0.06 GB | Metal 驱动分配: 1.08 GB
Epoch 1/50, Loss: 1.0791, Norm Ratio: 0.2363, Avg Sink Score: 0.2882, Avg Sink
Rate: 0.3833
Epoch 2/50, Loss: 0.9322, Norm Ratio: 0.2950, Avg Sink Score: 0.2736, Avg Sink
Rate: 0.2903
Epoch 3/50, Loss: 0.8603, Norm Ratio: 0.4350, Avg Sink Score: 0.2572, Avg Sink
Rate: 0.1875
Epoch 4/50, Loss: 0.8075, Norm Ratio: 0.4584, Avg Sink Score: 0.2414, Avg Sink
Rate: 0.1364
Epoch 5/50, Loss: 0.7707, Norm Ratio: 0.5010, Avg Sink Score: 0.2263, Avg Sink
Rate: 0.1059
Epoch 6/50, Loss: 0.7786, Norm Ratio: 0.4951, Avg Sink Score: 0.2133, Avg Sink
Rate: 0.0857
Epoch 7/50, Loss: 0.7639, Norm Ratio: 0.5090, Avg Sink Score: 0.2016, Avg Sink
Rate: 0.0720
Epoch 8/50, Loss: 0.7322, Norm Ratio: 0.5403, Avg Sink Score: 0.1926, Avg Sink
Rate: 0.0621
Epoch 9/50, Loss: 0.7622, Norm Ratio: 0.5098, Avg Sink Score: 0.1844, Avg Sink
Rate: 0.0545
Epoch 10/50, Loss: 0.7029, Norm Ratio: 0.5579, Avg Sink Score: 0.1771, Avg Sink
Rate: 0.0486
Epoch 11/50, Loss: 0.6820, Norm Ratio: 0.5744, Avg Sink Score: 0.1702, Avg Sink
Rate: 0.0439
Epoch 12/50, Loss: 0.6731, Norm Ratio: 0.5831, Avg Sink Score: 0.1640, Avg Sink
Rate: 0.0400
Epoch 13/50, Loss: 0.6210, Norm Ratio: 0.6213, Avg Sink Score: 0.1583, Avg Sink
Rate: 0.0367
Epoch 14/50, Loss: 0.6425, Norm Ratio: 0.6033, Avg Sink Score: 0.1534, Avg Sink
Rate: 0.0340
Epoch 15/50, Loss: 0.6188, Norm Ratio: 0.6263, Avg Sink Score: 0.1487, Avg Sink
Rate: 0.0316
Epoch 16/50, Loss: 0.5912, Norm Ratio: 0.6490, Avg Sink Score: 0.1448, Avg Sink
```

Rate: 0.0295

Epoch 17/50, Loss: 0.6237, Norm Ratio: 0.6166, Avg Sink Score: 0.1409, Avg Sink Rate: 0.0277

Epoch 18/50, Loss: 0.5706, Norm Ratio: 0.6545, Avg Sink Score: 0.1374, Avg Sink Rate: 0.0261

Epoch 19/50, Loss: 0.5770, Norm Ratio: 0.6559, Avg Sink Score: 0.1340, Avg Sink Rate: 0.0247

Epoch 20/50, Loss: 0.5845, Norm Ratio: 0.6527, Avg Sink Score: 0.1309, Avg Sink Rate: 0.0234

Epoch 21/50, Loss: 0.5433, Norm Ratio: 0.6775, Avg Sink Score: 0.1279, Avg Sink Rate: 0.0222

Epoch 22/50, Loss: 0.5647, Norm Ratio: 0.6624, Avg Sink Score: 0.1251, Avg Sink Rate: 0.0212

Epoch 23/50, Loss: 0.5379, Norm Ratio: 0.6831, Avg Sink Score: 0.1225, Avg Sink Rate: 0.0202

Epoch 24/50, Loss: 0.5074, Norm Ratio: 0.7029, Avg Sink Score: 0.1201, Avg Sink Rate: 0.0194

Epoch 25/50, Loss: 0.5467, Norm Ratio: 0.6778, Avg Sink Score: 0.1178, Avg Sink Rate: 0.0186

Epoch 26/50, Loss: 0.5018, Norm Ratio: 0.7081, Avg Sink Score: 0.1156, Avg Sink Rate: 0.0178

Epoch 27/50, Loss: 0.5041, Norm Ratio: 0.7057, Avg Sink Score: 0.1136, Avg Sink Rate: 0.0171

Epoch 28/50, Loss: 0.5030, Norm Ratio: 0.7082, Avg Sink Score: 0.1115, Avg Sink Rate: 0.0165

Epoch 29/50, Loss: 0.4671, Norm Ratio: 0.7335, Avg Sink Score: 0.1096, Avg Sink Rate: 0.0159

Epoch 30/50, Loss: 0.4845, Norm Ratio: 0.7197, Avg Sink Score: 0.1078, Avg Sink Rate: 0.0154

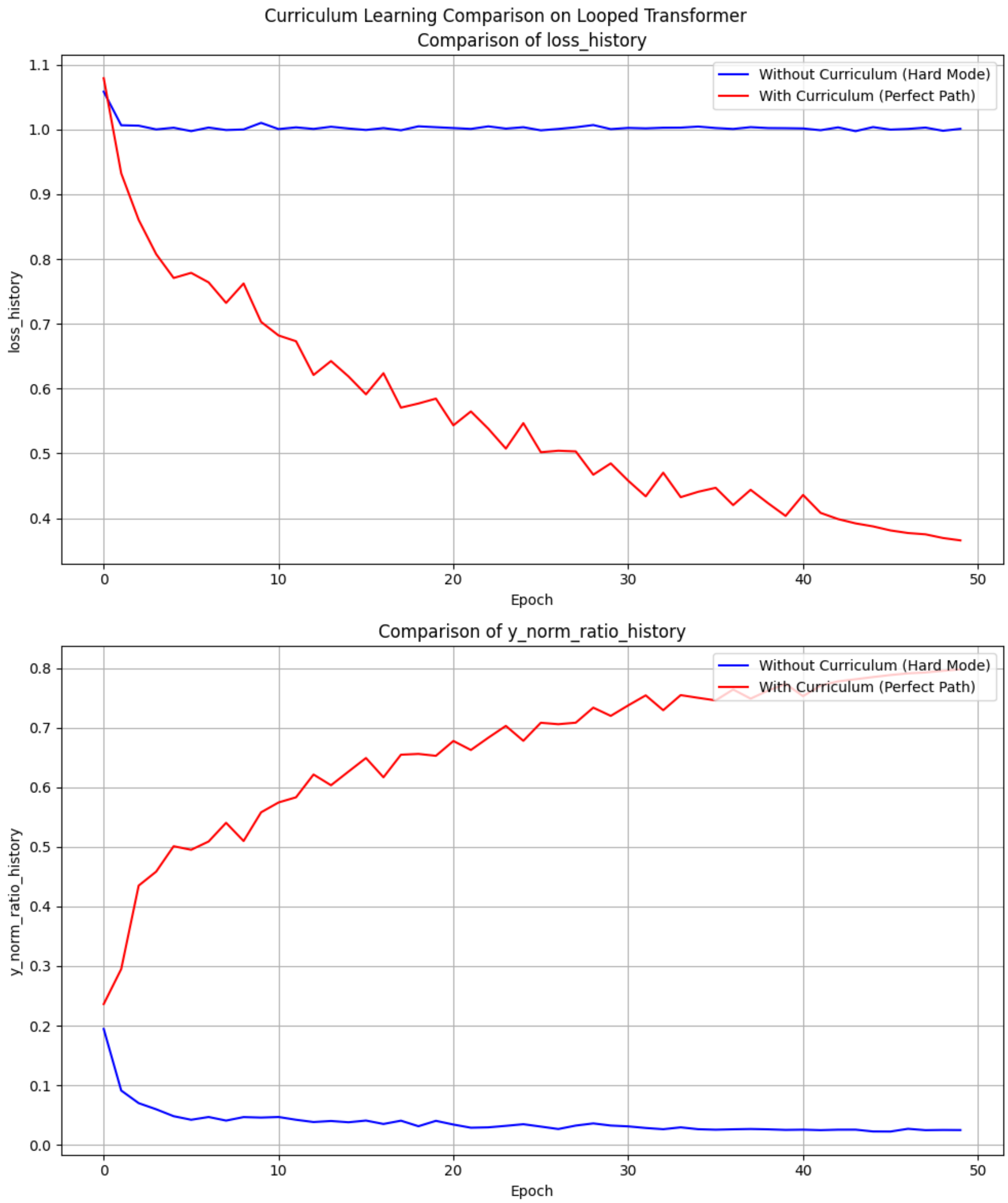
Epoch 31/50, Loss: 0.4578, Norm Ratio: 0.7373, Avg Sink Score: 0.1060, Avg Sink Rate: 0.0149

Epoch 32/50, Loss: 0.4337, Norm Ratio: 0.7541, Avg Sink Score: 0.1044, Avg Sink Rate: 0.0144

Epoch 33/50, Loss: 0.4702, Norm Ratio: 0.7292, Avg Sink Score: 0.1028, Avg Sink Rate: 0.0140

Epoch 34/50, Loss: 0.4324, Norm Ratio: 0.7545, Avg Sink Score: 0.1013, Avg Sink Rate: 0.0135

```
Epoch 35/50, Loss: 0.4407, Norm Ratio: 0.7499, Avg Sink Score: 0.0999, Avg Sink
Rate: 0.0131
Epoch 36/50, Loss: 0.4469, Norm Ratio: 0.7456, Avg Sink Score: 0.0986, Avg Sink
Rate: 0.0128
Epoch 37/50, Loss: 0.4202, Norm Ratio: 0.7639, Avg Sink Score: 0.0972, Avg Sink
Rate: 0.0124
Epoch 38/50, Loss: 0.4438, Norm Ratio: 0.7486, Avg Sink Score: 0.0959, Avg Sink
Rate: 0.0121
Epoch 39/50, Loss: 0.4230, Norm Ratio: 0.7621, Avg Sink Score: 0.0947, Avg Sink
Rate: 0.0118
Epoch 40/50, Loss: 0.4035, Norm Ratio: 0.7733, Avg Sink Score: 0.0935, Avg Sink
Rate: 0.0115
Curriculum ended at epoch 41, step 1.
Epoch 41/50, Loss: 0.4358, Norm Ratio: 0.7527, Avg Sink Score: 0.0923, Avg Sink
Rate: 0.0112
Epoch 42/50, Loss: 0.4082, Norm Ratio: 0.7704, Avg Sink Score: 0.0912, Avg Sink
Rate: 0.0109
Epoch 43/50, Loss: 0.3986, Norm Ratio: 0.7775, Avg Sink Score: 0.0902, Avg Sink
Rate: 0.0107
Epoch 44/50, Loss: 0.3920, Norm Ratio: 0.7811, Avg Sink Score: 0.0892, Avg Sink
Rate: 0.0104
Epoch 45/50, Loss: 0.3873, Norm Ratio: 0.7848, Avg Sink Score: 0.0883, Avg Sink
Rate: 0.0102
Epoch 46/50, Loss: 0.3810, Norm Ratio: 0.7882, Avg Sink Score: 0.0874, Avg Sink
Rate: 0.0099
Epoch 47/50, Loss: 0.3771, Norm Ratio: 0.7910, Avg Sink Score: 0.0865, Avg Sink
Rate: 0.0097
Epoch 48/50, Loss: 0.3750, Norm Ratio: 0.7924, Avg Sink Score: 0.0857, Avg Sink
Rate: 0.0095
Epoch 49/50, Loss: 0.3694, Norm Ratio: 0.7954, Avg Sink Score: 0.0850, Avg Sink
Rate: 0.0093
Epoch 50/50, Loss: 0.3657, Norm Ratio: 0.7977, Avg Sink Score: 0.0843, Avg Sink
Rate: 0.0091
Training completed in 3460.71 seconds.
[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 0.09 GB | Metal 驱动分配: 5.65 GB
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 0.07 GB | Metal 驱动分配: 2.09 GB
Experiment 2/2 completed.
```


Curriculum Learning Comparison on Looped Transformer
Comparison of loss_history
Without Curriculum (Hard Mode)
With Curriculum (Perfect Path)
loss_history
Epoch
Comparison of y_norm_ratio_history
Without Curriculum (Hard Mode)
With Curriculum (Perfect Path)
y_norm_ratio_history
Epoch

4.1.2 保存模型权重

[212]:
```
import torch
import os
os.makedirs('saved_checkpoints', exist_ok=True)
print("正在保存模型权重...")
for i, exp in enumerate(final_table.experiments):
    exp_name = final_table.init_parameters[i].get('experiment_name',␣
 ↪f'Experiment_{i}')
    # 清理文件名中的特殊字符
    safe_name = exp_name.replace(' ', '_').replace('(', '').replace(')', '')
    save_path = f'saved_checkpoints/{safe_name}.pth'
    checkpoint = {
        'model_state_dict': exp.model.state_dict(),
        'loss_history': getattr(exp, 'loss_history', []),
        'y_pred_norm_history': getattr(exp, 'y_pred_norm_history', []),
        'y_true_norm_history': getattr(exp, 'y_true_norm_history', []),
        'sink_score_history': getattr(exp, 'sink_score_history', []),
        'sink_rate_history': getattr(exp, 'sink_rate_history', []),
        'residual_gate_history': getattr(exp, 'residual_gate_history', []),
    }
    if hasattr(exp, 'optimizer'):
        checkpoint['optimizer_state_dict'] = exp.optimizer.state_dict()
    torch.save(checkpoint, save_path)
    print(f" Checkpoint 已安全保存至: {save_path}")
```

```
正在保存模型权重…
 Checkpoint 已安全保存至:
saved_checkpoints/Nonlinear_Without_Curriculum_Hard_Mode.pth
 Checkpoint 已安全保存至:
saved_checkpoints/Nonlinear_With_Curriculum_Perfect_Path.pth
```

4.1.3 加载模型、评估结果

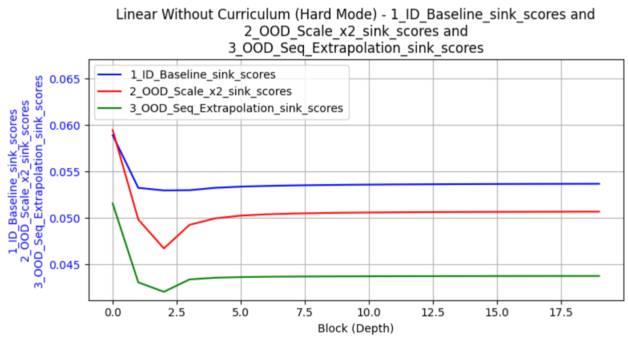
[ ]:
```
manual_config = dict(
    # 任务基础配置
    data_type='linear',
    d_x=20,
```

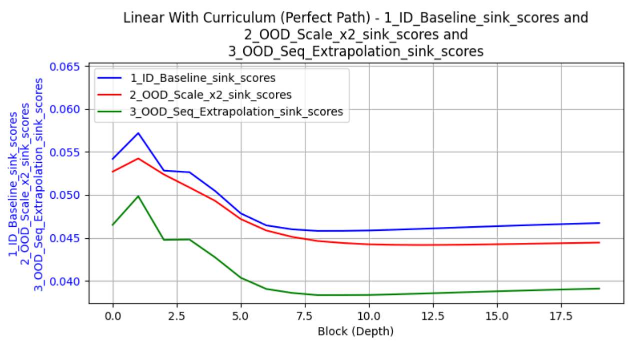
```
    seq_len=80,
    batch_size=64,

    # 模型架构 (完全对齐论文)
    num_blocks=20,
    num_eff=15,
    d_model=256,
    num_heads=8,
    pe_type=['learned_ape'],

    # 核心物理机制配置
    x_init='zero',              # 绝对零起点
    init_std='auto',            # 方差自适应 (1/sqrt(256))
    residual_gate=(1, 1),       # 标准注入
    residual_gate_type='fixed',

    # 优化器剥离外挂
    optimizer_type='adam',
    lr=1e-4,
    layer_weight_decay=1.0,     # 关闭层级惩罚
    seq_weight_decay=1.0,       # 关闭时间步惩罚
    scheduler_type=None,        # 恒定学习率

    # 运行控制
    epochs=50,
    steps_per_epoch=200,
    print_every=1,

    scheduled_training=True     # 开启 b 的 Kick-start
)

test_table = ExperimentTable(params_groups=[
    {
        'experiment_name': 'Linear Without Curriculum (Hard Mode)',
        'curriculum': {}, # 空字典关闭课程学习
```

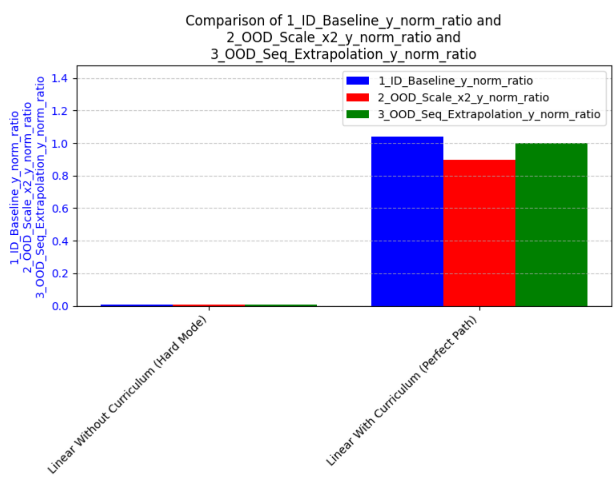
```
        'load_path': 'saved_checkpoints/Without_Curriculum_Hard_Mode_weights.
↪pth'
    },
    {
        'experiment_name': 'Linear With Curriculum (Perfect Path)',
        'curriculum': {
            'd_x': 5,
            'seq_len': 10,
            'duration_ratio': 0.8 # 前 40 个 epoch 完成课程
        },
        'load_path': 'saved_checkpoints/With_Curriculum_Perfect_Path_weights.
↪pth'
    }
], manual=manual_config)

results_lists = [
    (['1_ID_Baseline_sink_scores','2_OOD_Scale_x2_sink_scores',␣
↪'3_OOD_Seq_Extrapolation_sink_scores'], 'block'),
    (['1_ID_Baseline_y_norm_ratio', '2_OOD_Scale_x2_y_norm_ratio',␣
↪'3_OOD_Seq_Extrapolation_y_norm_ratio'], 'experiment'),
    (['1_ID_Baseline_loss', '2_OOD_Scale_x2_loss',␣
↪'3_OOD_Seq_Extrapolation_loss'], 'experiment')
]

eval_configs = [
    {'eval_name': '1_ID_Baseline', 'ood_kwargs': {}}, # 分布内基准
    {'eval_name': '2_OOD_Scale_x2', 'ood_kwargs': {'x_scale': 2.0}}, # X 尺度放
大（测试抗干扰）
    {'eval_name': '3_OOD_Seq_Extrapolation', 'ood_kwargs': {'seq_len_scale': 1.
↪2}} # 序列长度外推（测试算法泛化）
]
test_table.run(result_lists=results_lists, modes=['evaluate'],␣
↪eval_configs=eval_configs)
test_table.plot(compare_experiments=False, subplot_shape=(-1,1),suptitle='')
```

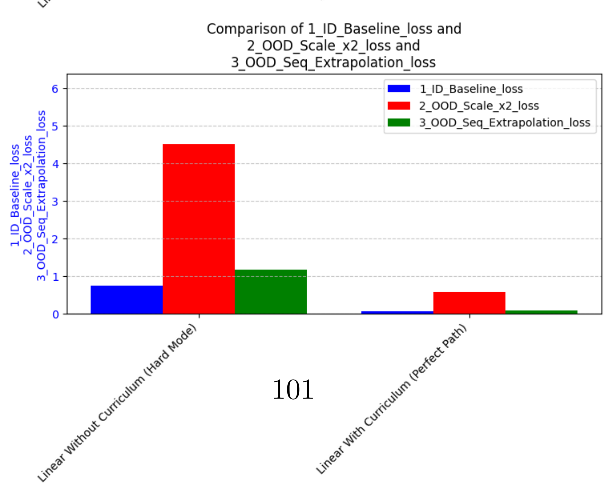
```
检测到纯权重 state_dict 文件（或历史版本），直接加载 model_state_dict…
```

```
  成功从 saved_checkpoints/Without_Curriculum_Hard_Mode_weights.pth 恢复实验状态。
Linear Without Curriculum (Hard Mode) initialized in 0.06 seconds. Using device:
mps
检测到纯权重 state_dict 文件（或历史版本），直接加载 model_state_dict…
  成功从 saved_checkpoints/With_Curriculum_Perfect_Path_weights.pth 恢复实验状态。
Linear With Curriculum (Perfect Path) initialized in 0.03 seconds. Using device:
mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.11 GB | Metal 驱动分配: 2.07 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.11 GB | Metal 驱动分配: 2.07 GB
[1_ID_Baseline] Loss: 0.7418, Norm Ratio: 0.0097
[2_OOD_Scale_x2] Loss: 4.5082, Norm Ratio: 0.0054
[3_OOD_Seq_Extrapolation] Loss: 1.1665, Norm Ratio: 0.0102
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.13 GB | Metal 驱动分配: 2.07 GB
Experiment 1/2 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 0.14 GB | Metal 驱动分配: 2.07 GB
[1_ID_Baseline] Loss: 0.0728, Norm Ratio: 1.0402
[2_OOD_Scale_x2] Loss: 0.5680, Norm Ratio: 0.8993
[3_OOD_Seq_Extrapolation] Loss: 0.0907, Norm Ratio: 0.9985
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 0.16 GB | Metal 驱动分配: 2.07 GB
Experiment 2/2 completed.
```


Linear Without Curriculum (Hard Mode) - 1_ID_Baseline_sink_scores and 2_OOD_Scale_x2_sink_scores and 3_OOD_Seq_Extrapolation_sink_scores
1_ID_Baseline_sink_scores
2_OOD_Scale_x2_sink_scores
3_OOD_Seq_Extrapolation_sink_scores
Block (Depth)


Linear With Curriculum (Perfect Path) - 1_ID_Baseline_sink_scores and 2_OOD_Scale_x2_sink_scores and 3_OOD_Seq_Extrapolation_sink_scores
1_ID_Baseline_sink_scores
2_OOD_Scale_x2_sink_scores
3_OOD_Seq_Extrapolation_sink_scores
Block (Depth)


Comparison of 1_ID_Baseline_y_norm_ratio and 2_OOD_Scale_x2_y_norm_ratio and 3_OOD_Seq_Extrapolation_y_norm_ratio
1_ID_Baseline_y_norm_ratio
2_OOD_Scale_x2_y_norm_ratio
3_OOD_Seq_Extrapolation_y_norm_ratio
Linear Without Curriculum (Hard Mode)
Linear With Curriculum (Perfect Path)


Comparison of 1_ID_Baseline_loss and 2_OOD_Scale_x2_loss and 3_OOD_Seq_Extrapolation_loss
1_ID_Baseline_loss
2_OOD_Scale_x2_loss
3_OOD_Seq_Extrapolation_loss
Linear Without Curriculum (Hard Mode)
Linear With Curriculum (Perfect Path)

4.2 nonlinear

4.2.1 任务配置

[60]:
```
manual_config = dict(
    # 任务基础配置
    data_type='nonlinear',
    function_callable=lambda x: 2**0.5 *F.relu(x),
    max_seq_len=200,
    d_x=20,
    seq_len=80,
    batch_size=64,

    # 模型架构 (完全对齐论文)
    num_blocks=20,
    num_eff=15,
    d_model=256,
    num_heads=8,
    pe_type=['learned_ape'],

    # 核心物理机制配置
    x_init='zero',              # 绝对零起点
    init_std='auto',            # 方差自适应 (1/sqrt(256))
    residual_gate=(1, 1),       # 标准注入
    residual_gate_type='fixed',

    # 优化器剥离外挂
    optimizer_type='adam',
    lr=1e-4,
    layer_weight_decay=1.0,     # 关闭层级惩罚
    seq_weight_decay=1.0,       # 关闭时间步惩罚
    scheduler_type=None,        # 恒定学习率

    # 运行控制
    epochs=50,
```

```
    steps_per_epoch=200,
    print_every=5,

    scheduled_training=True,      # 开启 b 的 Kick-start
    save_path='auto',
    load_path='auto'
)

param_groups = [
    {
        'experiment_name': 'Nonlinear Without Curriculum (Hard Mode)',
        'curriculum': {} # 空字典关闭课程学习
    },
    {
        'experiment_name': 'Nonlinear With Curriculum (Perfect Path)',
        'curriculum': {
            'd_x': 5,
            'seq_len': 10,
            'duration_ratio': 0.8 # 前 40 个 epoch 完成课程
        }
    }
]
results_lists = [
    (['1_ID_Baseline_sink_scores','2_OOD_Scale_x2_sink_scores',␣
 ↪'3_OOD_Seq_Extrapolation_sink_scores'], 'block'),
    (['1_ID_Baseline_y_norm_ratio', '2_OOD_Scale_x2_y_norm_ratio',␣
 ↪'3_OOD_Seq_Extrapolation_y_norm_ratio'], 'experiment'),
    (['1_ID_Baseline_loss', '2_OOD_Scale_x2_loss',␣
 ↪'3_OOD_Seq_Extrapolation_loss'], 'experiment')
]

eval_configs = [
    {'eval_name': '1_ID_Baseline', 'ood_kwargs': {}}, # 分布内基准
    {'eval_name': '2_OOD_Scale_x2', 'ood_kwargs': {'x_scale': 2.0}}, # X 尺度放
大 (测试抗干扰)
```

```
    {'eval_name': '3_OOD_Seq_Extrapolation', 'ood_kwargs': {'seq_len_scale': 1.
 ↪2}} # 序列长度外推（测试算法泛化）
]
```

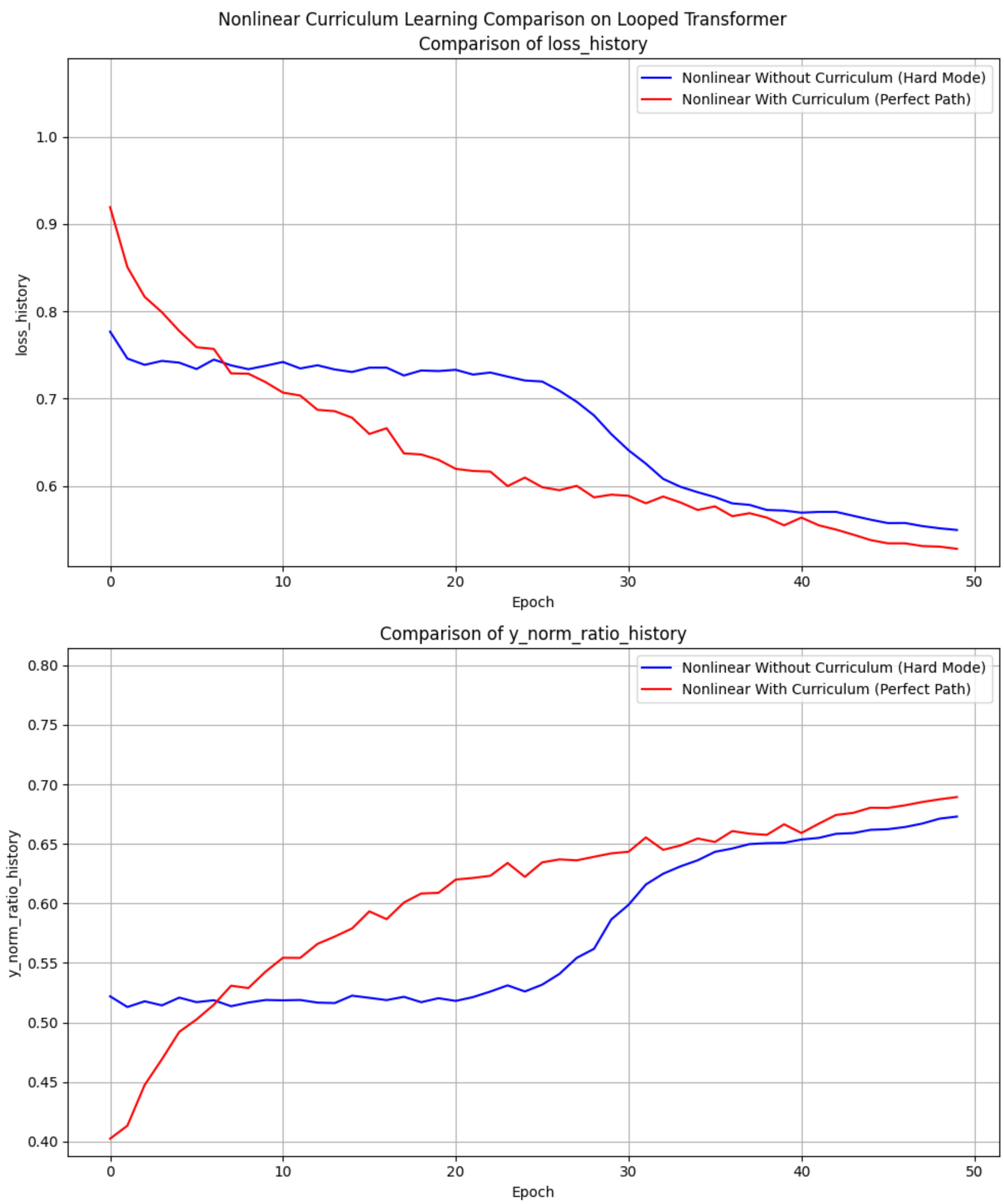
4.2.2 训练

[26]:
```
nonlinear_table = ExperimentTable(params_groups=param_groups,␣
 ↪manual=manual_config)
nonlinear_table.run(result_lists=[(['loss_history'], 'epoch'),␣
 ↪(['y_norm_ratio_history'], 'epoch')])
nonlinear_table.plot(figure_size=(10, 12), compare_experiments=True,␣
 ↪subplot_shape=(-1, 1), suptitle='Nonlinear Curriculum Learning Comparison on␣
 ↪Looped Transformer')
```

```
  找不到文件 saved_checkpoints/Nonlinear_Without_Curriculum_Hard_Mode.pth，跳过加
载。
Nonlinear Without Curriculum (Hard Mode) initialized in 0.49 seconds. Using
device: mps
  找不到文件 saved_checkpoints/Nonlinear_With_Curriculum_Perfect_Path.pth，跳过加
载。
Nonlinear With Curriculum (Perfect Path) initialized in 0.02 seconds. Using
device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.02 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.02 GB
Epoch 5/50, Loss: 0.7411, Norm Ratio: 0.5209, Avg Sink Score: 0.0620, Avg Sink
Rate: 0.0000
Epoch 10/50, Loss: 0.7376, Norm Ratio: 0.5189, Avg Sink Score: 0.0683, Avg Sink
Rate: 0.0000
Epoch 15/50, Loss: 0.7305, Norm Ratio: 0.5225, Avg Sink Score: 0.0745, Avg Sink
Rate: 0.0000
Epoch 20/50, Loss: 0.7315, Norm Ratio: 0.5204, Avg Sink Score: 0.0926, Avg Sink
Rate: 0.0000
Epoch 25/50, Loss: 0.7207, Norm Ratio: 0.5260, Avg Sink Score: 0.0901, Avg Sink
Rate: 0.0000
Epoch 30/50, Loss: 0.6593, Norm Ratio: 0.5865, Avg Sink Score: 0.0537, Avg Sink
Rate: 0.0000
Epoch 35/50, Loss: 0.5928, Norm Ratio: 0.6362, Avg Sink Score: 0.0412, Avg Sink
```

```
Rate: 0.0000
Epoch 40/50, Loss: 0.5718, Norm Ratio: 0.6509, Avg Sink Score: 0.0415, Avg Sink
Rate: 0.0000
Epoch 45/50, Loss: 0.5614, Norm Ratio: 0.6618, Avg Sink Score: 0.0383, Avg Sink
Rate: 0.0000
Epoch 50/50, Loss: 0.5496, Norm Ratio: 0.6730, Avg Sink Score: 0.0425, Avg Sink
Rate: 0.0000
Training completed in 5794.94 seconds.
 Checkpoint 已安全保存至:
saved_checkpoints/Nonlinear_Without_Curriculum_Hard_Mode.pth
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.23 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.01 GB | Metal 驱动分配: 1.05 GB
Experiment 1/2 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 0.02 GB | Metal 驱动分配: 1.07 GB
Epoch 5/50, Loss: 0.7774, Norm Ratio: 0.4921, Avg Sink Score: 0.1907, Avg Sink
Rate: 0.1118
Epoch 10/50, Loss: 0.7188, Norm Ratio: 0.5427, Avg Sink Score: 0.1222, Avg Sink
Rate: 0.0000
Epoch 15/50, Loss: 0.6781, Norm Ratio: 0.5789, Avg Sink Score: 0.0782, Avg Sink
Rate: 0.0000
Epoch 20/50, Loss: 0.6299, Norm Ratio: 0.6089, Avg Sink Score: 0.0646, Avg Sink
Rate: 0.0000
Epoch 25/50, Loss: 0.6095, Norm Ratio: 0.6223, Avg Sink Score: 0.0571, Avg Sink
Rate: 0.0000
Epoch 30/50, Loss: 0.5900, Norm Ratio: 0.6420, Avg Sink Score: 0.0521, Avg Sink
Rate: 0.0000
Epoch 35/50, Loss: 0.5725, Norm Ratio: 0.6545, Avg Sink Score: 0.0479, Avg Sink
Rate: 0.0000
Epoch 40/50, Loss: 0.5549, Norm Ratio: 0.6664, Avg Sink Score: 0.0431, Avg Sink
Rate: 0.0000
Curriculum ended at epoch 41, step 1.
Epoch 45/50, Loss: 0.5380, Norm Ratio: 0.6804, Avg Sink Score: 0.0437, Avg Sink
Rate: 0.0000
Epoch 50/50, Loss: 0.5280, Norm Ratio: 0.6893, Avg Sink Score: 0.0466, Avg Sink
Rate: 0.0000
Training completed in 5309.81 seconds.
```

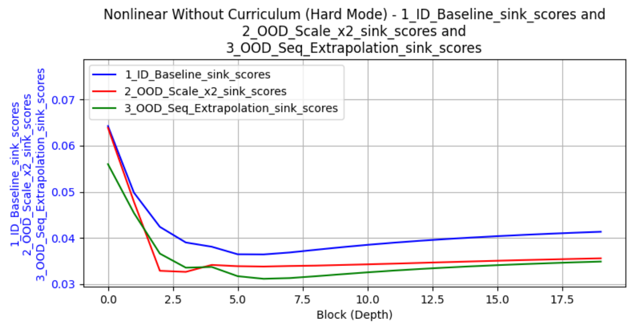
```
 Checkpoint 已安全保存至:
saved_checkpoints/Nonlinear_With_Curriculum_Perfect_Path.pth
[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 0.04 GB | Metal 驱动分配: 5.62 GB
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.07 GB
Experiment 2/2 completed.
```

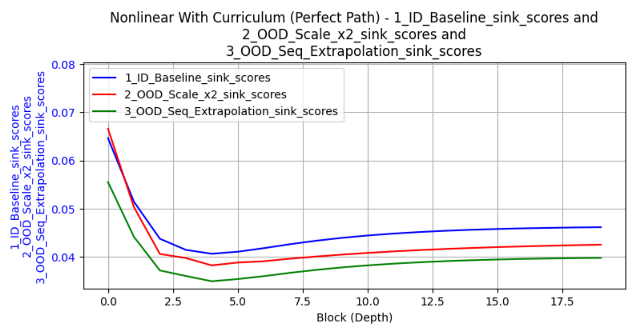


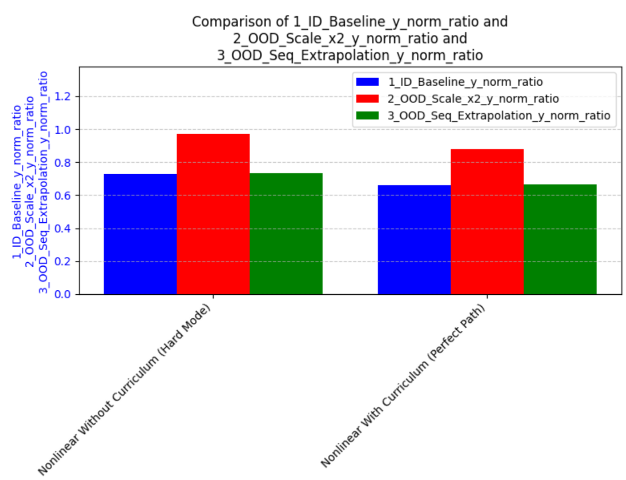
4.2.3 评估

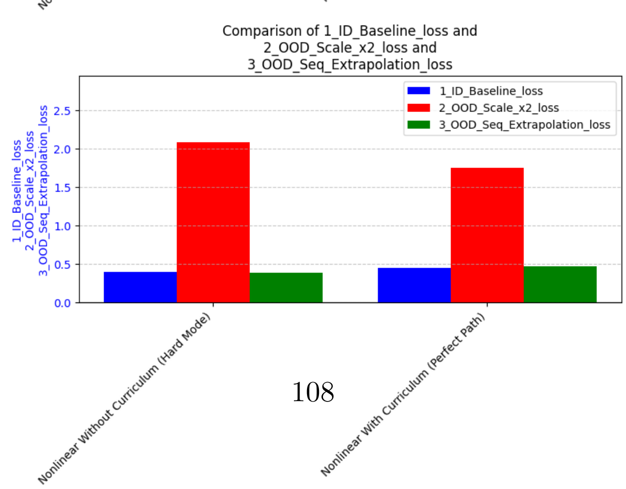
[61]:
```
test_table = ExperimentTable(params_groups=param_groups, manual=manual_config)
test_table.run(result_lists=results_lists, modes=['evaluate'],␣
↪eval_configs=eval_configs)
test_table.plot(compare_experiments=False, subplot_shape=(-1,1),suptitle='')
```

```
成功从 saved_checkpoints/Nonlinear_Without_Curriculum_Hard_Mode.pth 恢复实验状态。

Nonlinear Without Curriculum (Hard Mode) initialized in 0.10 seconds. Using
device: mps
成功从 saved_checkpoints/Nonlinear_With_Curriculum_Perfect_Path.pth 恢复实验状态。

Nonlinear With Curriculum (Perfect Path) initialized in 0.03 seconds. Using
device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.02 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.01 GB | Metal 驱动分配: 0.02 GB
[1_ID_Baseline] Loss: 0.4018, Norm Ratio: 0.7298
[2_OOD_Scale_x2] Loss: 2.0818, Norm Ratio: 0.9717
[3_OOD_Seq_Extrapolation] Loss: 0.3924, Norm Ratio: 0.7338
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.02 GB | Metal 驱动分配: 1.00 GB
Experiment 1/2 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 1.02 GB
[1_ID_Baseline] Loss: 0.4543, Norm Ratio: 0.6583
[2_OOD_Scale_x2] Loss: 1.7523, Norm Ratio: 0.8792
[3_OOD_Seq_Extrapolation] Loss: 0.4746, Norm Ratio: 0.6668
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 0.05 GB | Metal 驱动分配: 1.00 GB
Experiment 2/2 completed.

No Lorenz data cache found.
```


Nonlinear Without Curriculum (Hard Mode) - 1_ID_Baseline_sink_scores and 2_OOD_Scale_x2_sink_scores and 3_OOD_Seq_Extrapolation_sink_scores
1_ID_Baseline_sink_scores
2_OOD_Scale_x2_sink_scores
3_OOD_Seq_Extrapolation_sink_scores
1_ID_Baseline_sink_scores 2_OOD_Scale_x2_sink_scores 3_OOD_Seq_Extrapolation_sink_scores
0.07
0.06
0.05
0.04
0.03
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Block (Depth)


Nonlinear With Curriculum (Perfect Path) - 1_ID_Baseline_sink_scores and 2_OOD_Scale_x2_sink_scores and 3_OOD_Seq_Extrapolation_sink_scores
1_ID_Baseline_sink_scores
2_OOD_Scale_x2_sink_scores
3_OOD_Seq_Extrapolation_sink_scores
1_ID_Baseline_sink_scores 2_OOD_Scale_x2_sink_scores 3_OOD_Seq_Extrapolation_sink_scores
0.08
0.07
0.06
0.05
0.04
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Block (Depth)


Comparison of 1_ID_Baseline_y_norm_ratio and 2_OOD_Scale_x2_y_norm_ratio and 3_OOD_Seq_Extrapolation_y_norm_ratio
1_ID_Baseline_y_norm_ratio
2_OOD_Scale_x2_y_norm_ratio
3_OOD_Seq_Extrapolation_y_norm_ratio
1_ID_Baseline_y_norm_ratio 2_OOD_Scale_x2_y_norm_ratio 3_OOD_Seq_Extrapolation_y_norm_ratio
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Nonlinear Without Curriculum (Hard Mode)
Nonlinear With Curriculum (Perfect Path)


Comparison of 1_ID_Baseline_loss and 2_OOD_Scale_x2_loss and 3_OOD_Seq_Extrapolation_loss
1_ID_Baseline_loss
2_OOD_Scale_x2_loss
3_OOD_Seq_Extrapolation_loss
1_ID_Baseline_loss 2_OOD_Scale_x2_loss 3_OOD_Seq_Extrapolation_loss
2.5
2.0
1.5
1.0
0.5
0.0
Nonlinear Without Curriculum (Hard Mode)
Nonlinear With Curriculum (Perfect Path)

4.3 lorenz attractor

4.3.1 数据集生成

[90]:
```
create_lorenz_pool(pool_size=200000, traj_len=1000, dt=0.01, sigma=10.0, beta=8/
↪3, rho=28.0, save_path='auto')
```

未检测到历史文件，正在创建全新的离线数据池…
数据池已安全写入本地：data/lorenz/length_1000_dt0.01_sigma10.0_beta2.7_rho28.0.pth

4.3.2 Curriculum 对比实验

任务配置

[ ]:
```
MY_LORENZ_POOL_PATH = 'data/lorenz/length_1000_dt0.01_sigma10.0_beta2.7_rho28.0.
↪pth'

manual_config = dict(
    data_type='lorenz',
    d_x=3,                          # 洛伦兹状态空间的绝对物理维度 (X, Y, Z)
    d_y=3,                          # 预测目标同样是完整的三维下一时刻状态
    max_seq_len=500,
    seq_len=300,                     # 上下文包含 150 个 (x, y) 对
    batch_size=64,

    # 模型架构 (完全对齐论文与你的默认优秀设定)
    num_blocks=20,
    num_eff=15,
    d_model=256,
    num_heads=8,
    pe_type=['learned_ape'],        # 使用你在线性任务中表现优异的 Learned APE

    # 核心物理机制配置
    x_init='zero',                  # 绝对零起点，迫使模型完全依赖 Prompt 的 ICL 驱动
    init_std='auto',                # 方差自适应 (1/sqrt(256))
    residual_gate=(1, 1),           # 标准残差注入
    residual_gate_type='fixed',
```

```
    # 接入洛伦兹数据代入的特定物理参数
    lorenz_kwargs=dict(dt=0.01, burn_in=500),
    load_lorenz_from=MY_LORENZ_POOL_PATH,    # 载入你的高密度离线系综池

    # 优化器剥离外挂
    optimizer_type='adam',
    lr=1e-4,
    layer_weight_decay=1.0,      # 关闭层级惩罚
    seq_weight_decay=1.0,        # 关闭时间步惩罚
    scheduler_type=None,         # 恒定学习率

    # 运行控制
    epochs=50,
    steps_per_epoch=50,
    print_every=2,

    scheduled_training=True,      # 开启层数 b 的 Kick-start 渐进
    save_path='auto',
    load_path='auto'
)

param_groups = [
    {
        'experiment_name': 'Lorenz Without Curriculum (Hard Mode)',
        'curriculum': {}          # 空字典关闭课程学习，一上来就啃 80 步长序列
    },
    {
        'experiment_name': 'Lorenz With Curriculum (Perfect Path)',
        #  【理论调整点】: 删除了带有隐患的 'd_x' 渐变，仅对时序步长进行课程演进
        'curriculum': {
            'seq_len': 20,        # 课程从极短的 20 步（10 对点）开始，降低初始积
分关联难度
            'duration_ratio': 0.8 # 前 30 个 epoch 随着线性进度逐步拉长到 300 步
        }
    }
```

```python
]

results_lists = [
    (['1_ID_Baseline_sink_scores','2_OOD_Param_Shift_sink_scores',␣
 ↪'3_OOD_Seq_Extrapolation_sink_scores'], 'block'),
    (['1_ID_Baseline_y_norm_ratio', '2_OOD_Param_Shift_y_norm_ratio',␣
 ↪'3_OOD_Seq_Extrapolation_y_norm_ratio'], 'experiment'),
    (['1_ID_Baseline_loss', '2_OOD_Param_Shift_loss',␣
 ↪'3_OOD_Seq_Extrapolation_loss'], 'experiment')
]

# 针对混沌特征量身定制的 OOD 测试集
eval_configs = [
    # 1. 分布内基准 (使用在线即时生成，检验模型对流形任一位置的单步预测)
    {'eval_name': '1_ID_Baseline', 'ood_kwargs': {}},

    # 2. 物理流形参数偏移 (Parameter Shift)
    # 将瑞利数 rho 向上偏移 5.0 (变成 33.0)，测试网络是否掌握了隐式微分算子，还是只会背吸引子几何外形
    {'eval_name': '2_OOD_Param_Shift', 'ood_kwargs': {'rho_shift': 5.0}},

    # 3. 混沌时序长度外推 (Sequence Extrapolation)
    # 将序列长度外推 20%，跨入更长历史的确定性窗口，测试注意力机制的泛化边界
    {'eval_name': '3_OOD_Seq_Extrapolation', 'ood_kwargs': {'seq_len_scale': 1.
 ↪2}}
]
```

训练

[ ]:
```python
lorenz_table = ExperimentTable(params_groups=param_groups, manual=manual_config)
lorenz_table.run(result_lists=[(['loss_history'], 'epoch'),␣
 ↪(['y_norm_ratio_history'], 'epoch')])
lorenz_table.plot(subplot_shape=(-1, 1),
                  suptitle='Lorenz Attractor Dynamic Curriculum Learning␣
 ↪Comparison')
```

找不到文件 saved_checkpoints/Lorenz_Without_Curriculum_Hard_Mode.pth，跳过加载。
Lorenz Without Curriculum (Hard Mode) initialized in 0.07 seconds. Using device:

mps

找不到文件 saved_checkpoints/Lorenz_With_Curriculum_Perfect_Path.pth，跳过加载。
Lorenz With Curriculum (Perfect Path) initialized in 0.02 seconds. Using device: mps

[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.02 GB

[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.02 GB

Loading Lorenz data from data/lorenz/length_1000_dt0.01_sigma10.0_beta2.7_rho28.0.pth…

Epoch 2/50, Loss: 8.6358, Norm Ratio: 0.5641, Avg Sink Score: 0.0232, Avg Sink Rate: 0.0000

Epoch 4/50, Loss: 5.7645, Norm Ratio: 0.6464, Avg Sink Score: 0.0221, Avg Sink Rate: 0.0000

Epoch 6/50, Loss: 4.0042, Norm Ratio: 0.6990, Avg Sink Score: 0.0215, Avg Sink Rate: 0.0000

Epoch 8/50, Loss: 3.2121, Norm Ratio: 0.7480, Avg Sink Score: 0.0219, Avg Sink Rate: 0.0000

Epoch 10/50, Loss: 2.7072, Norm Ratio: 0.7896, Avg Sink Score: 0.0229, Avg Sink Rate: 0.0000

Epoch 12/50, Loss: 1.9884, Norm Ratio: 0.8238, Avg Sink Score: 0.0228, Avg Sink Rate: 0.0000

Epoch 14/50, Loss: 1.4649, Norm Ratio: 0.8543, Avg Sink Score: 0.0243, Avg Sink Rate: 0.0000

Epoch 16/50, Loss: 1.0976, Norm Ratio: 0.8794, Avg Sink Score: 0.0235, Avg Sink Rate: 0.0000

Epoch 18/50, Loss: 0.8508, Norm Ratio: 0.8959, Avg Sink Score: 0.0237, Avg Sink Rate: 0.0000

Epoch 20/50, Loss: 0.6910, Norm Ratio: 0.9148, Avg Sink Score: 0.0223, Avg Sink Rate: 0.0000

Epoch 22/50, Loss: 0.4726, Norm Ratio: 0.9348, Avg Sink Score: 0.0212, Avg Sink Rate: 0.0000

Epoch 24/50, Loss: 0.3905, Norm Ratio: 0.9412, Avg Sink Score: 0.0219, Avg Sink Rate: 0.0000

Epoch 26/50, Loss: 0.3619, Norm Ratio: 0.9487, Avg Sink Score: 0.0212, Avg Sink Rate: 0.0000

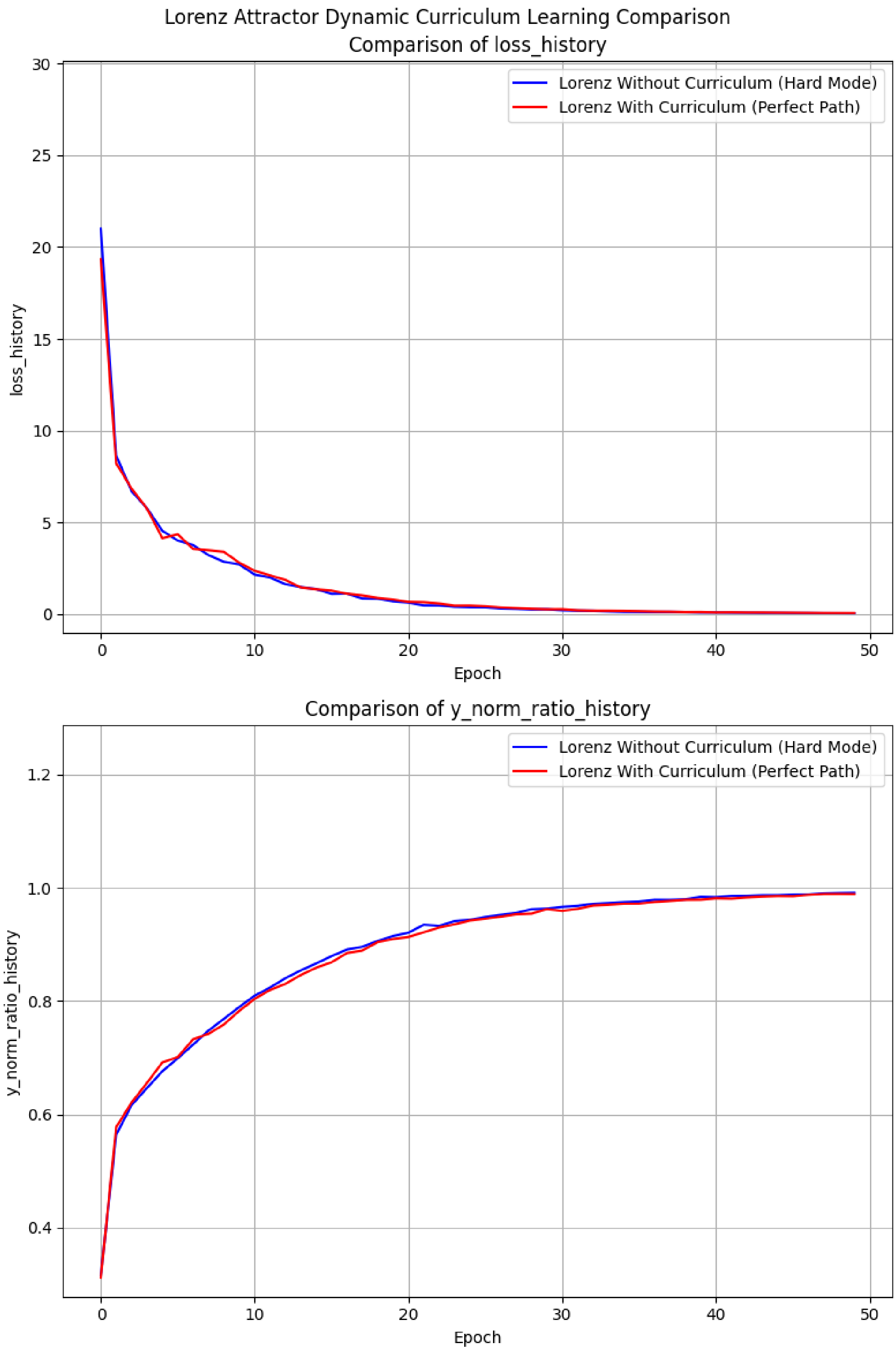
Epoch 28/50, Loss: 0.2792, Norm Ratio: 0.9556, Avg Sink Score: 0.0210, Avg Sink Rate: 0.0000

Epoch 30/50, Loss: 0.2572, Norm Ratio: 0.9632, Avg Sink Score: 0.0231, Avg Sink

```
Rate: 0.0000
Epoch 32/50, Loss: 0.1830, Norm Ratio: 0.9680, Avg Sink Score: 0.0247, Avg Sink
Rate: 0.0000
Epoch 34/50, Loss: 0.1476, Norm Ratio: 0.9732, Avg Sink Score: 0.0215, Avg Sink
Rate: 0.0000
Epoch 36/50, Loss: 0.1190, Norm Ratio: 0.9759, Avg Sink Score: 0.0225, Avg Sink
Rate: 0.0000
Epoch 38/50, Loss: 0.1158, Norm Ratio: 0.9787, Avg Sink Score: 0.0223, Avg Sink
Rate: 0.0000
Epoch 40/50, Loss: 0.0826, Norm Ratio: 0.9838, Avg Sink Score: 0.0228, Avg Sink
Rate: 0.0000
Epoch 42/50, Loss: 0.0744, Norm Ratio: 0.9856, Avg Sink Score: 0.0222, Avg Sink
Rate: 0.0000
Epoch 44/50, Loss: 0.0618, Norm Ratio: 0.9870, Avg Sink Score: 0.0236, Avg Sink
Rate: 0.0000
Epoch 46/50, Loss: 0.0562, Norm Ratio: 0.9881, Avg Sink Score: 0.0220, Avg Sink
Rate: 0.0000
Epoch 48/50, Loss: 0.0479, Norm Ratio: 0.9902, Avg Sink Score: 0.0219, Avg Sink
Rate: 0.0000
Epoch 50/50, Loss: 0.0425, Norm Ratio: 0.9912, Avg Sink Score: 0.0221, Avg Sink
Rate: 0.0000
Training completed in 9046.99 seconds.
  Checkpoint 已安全保存至: saved_checkpoints/Lorenz_Without_Curriculum_Hard_Mode.
  ↪pth
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.06 GB | Metal 驱动分配: 10.41␣
  ↪GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.17 GB | Metal 驱动分配: 1.18 GB
Experiment 1/2 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 0.18 GB | Metal 驱动分配: 1.20 GB
Epoch 2/50, Loss: 8.2145, Norm Ratio: 0.5773, Avg Sink Score: 0.1383, Avg Sink
Rate: 0.0000
Epoch 4/50, Loss: 5.7493, Norm Ratio: 0.6555, Avg Sink Score: 0.1041, Avg Sink
Rate: 0.0000
Epoch 6/50, Loss: 4.3482, Norm Ratio: 0.7012, Avg Sink Score: 0.0831, Avg Sink
Rate: 0.0000
```

```
Epoch 8/50, Loss: 3.4763, Norm Ratio: 0.7422, Avg Sink Score: 0.0709, Avg Sink
Rate: 0.0000
Epoch 10/50, Loss: 2.7985, Norm Ratio: 0.7827, Avg Sink Score: 0.0623, Avg Sink
Rate: 0.0000
Epoch 12/50, Loss: 2.1080, Norm Ratio: 0.8196, Avg Sink Score: 0.0540, Avg Sink
Rate: 0.0000
Epoch 14/50, Loss: 1.4497, Norm Ratio: 0.8460, Avg Sink Score: 0.0497, Avg Sink
Rate: 0.0000
Epoch 16/50, Loss: 1.2825, Norm Ratio: 0.8685, Avg Sink Score: 0.0449, Avg Sink
Rate: 0.0000
Epoch 18/50, Loss: 1.0158, Norm Ratio: 0.8892, Avg Sink Score: 0.0417, Avg Sink
Rate: 0.0000
Epoch 20/50, Loss: 0.7970, Norm Ratio: 0.9096, Avg Sink Score: 0.0383, Avg Sink
Rate: 0.0000
Epoch 22/50, Loss: 0.6451, Norm Ratio: 0.9214, Avg Sink Score: 0.0362, Avg Sink
Rate: 0.0000
Epoch 24/50, Loss: 0.4623, Norm Ratio: 0.9355, Avg Sink Score: 0.0332, Avg Sink
Rate: 0.0000
Epoch 26/50, Loss: 0.4306, Norm Ratio: 0.9455, Avg Sink Score: 0.0319, Avg Sink
Rate: 0.0000
Epoch 28/50, Loss: 0.3244, Norm Ratio: 0.9533, Avg Sink Score: 0.0297, Avg Sink
Rate: 0.0000
Epoch 30/50, Loss: 0.2559, Norm Ratio: 0.9622, Avg Sink Score: 0.0283, Avg Sink
Rate: 0.0000
Epoch 32/50, Loss: 0.2003, Norm Ratio: 0.9628, Avg Sink Score: 0.0272, Avg Sink
Rate: 0.0000
Epoch 34/50, Loss: 0.1793, Norm Ratio: 0.9700, Avg Sink Score: 0.0263, Avg Sink
Rate: 0.0000
Epoch 36/50, Loss: 0.1559, Norm Ratio: 0.9720, Avg Sink Score: 0.0251, Avg Sink
Rate: 0.0000
Epoch 38/50, Loss: 0.1283, Norm Ratio: 0.9769, Avg Sink Score: 0.0244, Avg Sink
Rate: 0.0000
Epoch 40/50, Loss: 0.1124, Norm Ratio: 0.9791, Avg Sink Score: 0.0236, Avg Sink
Rate: 0.0000
Curriculum ended at epoch 41, step 1.
Epoch 42/50, Loss: 0.0985, Norm Ratio: 0.9809, Avg Sink Score: 0.0233, Avg Sink
Rate: 0.0000
```

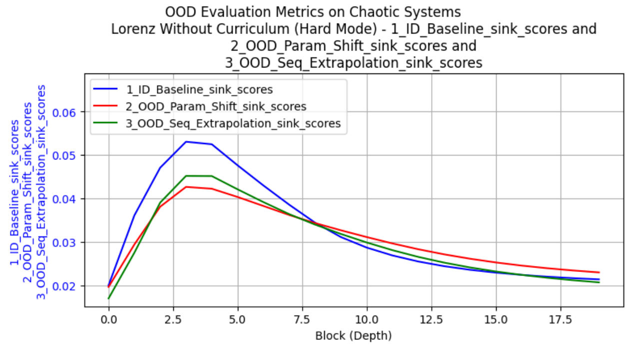
```
Epoch 44/50, Loss: 0.0881, Norm Ratio: 0.9843, Avg Sink Score: 0.0233, Avg Sink
Rate: 0.0000
Epoch 46/50, Loss: 0.0754, Norm Ratio: 0.9851, Avg Sink Score: 0.0228, Avg Sink
Rate: 0.0000
Epoch 48/50, Loss: 0.0610, Norm Ratio: 0.9889, Avg Sink Score: 0.0231, Avg Sink
Rate: 0.0000
Epoch 50/50, Loss: 0.0577, Norm Ratio: 0.9888, Avg Sink Score: 0.0228, Avg Sink
Rate: 0.0000
Training completed in 4855.19 seconds.
 Checkpoint 已安全保存至: saved_checkpoints/Lorenz_With_Curriculum_Perfect_Path.
 ↪pth
[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 0.23 GB | Metal 驱动分配: 25.95␣
 ↪GB
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 0.34 GB | Metal 驱动分配: 17.32 GB
Experiment 2/2 completed.

Lorenz data cache cleared.
```

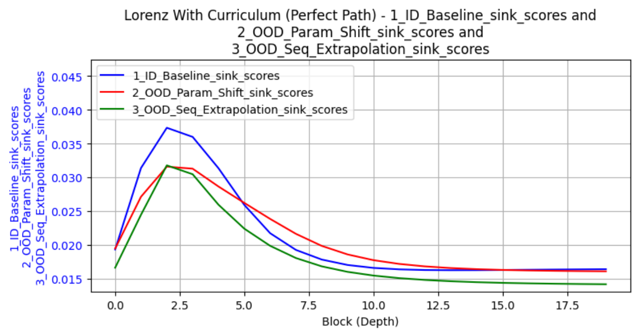

Lorenz Attractor Dynamic Curriculum Learning Comparison
Comparison of loss_history
Lorenz Without Curriculum (Hard Mode)
Lorenz With Curriculum (Perfect Path)
loss_history
Epoch
Comparison of y_norm_ratio_history
Lorenz Without Curriculum (Hard Mode)
Lorenz With Curriculum (Perfect Path)
y_norm_ratio_history
Epoch

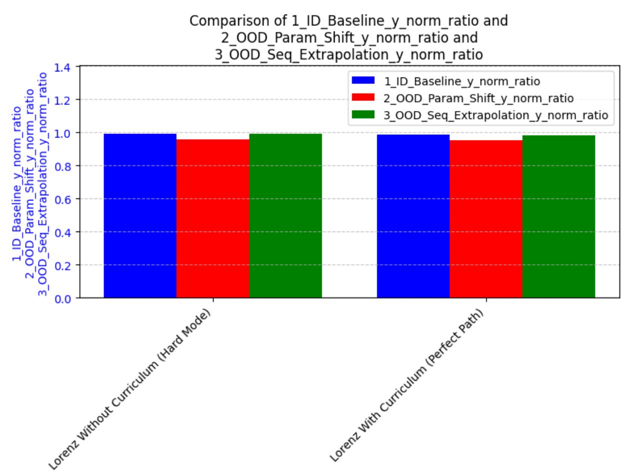
评估

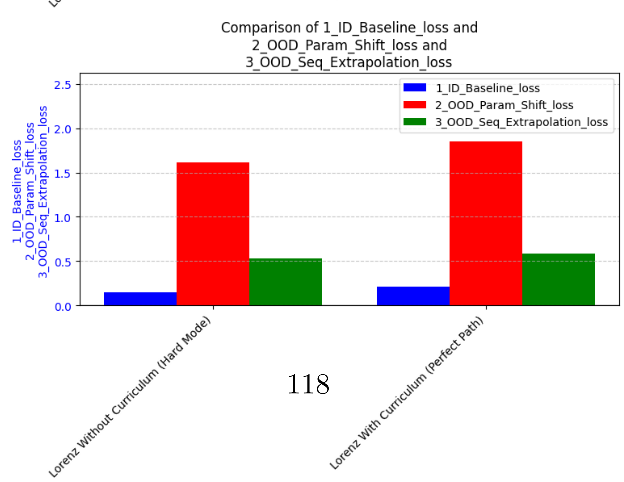
```
[ ]: test_table = ExperimentTable(params_groups=param_groups, manual=manual_config)
test_table.run(result_lists=results_lists, modes=['evaluate'],␣
 ↪eval_configs=eval_configs, parallel_workers=2)
test_table.plot(compare_experiments=False, subplot_shape=(-1,1), suptitle='OOD␣
 ↪Evaluation Metrics on Chaotic Systems')
```

```
 成功从 saved_checkpoints/Lorenz_Without_Curriculum_Hard_Mode.pth 恢复实验状态。
Lorenz Without Curriculum (Hard Mode) initialized in 0.12 seconds. Using device:
mps
 成功从 saved_checkpoints/Lorenz_With_Curriculum_Perfect_Path.pth 恢复实验状态。
Lorenz With Curriculum (Perfect Path) initialized in 0.07 seconds. Using device:
mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.34 GB | Metal 驱动分配: 8.13 GB
Running 2 experiments in parallel with 2 workers…
[Parallel 峰值显存检测] 显存占用 -> PyTorch 分配: 0.37 GB | Metal 驱动分配: 8.14 GB
[1_ID_Baseline] Loss: 0.2062, Norm Ratio: 0.9849
[1_ID_Baseline] Loss: 0.1474, Norm Ratio: 0.9924
[2_OOD_Param_Shift] Loss: 1.8554, Norm Ratio: 0.9527
[2_OOD_Param_Shift] Loss: 1.6128, Norm Ratio: 0.9595
[3_OOD_Seq_Extrapolation] Loss: 0.5308, Norm Ratio:
0.9916[3_OOD_Seq_Extrapolation] Loss: 0.5813, Norm Ratio: 0.9828

No Lorenz data cache found.
```


OOD Evaluation Metrics on Chaotic Systems
Lorenz Without Curriculum (Hard Mode) - 1_ID_Baseline_sink_scores and
2_OOD_Param_Shift_sink_scores and
3_OOD_Seq_Extrapolation_sink_scores
1_ID_Baseline_sink_scores
2_OOD_Param_Shift_sink_scores
3_OOD_Seq_Extrapolation_sink_scores
Block (Depth)


Lorenz With Curriculum (Perfect Path) - 1_ID_Baseline_sink_scores and
2_OOD_Param_Shift_sink_scores and
3_OOD_Seq_Extrapolation_sink_scores
1_ID_Baseline_sink_scores
2_OOD_Param_Shift_sink_scores
3_OOD_Seq_Extrapolation_sink_scores
Block (Depth)


Comparison of 1_ID_Baseline_y_norm_ratio and
2_OOD_Param_Shift_y_norm_ratio and
3_OOD_Seq_Extrapolation_y_norm_ratio
1_ID_Baseline_y_norm_ratio
2_OOD_Param_Shift_y_norm_ratio
3_OOD_Seq_Extrapolation_y_norm_ratio
Lorenz Without Curriculum (Hard Mode)
Lorenz With Curriculum (Perfect Path)


Comparison of 1_ID_Baseline_loss and
2_OOD_Param_Shift_loss and
3_OOD_Seq_Extrapolation_loss
1_ID_Baseline_loss
2_OOD_Param_Shift_loss
3_OOD_Seq_Extrapolation_loss
Lorenz Without Curriculum (Hard Mode)
Lorenz With Curriculum (Perfect Path)

分析

1. curriculum learning 是否添加对最终损失情况影响不大（两实验均收敛良好），但能减少几乎一半的训练时间
2. 对于 OOD 评估，norm_ratio 几乎均维持在 0.95~1.0 的高水平，说明模型学到的规律具有较好的泛化能力，能适应物理参数的偏移和长序列外推。
3. attention sink 峰值在 3～4 层出现，后面层的 sink rate 反而有所下降

4.4 三大实验的统一分析

4.4.1 实验对象配置

[29]:
```
manual=dict(
    batch_size=64,
    num_blocks=20,
    num_eff=15,
    d_model=256,
    num_heads=8,
    pe_type=['learned_ape'],
    x_init='zero',
    init_std='auto',
    residual_gate=(1, 1),
    residual_gate_type='fixed',
    optimizer_type='adam',
    lr=1e-4,
    layer_weight_decay=1.0,
    seq_weight_decay=1.0,
    scheduler_type=None,
    scheduled_training=True
)
manual_linear = dict(
    data_type='linear',
    d_x=20,
    seq_len=80,
    epochs=50,
```

```
    steps_per_epoch=200,
    print_every=1
)
manual_nonlinear = dict(
    data_type='nonlinear',
    function_callable=lambda x: 2**0.5 *F.relu(x),
    max_seq_len=200,
    d_x=20,
    seq_len=80,
    epochs=50,
    steps_per_epoch=200,
    print_every=5,
    save_path='auto',
    load_path='auto'
)
MY_LORENZ_POOL_PATH = 'data/lorenz/length_1000_dt0.01_sigma10.0_beta2.7_rho28.0.
 ↪pth'
manual_lorenz = dict(
    data_type='lorenz',
    d_x=3,                          # 洛伦兹状态空间的绝对物理维度 (X, Y, Z)
    d_y=3,                          # 预测目标同样是完整的三维下一时刻状态
    max_seq_len=500,
    seq_len=300,                     # 上下文包含 150 个 (x, y) 对
    lorenz_kwargs=dict(dt=0.01, burn_in=500),
    load_lorenz_from=MY_LORENZ_POOL_PATH,    # 载入你的高密度离线系综池
    epochs=50,
    steps_per_epoch=50,
    print_every=2,
    save_path='auto',
    load_path='auto'
)
# 使用字典解包 ** 完美融合配置，避免 update() 返回 None 的灾难
params_groups = [
    {
        **manual_linear,
        'experiment_name': 'Linear Without Curriculum (Hard Mode)',
```

```
        'curriculum': {},
        'load_path': 'saved_checkpoints/Without_Curriculum_Hard_Mode_weights.
↪pth'
    },
    {
        **manual_linear,
        'experiment_name': 'Linear With Curriculum (Perfect Path)',
        'curriculum': {
            'd_x': 5,
            'seq_len': 10,
            'duration_ratio': 0.8
        },
        'load_path': 'saved_checkpoints/With_Curriculum_Perfect_Path_weights.
↪pth'
    },
    {
        **manual_nonlinear,
        'experiment_name': 'Nonlinear Without Curriculum (Hard Mode)',
        'curriculum': {}
    },
    {
        **manual_nonlinear,
        'experiment_name': 'Nonlinear With Curriculum (Perfect Path)',
        'curriculum': {
            'd_x': 5,
            'seq_len': 10,
            'duration_ratio': 0.8
        }
    },
    {
        **manual_lorenz,
        'experiment_name': 'Lorenz Without Curriculum (Hard Mode)',
        'curriculum': {}
    },
    {
        **manual_lorenz,
```

```
        'experiment_name': 'Lorenz With Curriculum (Perfect Path)',
        'curriculum': {
            'seq_len': 20,
            'duration_ratio': 0.8
        }
    }
]
eval_configs = [
    {'eval_name': '1_ID_Baseline', 'ood_kwargs': {}}, # 分布内基准
    {'eval_name': '2_OOD_Scale_x2_or_Param_Shift', 'ood_kwargs': {'x_scale': 2.
↪0, 'rho_shift': 5.0}}, # X 尺度放大（测试抗干扰）
    {'eval_name': '3_OOD_Seq_Extrapolation', 'ood_kwargs': {'seq_len_scale': 1.
↪2}} # 序列长度外推（测试算法泛化）
]
result_lists = [
    (['1_ID_Baseline_sink_scores','2_OOD_Scale_x2_or_Param_Shift_sink_scores',␣
↪'3_OOD_Seq_Extrapolation_sink_scores'], 'block'),
    (['1_ID_Baseline_y_norm_ratio',␣
↪'2_OOD_Scale_x2_or_Param_Shift_y_norm_ratio',␣
↪'3_OOD_Seq_Extrapolation_y_norm_ratio'], 'experiment'),
    (['1_ID_Baseline_y_cos', '2_OOD_Scale_x2_or_Param_Shift_y_cos',␣
↪'3_OOD_Seq_Extrapolation_y_cos'], 'experiment'),
    (['1_ID_Baseline_loss', '2_OOD_Scale_x2_or_Param_Shift_loss',␣
↪'3_OOD_Seq_Extrapolation_loss'], 'experiment')
]
```

4.4.2 运行实验

[ ]:
```
unified_table = ExperimentTable(params_groups=params_groups, manual=manual)
unified_table.run(result_lists=result_lists, modes=['evaluate'],␣
↪eval_configs=eval_configs, parallel_workers=1)
```

```
检测到纯权重 state_dict 文件（或历史版本），直接加载 model_state_dict…
 成功从 saved_checkpoints/Without_Curriculum_Hard_Mode_weights.pth 恢复实验状态。
Linear Without Curriculum (Hard Mode) initialized in 0.56 seconds. Using device:
mps
检测到纯权重 state_dict 文件（或历史版本），直接加载 model_state_dict…
```

成功从 saved_checkpoints/With_Curriculum_Perfect_Path_weights.pth 恢复实验状态。
Linear With Curriculum (Perfect Path) initialized in 0.03 seconds. Using device: mps
成功从 saved_checkpoints/Nonlinear_Without_Curriculum_Hard_Mode.pth 恢复实验状态。

Nonlinear Without Curriculum (Hard Mode) initialized in 0.04 seconds. Using device: mps
成功从 saved_checkpoints/Nonlinear_With_Curriculum_Perfect_Path.pth 恢复实验状态。

Nonlinear With Curriculum (Perfect Path) initialized in 0.03 seconds. Using device: mps
成功从 saved_checkpoints/Lorenz_Without_Curriculum_Hard_Mode.pth 恢复实验状态。
Lorenz Without Curriculum (Hard Mode) initialized in 0.03 seconds. Using device: mps
成功从 saved_checkpoints/Lorenz_With_Curriculum_Perfect_Path.pth 恢复实验状态。
Lorenz With Curriculum (Perfect Path) initialized in 0.03 seconds. Using device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.02 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.02 GB
[1_ID_Baseline] [Linear Without Curriculum (Hard Mode)]Loss: 0.7418, Norm Ratio: 0.0097, Norm Cosine Similarity: 0.2244
[2_OOD_Scale_x2_or_Param_Shift] [Linear Without Curriculum (Hard Mode)]Loss: 4.5082, Norm Ratio: 0.0054, Norm Cosine Similarity: 0.0313
[3_OOD_Seq_Extrapolation] [Linear Without Curriculum (Hard Mode)]Loss: 1.1665, Norm Ratio: 0.0102, Norm Cosine Similarity: -0.0405
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.00 GB
Experiment 1/6 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.02 GB
[1_ID_Baseline] [Linear With Curriculum (Perfect Path)]Loss: 0.0728, Norm Ratio: 1.0402, Norm Cosine Similarity: 0.9538
[2_OOD_Scale_x2_or_Param_Shift] [Linear With Curriculum (Perfect Path)]Loss: 0.5680, Norm Ratio: 0.8993, Norm Cosine Similarity: 0.9356
[3_OOD_Seq_Extrapolation] [Linear With Curriculum (Perfect Path)]Loss: 0.0907, Norm Ratio: 0.9985, Norm Cosine Similarity: 0.9610
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.00 GB

```
Experiment 2/6 completed.

[Experiment 3 训练前] 显存占用 -> PyTorch 分配: 0.01 GB | Metal 驱动分配: 0.02 GB
[1_ID_Baseline] [Nonlinear Without Curriculum (Hard Mode)]Loss: 0.4018, Norm
Ratio: 0.7298, Norm Cosine Similarity: 0.8001
[2_OOD_Scale_x2_or_Param_Shift] [Nonlinear Without Curriculum (Hard Mode)]Loss:
2.0818, Norm Ratio: 0.9717, Norm Cosine Similarity: 0.6234
[3_OOD_Seq_Extrapolation] [Nonlinear Without Curriculum (Hard Mode)]Loss:
0.3924, Norm Ratio: 0.7338, Norm Cosine Similarity: 0.8366
[Experiment 3 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.00 GB
Experiment 3/6 completed.

[Experiment 4 训练前] 显存占用 -> PyTorch 分配: 0.01 GB | Metal 驱动分配: 0.02 GB
[1_ID_Baseline] [Nonlinear With Curriculum (Perfect Path)]Loss: 0.4543, Norm
Ratio: 0.6583, Norm Cosine Similarity: 0.7755
[2_OOD_Scale_x2_or_Param_Shift] [Nonlinear With Curriculum (Perfect Path)]Loss:
1.7523, Norm Ratio: 0.8792, Norm Cosine Similarity: 0.6576
[3_OOD_Seq_Extrapolation] [Nonlinear With Curriculum (Perfect Path)]Loss:
0.4746, Norm Ratio: 0.6668, Norm Cosine Similarity: 0.8015
[Experiment 4 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.00 GB
Experiment 4/6 completed.

[Experiment 5 训练前] 显存占用 -> PyTorch 分配: 0.01 GB | Metal 驱动分配: 0.02 GB
[1_ID_Baseline] [Lorenz Without Curriculum (Hard Mode)]Loss: 0.1474, Norm Ratio:
0.9924, Norm Cosine Similarity: 0.9997
[2_OOD_Scale_x2_or_Param_Shift] [Lorenz Without Curriculum (Hard Mode)]Loss:
1.6128, Norm Ratio: 0.9595, Norm Cosine Similarity: 0.9985
[3_OOD_Seq_Extrapolation] [Lorenz Without Curriculum (Hard Mode)]Loss: 0.5308,
Norm Ratio: 0.9916, Norm Cosine Similarity: 0.9990
[Experiment 5 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.00 GB
Experiment 5/6 completed.

[Experiment 6 训练前] 显存占用 -> PyTorch 分配: 0.01 GB | Metal 驱动分配: 0.02 GB
[1_ID_Baseline] [Lorenz With Curriculum (Perfect Path)]Loss: 0.2062, Norm Ratio:
0.9849, Norm Cosine Similarity: 0.9997
[2_OOD_Scale_x2_or_Param_Shift] [Lorenz With Curriculum (Perfect Path)]Loss:
1.8554, Norm Ratio: 0.9527, Norm Cosine Similarity: 0.9984
```

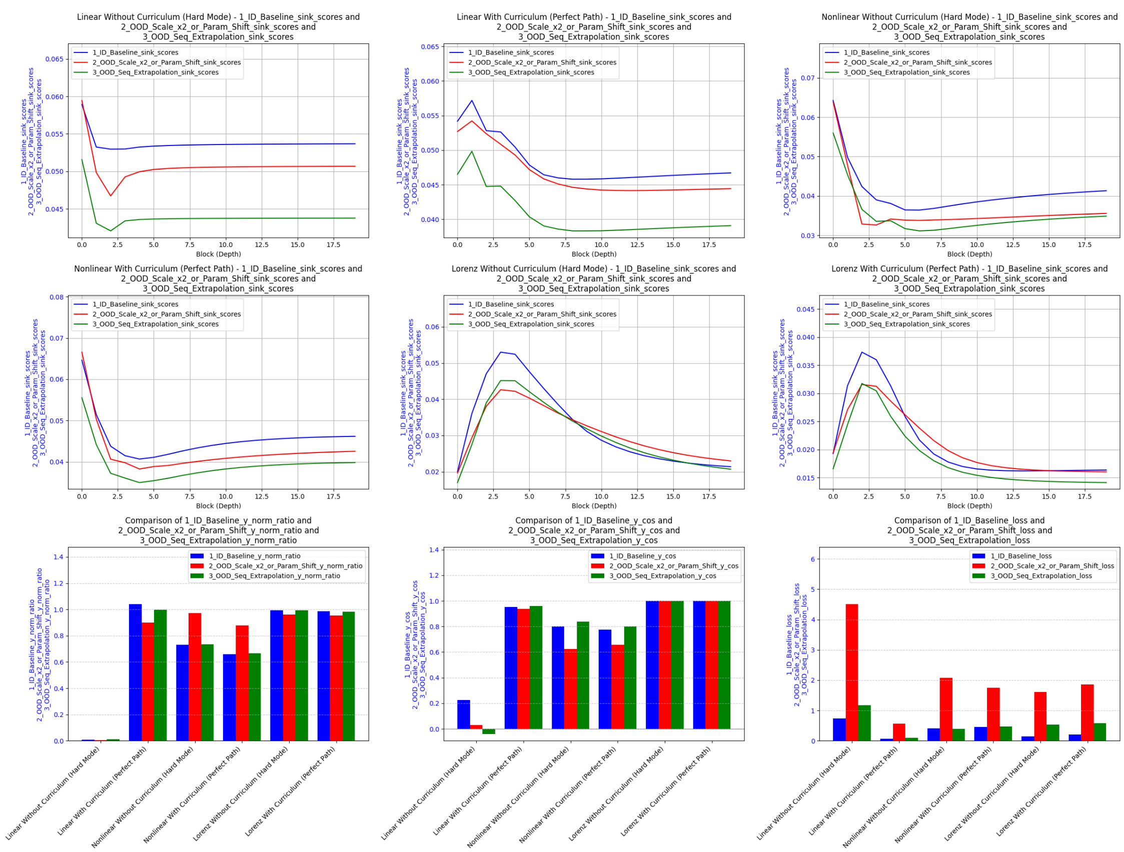
```
[3_OOD_Seq_Extrapolation] [Lorenz With Curriculum (Perfect Path)]Loss: 0.5813,
Norm Ratio: 0.9828, Norm Cosine Similarity: 0.9990
[Experiment 6 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.00 GB
Experiment 6/6 completed.
```

4.4.3 画图

折线图/柱状图

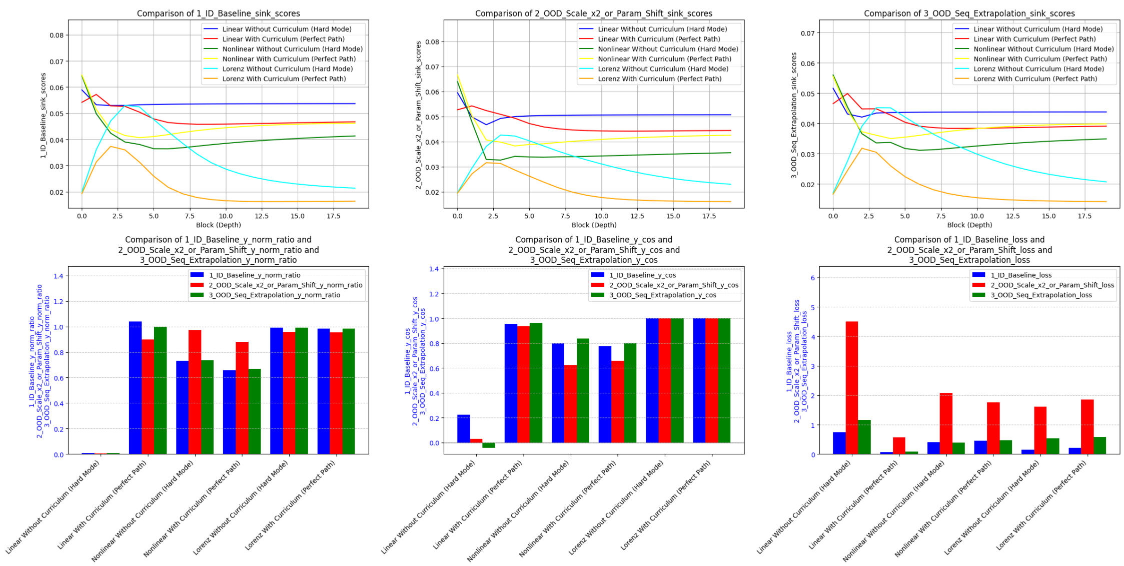
[ ]:
```python
unified_table.plot(compare_experiments=False, subplot_shape=(3,-1),
    suptitle=None)
```

```
No Lorenz data cache found.
```



[ ]:
```python
unified_table.plot(compare_experiments=True, subplot_shape=(2,-1),
    suptitle=None)
```

```
No Lorenz data cache found.
```



散点图

函数定义

[37]:
```python
def plot_unified_dots(target_table):
    """
    将所有任务、所有场景的 18 个点绘制在统一的对数流形空间中。
    一式两份：左图按 OOD 场景着色（红黄蓝），右图按系统 Data Type 着色（红黄蓝）。
    """
    all_points = []
    # 1. 采用你更新后的 OOD 场景名称
    eval_names = ['1_ID_Baseline', '2_OOD_Scale_x2_or_Param_Shift', '3_OOD_Seq_Extrapolation']

    for i, exp in enumerate(target_table.experiments):
        res = exp.eval_results
        data_type = target_table.train_parameters[i].get('data_type', 'unknown')

        for eval_name in eval_names:
            loss_key = f"{eval_name}_loss"
            ratio_key = f"{eval_name}_y_norm_ratio"
            y_true_key = f"{eval_name}_y_true_norm"
```

```
        if loss_key in res and ratio_key in res and y_true_key in res:
            L_val = res[loss_key]
            L_normalized = L_val / (res[y_true_key] ** 2 + 1e-8)  # 加小常数
避免除零
            R_val = res[ratio_key]

            # 确保进入对数空间的数据严格为正数
            if (L_normalized is not None and R_val is not None and
                not np.isnan(L_normalized) and not np.isnan(R_val) and
                L_normalized > 0 and R_val > 1e-5):
                all_points.append({
                    'R': R_val,
                    'L': L_normalized,
                    'OOD': eval_name,
                    'DataType': data_type
                })

    if len(all_points) < 2:
        print(" 错误：有效数据点不足，请先确认 scale_table 是否完整跑完了␣
↪evaluate 流程。")
        return

    # 2. 全局散点图
    Rs = np.array([p['R'] for p in all_points])
    Ls = np.array([p['L'] for p in all_points])

    log_R = np.log(Rs)
    log_L = np.log(Ls)

    print(f"[全局散点图解析成功] 当前激活样本量：{len(all_points)} 个点")

    # 3. 画布构建
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 6))

    #  严格定义红黄蓝（RYB）高亮色板
```

```python
    ryb = {
        'red': '#E41A1C',      # 强对比红
        'yellow': '#FFCC00',  # 高可视度金黄
        'blue': '#377EB8'      # 科技感深蓝
    }

    # ==================== 左图: 按 OOD 类型着色 (红黄蓝) ====================
    ood_map = {
        '1_ID_Baseline': ('1_ID_Baseline', ryb['blue']),
        '2_OOD_Scale_x2_or_Param_Shift': ('2_OOD_Scale_x2_or_Param_Shift', ryb['yellow']),
        '3_OOD_Seq_Extrapolation': ('3_OOD_Seq_Extrapolation', ryb['red'])
    }

    for ood_cat, (label, color) in ood_map.items():
        pts = [p for p in all_points if p['OOD'] == ood_cat]
        if pts:
            ax1.scatter([p['R'] for p in pts], [p['L'] for p in pts],
                        color=color, label=label, s=120, edgecolor='k', alpha=0.9, zorder=3)
    ax1.set_title('Color Coded by OOD Evaluation Type', fontsize=12, fontweight='bold', pad=10)

    # ==================== 右图: 按 Data Type 着色 (红黄蓝) ====================
    dtype_map = {
        'linear': ('LINEAR', ryb['blue']),
        'nonlinear': ('NONLINEAR', ryb['yellow']),
        'lorenz': ('LORENZ', ryb['red'])
    }

    for dtype_cat, (label, color) in dtype_map.items():
        pts = [p for p in all_points if p['DataType'] == dtype_cat]
        if pts:
            ax2.scatter([p['R'] for p in pts], [p['L'] for p in pts],
                        color=color, label=label, s=120, edgecolor='k', alpha=0.9, zorder=3)
```

```python
    ax2.set_title('Color Coded by System Data Type', fontsize=12, fontweight='bold', pad=10)

    # ==================== 统一的双对数坐标物理美学配置 ====================
    for ax in [ax1, ax2]:
        ax.set_xscale('log')
        ax.set_yscale('log')
        ax.set_xlabel('Norm Ratio $R$ ($||\\hat{y}|| / ||y||$)', fontsize=11)
        ax.set_ylabel('Loss $L$', fontsize=11)
        ax.grid(True, which="both", linestyle="--", alpha=0.4, color='gray')
        ax.legend(fontsize=9, loc='best', frameon=True, facecolor='white', edgecolor='gainsboro')

    plt.suptitle("Unified Dots: 18 Checkpoints Log-Log Mapping", fontsize=14, fontweight='bold', y=1.02)
    plt.tight_layout()
    plt.show()
```

图

[ ]:
```python
plot_unified_dots(unified_table)
```

[全局散点图解析成功] 当前激活样本量: 18 个点